Introduction

Thomas H. Jørgensen

2026

Outline

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- **Teacher:** Thomas H. Jørgensen
  (Associate Professor, University of Copenhagen)

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  Dynamic programming in Python
- **Prerequisite:**
  Introduction to Programming and Numerical Analysis

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- **Tools:**
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- **Prerequisite:**
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- Plan for today:
  1. Course description
  2. Programming in Python
  3. Introduction to Dynamic Programming

Teaching Methods

- **Lectures:** Tuesday 10-13
  $\sim$2 hours lecture
  $\sim$1 hour of problem-solving (e.g. programming)

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  2. Discussions of simple *models* to investigate central mechanisms
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  I will provide code.
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- **Material:**
  `https://github.com/ThomasHJorgensen/HouseholdBehaviorCourse`

Course Plan

Introduction

1. Introduction
2. Dynamic Programming and Structural Estimation

Part 1: Labor Supply

3. Static and Dynamic labor supply
.. *No lecture: Work on assignment / read papers*
4. Dynamic labor supply and human capital
5. Career costs of children
6. Household Labor Supply and Taxation
7. Household Labor Supply and Child-Related Transfers

Part 2: Family Formation and Dissolution

8. Models of Household Behavior
9. Divorce Law and Intra-Household Bargaining
10. Marriage and Divorce
11. Fertility and labor supply
12. buffer

Outroduction

13. Children and Time Allocation

Assignments and Exam

- **Exam** (Portfolio):
  1. 3 individual assignments.
     + peer feedback.
  2. 24 hour individual take-home exam.
     Model formulation, code modification, simulations, economic interpretations.

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- 3 individual **assignments** (hand-in on Absalon)
  1. Based on our dynamic **labor supply** model
     Deadline: March 14
  2. Based on our dynamic **household** model
     Deadline: April 10
  3. **Free**: Formulate a research question + model + data.
     Deadline: May 8
     Deadline for peer feedback: May 15
- All feedback can be used to improve assignments before exam date

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Models! What are they good for!?

- We will use **empirical regularities as motivation**

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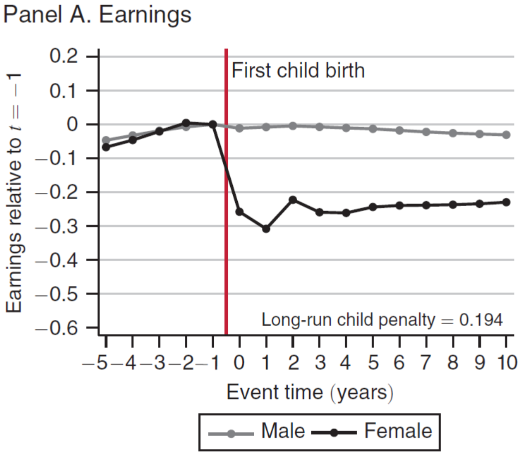
Example: "Child penalty" (Kleven, Landais and Søgaard, 2019)



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  Example: "Career costs of children" (Adda, Dustmann and Stevens, 2017)
  "We estimate a dynamic life cycle model of labor supply, fertility, and savings, incorporating occupational choices, with specific wage paths and skill atrophy that vary over the career. This allows us to understand the trade-off between occupational choice and desired fertility, as well as sorting both into the labor market and across occupations. We **quantify** the life cycle career costs associated with children, how they **decompose** into loss of skills during interruptions, lost earnings opportunities, and selection into more child-friendly occupations. We analyze the **long-run effects of policies** that encourage fertility and show that they are considerably smaller than short-run effects."

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  Equations that describe a small part of reality...

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> *"All models are wrong, but some are useful"*
>
> *George E.P. Box*

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  $\Rightarrow$ policy advice can not rely on estimated behavioral rules
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Models! What are they good for!?

- Why ever use a model then?

- **The Lucas critique:** *Behavioral rules change with policy*
  $\Rightarrow$ policy advice can not rely on estimated behavioral rules (reduced-form estimates using existing variation)

  $\Rightarrow$ we need to estimate *structural (deep) parameters*

  "Invariance of parameters in an economic model is not, of course, a property which can be assured in advance, but it seems reasonable to hope that neither tastes nor technology vary systematically with variations in counter-cyclical policies." (Lucas, 1977)

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  The usefulness of a model lies in which types of mechanisms are included and thus which *counterfactual* scenarios the model can inform about.

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  Implications of not-yet implemented policy reforms (e.g. child subsidies)
  Responses to changes in the economic environment (e.g. wages)
  Quantifying the importance of mechanisms (e.g. risk)

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  But it might be able to predict how changes in the *early retirement scheme* changes *labor supply at older ages*.

→ Tight link between **research question** and model formulation.

Learning Objectives

See all at the Course page

- **Knowledge:**
  Define, formulate and interpret *models* of household behaviour
  Account for backwards induction and how to *solve* dynamic programming models

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- **Skills:**
  Analyze *counterfactual* policy reform simulations from simple and more complex models of household behavior

- **Competences:**
  Discuss and evaluate *research* on household behavior over the life cycle
  Modify computer *code* to analyze small changes to simple models

How to read a research paper (in this course)

- Each lecture will be based on 1 mandatory (*) **research paper**
  What is the main *research question*?
  What is the *(empirical) motivation*?
  What are the central *mechanisms in the model*?
  What is the *simplest model* in which we could capture these?
  Challenging: Research frontier.

How to read a research paper (in this course)

- Each lecture will be based on 1 mandatory (*) **research paper**
  What is the main *research question*?
  What is the *(empirical) motivation*?
  What are the central *mechanisms in the model*?
  What is the *simplest model* in which we could capture these?
  Challenging: Research frontier.

- **How to read** a research paper in this course?
  - **Focus** on the questions above
    How do the questions interact and inform each other?
  - Try **not to get stuck** in too many details!
    (we can discuss some in class if you want)
  - Research papers often include many **"bells and whistles"**
  - Read ~**40 min before** each lecture.
    See reading-guide for each lecture

Outline

1. This Course
2. Models
3. Programming in Python
4. Dynamic Programming

Programming in Python

- **Only very simple**/unrealistic models can be solved with pen/paper
  $\rightarrow$ We need numerical methods

Programming in Python

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- **Purpose** of you coding:
  Learn by implementing!
  Appreciate bottlenecks and challenges
  Better understanding of frontier research
  Set you free...!

Programming in Python

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  $\rightarrow$ We need numerical methods
- **Purpose** of you coding:
  Learn by implementing!
  Appreciate bottlenecks and challenges
  Better understanding of frontier research
  Set you free...!
- **Goal** is to keep things simple!
  No fancy numerical tricks (I have resources if interested)
  Code should be "intuitive"
  $\rightarrow$ slow...

Programming in Python

- **Setup** *Visual Studio Code* as
  Introduction to Programming and Numerical Analysis
  `https://sites.google.com/view/numeconcph-introprog/`
- **Installation** guide:
  `https://sites.google.com/view/numeconcph-introprog/guides/installation`
- **Packages** (all by Jeppe Druedahl):
  EconModel
  consav
  See "`01. Introduction to EconModel and consav.ipynb`"

Outline

Introduction to Dynamic Programming (DP)

- **Agents maximize** expected discounted sum of utility throughout life
  Maximize wrt. $\{\mathcal{C}_t\}_{t=1}^{T}$
  Forward looking $\rightarrow$ dynamic
  Assume *optimal* behavior in all periods

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- **Agents maximize** expected discounted sum of utility throughout life
  Maximize wrt. $\{\mathcal{C}_t\}_{t=1}^{T}$
  Forward looking $\rightarrow$ dynamic
  Assume *optimal* behavior in all periods
- **Bellman Equation**:

$$V_t(\mathcal{S}_t) = \max_{\mathcal{C}_t} U(\mathcal{C}_t, \mathcal{S}_t) + \beta \mathbb{E}[V_{t+1}(\mathcal{S}_{t+1}) | \mathcal{C}_t, \mathcal{S}_t]$$

$$\text{s.t.}$$

$$\mathcal{S}_{t+1} \sim F(\mathcal{C}_t, \mathcal{S}_t)$$

  - $V_t(\mathcal{S}_t)$: Indirect utility, Value today of *states*, $\mathcal{S}_t$ (all relevant info).
  - $U(\mathcal{C}_t, \mathcal{S}_t)$: flow-utility
  - $\beta \mathbb{E}[V_{t+1}(\mathcal{S}_{t+1}) | \mathcal{C}_t, \mathcal{S}_t]$: expected discounted value of next-period
  - $\mathcal{S}_{t+1} \sim F(\mathcal{C}_t, \mathcal{S}_t)$: transition *density* of states (fcn of $\mathcal{C}_t$!)
    (there might be other constraints)

Backwards Induction

- Solved by **backwards induction**

1. Start with last/terminal period, T (no future)

$$V_T(\mathcal{S}_T) = \max_{\mathcal{C}_T} U(\mathcal{C}_T, \mathcal{S}_T)$$

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2. Go to second-to-last (we now know period-T optimal behavior for all $\mathcal{S}_T$)

$$V_{T-1}(\mathcal{S}_{T-1}) = \max_{\mathcal{C}_{T-1}} U(\mathcal{C}_{T-1}, \mathcal{S}_{T-1}) + \beta \mathbb{E}[V_T(\mathcal{S}_T) | \mathcal{C}_{T-1}, \mathcal{S}_{T-1}]$$

s.t.

$$\mathcal{S}_T \sim F(\mathcal{C}_{T-1}, \mathcal{S}_{T-1})$$

Backwards Induction

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3. Continue backwards...

Introduction to Numerical Dynamic Programming (DP)

- **On a computer**, everything is discrete $\rightarrow$ arrays $+$ loops

Introduction to Numerical Dynamic Programming (DP)

- **On a computer**, everything is discrete $\rightarrow$ arrays + loops
- ***Numerical*** Dynamic Programming
  - Backwards induction on arrays
  - Grids
  - Interpolation
  - Integration
- See "`02. Consumption-Saving Model.ipynb`"

Next Time

- **Next time:**
  Dynamic programming with *uncertainty*
  Structural *estimation*

- **Literature:**
  Gourinchas and Parker (2002): "Consumption Over the Life Cycle"
  - **Read** before lecture
  - **Reading guide:**
    Section 1: Introduction – *Key* (page 50 is not that important)
    Section 2: Model – *Key*, we will discuss. Do not get stuck.
    Section 3: Estimation method (SMM). *Key*, we will discuss.
    Section 4: First stage calibrations. Skim fast.
    Section 5: Data. Skim fast.
    Section 6: Results. Focus on 6.1. Figures 5 and 7 are main results.

References I

ADDA, J., C. DUSTMANN AND K. STEVENS (2017): "The Career Costs of Children," *Journal of Political Economy*, 125(2), 293–337.

GOURINCHAS, P.-O. AND J. A. PARKER (2002): "Consumption Over the Life Cycle," *Econometrica*, 70(1), 47–89.

KLEVEN, H. J., C. LANDAIS AND J. E. SØGAARD (2019): "Children and gender inequality: Evidence from Denmark," *American Economic Journal: Applied Economics*, 11, 181–209.

Dynamic Programming and Structural Estimation

Thomas H. Jørgensen

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Stochastic Dynamic Programming

- **Last time:** Dynamic Programming
  Backwards induction
  Grids
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    Future income is uncertain
    + Another state variable: Permanent income
    + "Normalization" of one state variable.

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  - **Estimation:**
    Simulated Method of Moments (SMM/SMD)
    Relate to "reduced-form"
    Can we combine approaches?
    See Eisenhauer, Heckman and Mosso (2015) Todd and Wolpin (2023)

Stochastic Dynamic Programming

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    See Eisenhauer, Heckman and Mosso (2015) Todd and Wolpin (2023)
- **Example:** Buffer Stock model of Deaton (1991); Carroll (1992)
  - Estimated in Gourinchas and Parker (2002)

Gourinchas and Parker (2002)

- **Research question:** “Which savings motives dominate across life?”

Gourinchas and Parker (2002)

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- **Approach:**
  1. **Estimate model** with 2+ motives:
     Buffer-stock motive: Income risk while working.
     Life cycle motive: Consumption in retirement.

Gourinchas and Parker (2002)

- **Research question:** "Which savings motives dominate across life?"
- **Approach:**
  1. **Estimate model** with 2+ motives:
     Buffer-stock motive: Income risk while working.
     Life cycle motive: Consumption in retirement.
  2. **Quantify importance** of these motives over life
     Counterfactual simulations

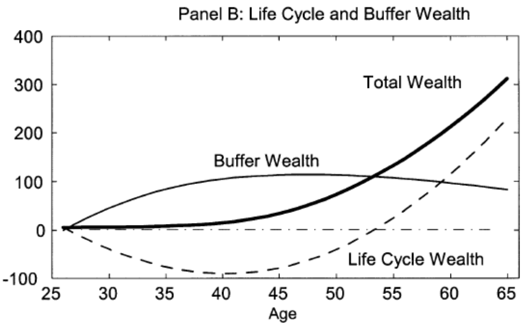

FIGURE 7.—The role of risk in saving and wealth accumulation.

Outline

1. Introduction
2. Stochastic DP
3. Structural Estimation

Buffer-stock model (Deaton-Carroll) Bellman equation

- **Simplest version** of the buffer-stock model is

$$
\begin{aligned}
V_t(M_t, P_t) &= \max_{C_t} \frac{C_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t\left[V_{t+1}(M_{t+1}, P_{t+1})\right] \\
&\text{s.t.} \\
A_t &= M_t - C_t && \text{(assets)} \\
M_{t+1} &= RA_t + Y_{t+1} && \text{(resources/cash-on-hand)} \\
Y_{t+1} &= P_{t+1}\xi_{t+1} && \text{(income)} \\
P_{t+1} &= GP_t\psi_{t+1} && \text{(perm. income)} \\
A_t &\geq 0, \forall t && \text{(no borrowing)}
\end{aligned}
$$

where $\mathbb{E}_t\left[V_{t+1}(M_{t+1}, P_{t+1})\right] = \mathbb{E}\left[V_{t+1}(M_{t+1}, P_{t+1}) | M_t, P_t, C_t\right]$ are expectations over perm. and trans. income shocks,

$$\log \xi_{t+1} \sim \mathcal{N}(\mu_\xi, \sigma_\xi^2), \ \log \psi_{t+1} \sim \mathcal{N}(\mu_\psi, \sigma_\psi^2)$$

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- Gourinchas and Parker (2002): "natural" borrowing constraint. mass-point at zero in trans. income shock distribution, ξ_{t+1}

Buffer-stock model (Deaton-Carroll) Bellman equation

- **Last period:** Everything is consumed,

$$C^{\star}_T(M_T, P_T) = M_T$$
$$V_T(M_T, P_T) = \frac{M_T^{1-\rho}}{1-\rho}$$

Buffer-stock model (Deaton-Carroll) Bellman equation

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$$C_T^{\star}(M_T, P_T) = M_T$$
$$V_T(M_T, P_T) = \frac{M_T^{1-\rho}}{1-\rho}$$

- **Gourinchas and Parker (2002)**: Retirement-periods
Assumes a linear post-retirement value (w. $P_{T+1} = P_T$)

$$V_{T+1}(M_{T+1}, P_{T+1}) = \kappa \cdot (M_{T+1} + h \cdot P_{T+1})$$

Motivated by a deterministic perfect credit market solution (estimate κ and h, through γ_0 and γ_1 – see e.g. Jørgensen and Tô, 2020)

  - They also allow for time-varying taste-shifters, $v_t(Z_t)$.

Normalization I

- **Defining** $c_t \equiv C_t/P_t$, $m_t \equiv M_t/P_t$ etc. implies

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A_t &= M_t - C_t \\
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a_t &= m_t - c_t
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$$a_t = m_t - c_t$$

and state transition

$$M_{t+1} = RA_t + Y_{t+1}$$
$$M_{t+1}/P_{t+1} = RA_t/P_{t+1} + Y_{t+1}/P_{t+1}$$
$$m_{t+1} = Ra_tP_t/P_{t+1} + \xi_{t+1}$$
$$m_{t+1} = \frac{R}{G\psi_{t+1}}a_t + \xi_{t+1}$$

The **adjustment factor** $\frac{1}{G\psi_{t+1}}$ is due to changes in permanent income

Normalization II

- **Defining** $v_t(m_t) = V_t(M_t, P_t)/P_t^{1-\rho}$ implies

$$
\begin{aligned}
V_t(M_t, P_t) &= \max_{C_t} \frac{C_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[V_{t+1}(M_{t+1}, P_{t+1}) \right] \\
&= \max_{c_t} \frac{(c_t P_t)^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[V_{t+1}(M_{t+1}, P_{t+1}) \right] \Leftrightarrow \\
V_t(M_t, P_t)/P_t^{1-\rho} &= \max_{c_t} \frac{(c_t P_t)^{1-\rho}/P_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[V_{t+1}(M_{t+1}, P_{t+1})/P_t^{1-\rho} \right] \Leftrightarrow \\
v_t(m_t) &= \max_{c_t} \frac{{c_t}^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[\underbrace{V_{t+1}(M_{t+1}, P_{t+1})/P_{t+1}^{1-\rho}}_{=v_{t+1}(m_{t+1})} \cdot \underbrace{P_{t+1}^{1-\rho}/P_t^{1-\rho}}_{=(G\psi_{t+1})^{1-\rho}} \right] \\
&= \max_{c_t} \frac{{c_t}^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[(G\psi_{t+1})^{1-\rho} v_{t+1}(m_{t+1}) \right]
\end{aligned}
$$

Bellman equation in ratio form

$$
\begin{aligned}
v_t(m_t) &= \max_{c_t} \frac{c_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[(G\psi_{t+1})^{1-\rho} v_{t+1}(m_{t+1}) \right] \\
&\text{s.t.} \\
a_t &= m_t - c_t \\
m_{t+1} &= \frac{1}{G\psi_{t+1}} R a_t + \xi_{t+1} \\
a_t &\geq 0
\end{aligned}
$$

- **Benefit:** Dimensionality of state space reduced, $2 \rightarrow 1$.
  Can this always be done?

Bellman equation in ratio form

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m_{t+1} &= \frac{1}{G\psi_{t+1}} R a_t + \xi_{t+1} \\
a_t &\geq 0
\end{aligned}
$$

- **Benefit:** Dimensionality of state space reduced, $2 \to 1$.
  Can this always be done?
- No... Uses that utility is homothetic (budget constraint also important)

$$
V_T(M_T, P_T) = \frac{M_T^{1-\rho}}{1-\rho} = \frac{(m_T P_T)^{1-\rho}}{1-\rho} = \frac{m_T^{1-\rho}}{1-\rho} P_T^{1-\rho}
$$

such that $v_T(m_T) = V_T(M_T, P_T) / P_T^{1-\rho}$ holds!

Solving the model: Numerical Integration

- Solved by **backwards induction**
  Terminal period:

$$v_T(m_T) = \frac{m_T^{1-\rho}}{1-\rho}$$

For $t < T$:

$$v_t(m_t) \quad = \quad \max_{c_t} \frac{c_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}_t \left[(G\psi_{t+1})^{1-\rho} v_{t+1}(m_{t+1}) \right]$$

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$$\mathbb{E}_t[\bullet] = \int_{\psi_{t+1}} \int_{\xi_{t+1}} \left[(G\psi_{t+1})^{1-\rho} v_{t+1}(m_{t+1})\right] f(d\psi_{t+1}, d\xi_{t+1})$$

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- **Numerical Integration**: Discretize into sum (Gauss-Hermite)

$$\mathbb{E}_t[\bullet] \approx \sum_{j=1}^{J} \sum_{k=1}^{K} [(G\psi^{(j)})^{1-\rho} v_{t+1}(m^{(j,k)})] \omega_j \omega_k$$

  and interpolate $v_{t+1}(\bullet)$ for values $m^{(j,k)} = \frac{1}{G\psi^{(j)}} Ra_t + \xi^{(k)}$ of $\overrightarrow{m}$ grid.

Outline

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- **Frontier:** Use exogenous variation to estimate structural model.

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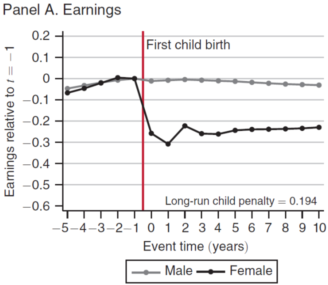
- **Reduced-form** to be *causal*: "statistical" assumptions
  - No self-selection (timing)
  - No anticipation effects.
  - Parallel trends.



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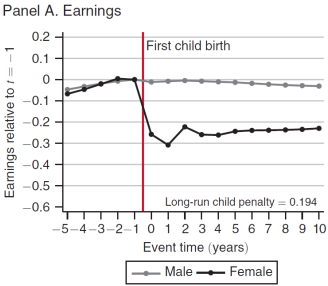

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  - No anticipation effects.
  - Parallel trends.

- **A model can allow** for these assumptions to be violated
  But only through the chosen functional forms and mechanisms
  - "Economic" assumptions
  - Easier to debate and improve upon (?)

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- **Example model:** Life-cycle buffer-stock model
  - States: M_{it}, P_{it}
  - Choice: C_{it}
- **Parameters** to estimate: $\theta = \{\beta, \rho\}$
  - Calibration: G, σ_ψ, σ_ξ, R, and λ ("known")

Simulated Method of Moments (SMM/SMD)

- $\Lambda^d = \frac{1}{N}\sum_{i=1}^{N}\Lambda_i^d$ are some **moments in the data**
  Could be avg., var, cov, regression-coefs, etc.

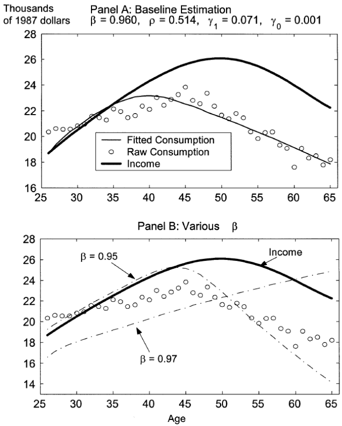

FIGURE 5.—The fitted consumption profile.

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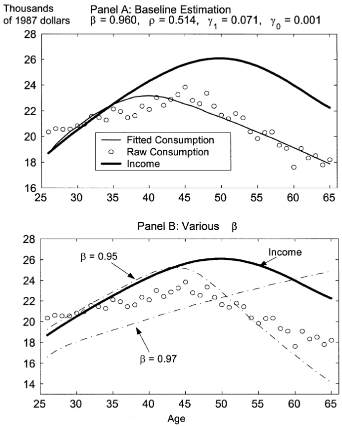

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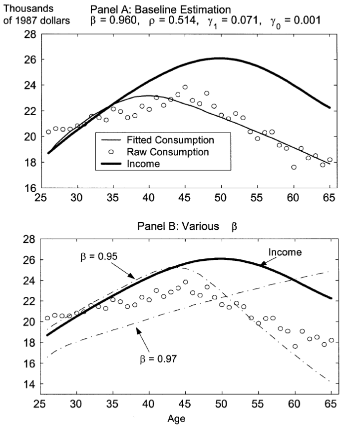

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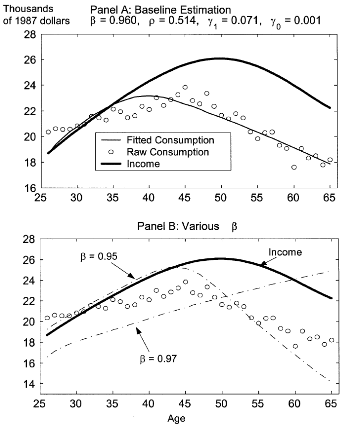
- **SMM** then is

$$\hat{\theta} = \arg\min_{\theta} g(\theta)' W g(\theta)$$

  where W is **weighting matrix**.



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Weighting Matrix, W

- Common weighting matrices, W, are
  (should be positive-definite)

  1. **Theoretically optimal**
     Inverse of covariance matrix of empirical moments
     Can cause problems in finite samples
  2. **Identity, I**
     Equal weighting.
     Does not take level-differences out of moments
  3. **Diagonal matrix** with *inverse* of empirical moment *variances*
     Removes "level" differences.
     Scales with uncertainty about empirical moments
     Popular
  4. **Freely chosen**
     Focus on fitting some specific dimensions of the data

Estimation experiment

1. **Solve** the buffer-stock model and **simulate** a full panel
2. Construct a **data set** from the simulated data
3. Try to **estimate** $\theta = \{\beta, \rho\}$
   using as moments the **average wealth for each age between 40 and 55**
   $\Lambda^d = (A_{40}, A_{41}, \ldots, A_{55})$

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- I will now describe how to calculate the objective function

$$Q(\theta) = \Big(\Lambda^d - \Lambda^m(\theta)\Big)' W \Big(\Lambda^d - \Lambda^m(\theta)\Big)$$

  for a given value of θ.
- This function should then be minimized to get

$$\hat{\theta} = \arg\min_{\theta} Q(\theta)$$

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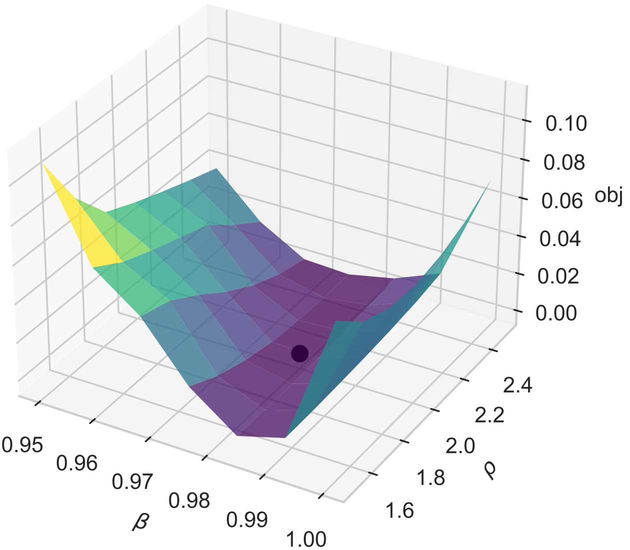

Indirect inference / minimum distance

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  - McFadden (1989): Method of Simulated Moments (MSM)
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- SMD/II rely on an **auxillary statistical model**
  - Let Λ^d be the parameters of the auxillary model when estimated on the *actual* data
  - Let $\Lambda_s(\theta)$ be the parameters of the auxillary model when estimated on *simulated* data
- **Note:** The auxillary statistical model is *misspecified* and its parameters are thus typically *not interpretable*

Simulation Pitfalls

- **FIX the seed (or draws!)**
- **Flat** objective function!
  - Discrete choices: Taking a mean of an **indicator function**
- **Gradient** based numerical optimization will likely FAIL!
  - Use, e.g., `scipy.optimize.minimize(fun , method='Nelder-Mead')` (Nelder-Mead)
  - Or some smoothing device (e.g. Logit)
- As $N, S \to \infty$ this problem vanishes
- The problem is also less severe around θ_0
- Continuous outcomes do not have this problem

Asymptotics

- **MSM** is **consistent** and **asymptotically normal** under standard assumptions
$$\sqrt{N}(\hat{\theta} - \theta_0) \to \mathcal{N}(0, (1 + S^{-1})V)$$
where θ_0 is vector of true parameters
- **Standard formulas for V:**
$$V = (G'WG)^{-1} G'W\Omega W'G(G'WG)^{-1}$$
where $G = -\frac{\partial \Lambda^m(\theta)}{\partial \theta}$ is the Jacobian of the objective function.
$\Omega = Var(\Lambda_i^d)$ is the variance of the (individual) moments in the data.
Remember: *Standard errors are large if large changes in θ imply small changes in the objective function*

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- **Problems:**
  1. The objective function might have multiple minima *(no global solver exists)*
  2. The objective function could be very flat in some directions *(increasing S might help)*

Robustness/Sensitivity

- **Curse of dimensionality and lack of identification**
  $\Rightarrow$ we cannot estimate all the parameters of the model
  $\Rightarrow$ *first step estimation/calibration is often necessary*
  1. Calculations on own data (e.g. exogenous processes)
  2. References to previous estimates
  3. Standard choices

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  2. References to previous estimates
  3. Standard choices
- **Robustness:** Can we vary the calibration choices without changing the result substantially?
  "Sensitivity to Calibration": (Jørgensen, 2023)
  "*How much does estimates of θ change when 1. step calibrations change?*"

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  Time-consuming!

- **Calibration:** "hand-held"
  Use a (small) grid of values for θ
  to minimize some moment(s). Sometimes eyeballing, sequentially for each parameter.
  Do not report standard errors
  Less time-consuming!
  "Illustrate proposed mechanism"

Next Time

- **Next time:**
  Static and dynamic labor supply
  Recap for some + new stuff for most.

- **Literature:**
  Keane (2011, sections 1–5): "Labor Supply and Taxes: A Survey"
  - **Read** before lecture
  - **Reading guide:**
    Section 1: short Introduction
    Section 2: Optimal Taxation, Motivation. Skim fast.
    Section 3: Basic model. *Key, focus here*.
    Section 4: Econometric issues. Skim.
    Section 5: Roadmap of empirical literature. *Short, read*.
    (Remaining: empirical literature.)

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Labor Supply: Static and Dynamic

Thomas H. Jørgensen

2025

Outline

1. Introduction
2. Static Labor Supply
3. Dynamic Labor Supply
4. Estimating Elasticities

Plan

- Simple static and dynamic labor supply models
  - recap for some
  - brings us to same page
  - illustrate numerical approach with closed form checks
- Keane (2011, sections 1–5)
  - same notation as him

Motivation: Labor supply Elasticities are Important!

Figure: Estimated Elasticities. Hicks, Men (Keane, 2011).

TABLE 1
OPTIMAL TOP BRACKET TAX RATES FOR DIFFERENT LABOR SUPPLY ELASTICITIES

Labor supply elasticity (e)	Optimal top-bracket tax rate (τ) =1/(1+a*e)		
	a = 1.50	a = 1.67	a = 2.0
2.0	25%	23%	20%
1.0	40%	37%	33%
0.67	50%	47%	43%
0.5	57%	54%	50%
0.3	69%	67%	63%
0.2	77%	75%	71%
0.1	87%	86%	83%
0.0	100%	100%	100%

Note: These rates assume the government places essentially no value on giving extra income to the top earners.

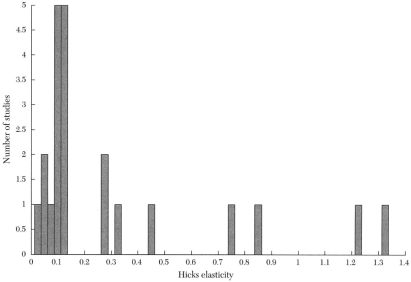

Figure 5. Distribution of Hicks Elasticity of Substitution Estimates

Note: The figure contains a frequency distribution of the twenty-two estimates of the Hicks elasticity of substitution discussed in this survey.

Outline

Static Setup

- Individuals maximize utility wrt. consumption and hours worked

$$\max_{C,h} U(C,h) = \frac{C^{1+\eta}}{1+\eta} - \beta \frac{h^{1+\gamma}}{1+\gamma}$$

where

- $\eta \leq 0$ is the CRRA coefficient
- $\gamma \geq 0$ is curvature in hours

Static Setup

- Individuals maximize utility wrt. consumption and hours worked

$$\max_{C,h} U(C,h) = \frac{C^{1+\eta}}{1+\eta} - \beta \frac{h^{1+\gamma}}{1+\gamma}$$

Substitution (gamma)
Income (eta)

where

  - $\eta \leq 0$ is the CRRA coefficient
  - $\gamma \geq 0$ is curvature in hours

- subject to the budget constraint

$$C = (1-\tau)wh + N$$

where

  - τ is the (flat) marginal tax rate
  - w is the hourly wage rate
  - N is non-labor income

Static Solution

- Insert budget constraint

$$\max_h \frac{((1-\tau)wh+N)^{1+\eta}}{1+\eta} - \beta\frac{h^{1+\gamma}}{1+\gamma}$$

- First order condition (FOC)

$$\begin{aligned}\frac{\partial U}{\partial h} &= (1-\tau)w((1-\tau)wh+N)^{\eta} - \beta h^{\gamma} \\ &= 0\end{aligned}$$

such that the MRS is

$$(1-\tau)w = \frac{\beta h^{\gamma}}{((1-\tau)wh+N)^{\eta}} \tag{1}$$

Static Elasticities

- **No analytic solution** for optimal hours, $h^{\star}(w, N)$.
  We can solve numerically!
- **Elasticities** can be derived analytically (see extra slides)!
  Can compare with numerical!

Static Elasticities

- **No analytic solution** for optimal hours, $h^\star(w, N)$.
  We can solve numerically!
- **Elasticities** can be derived analytically (see extra slides)!
  Can compare with numerical!
- **Slutsky** equation:

$$\frac{\partial h}{\partial w} = \left.\frac{\partial h}{\partial w}\right|_u + h\frac{\partial h}{\partial N}$$

- **In elasticities** (% change from 1% change)

$$\frac{w}{h}\frac{\partial h}{\partial w} = \frac{w}{h}\left.\frac{\partial h}{\partial w}\right|_u + \frac{wh}{N}\frac{N}{h}\frac{\partial h}{\partial N}$$

$$\underbrace{e_M}_{\text{marshall}} = \underbrace{e_H}_{\text{hicks}} + \underbrace{\frac{wh}{N}e_I}_{\text{income effect}}$$

Static Elasticities

- Letting $S = \frac{(1-\tau)wh}{(1-\tau)wh+N}$, we have (see extra slides)

$$e_M = \frac{\partial \log h}{\partial \log w} = \frac{1+\eta S}{\gamma - \eta S}$$

$$e_H = \left.\frac{\partial \log h}{\partial \log w}\right|_u = \frac{1}{\gamma - \eta S}$$

$$ie = \frac{\eta S}{\gamma - \eta S} < 0$$

- Since $\eta \leq 0$:

$$e_H > e_M$$

that is, "ignoring income effects gives a larger response".

Static Elasticities

- **Numerical "check"** of these results
  Simple setup
  Shows how to do it
  Can check results

Static Elasticities

- **Numerical "check"** of these results
  Simple setup
  Shows how to do it
  Can check results

1. **Solve** optimal labor supply, $h^\star(w, N)$
2. **Simulate baseline** labor supply for $w \rightarrow h_i(w, N)$
3. **Simulate alternative** with 1% higher wage $\rightarrow h_i(w(1+0.01), N)$
4. **Calculate** average pct change,

$$\frac{1}{n}\sum_{i=1}^{n}\frac{h_i(w(1+0.01), N) - h_i(w, N)}{h_i(w, N)} \times 100$$

- **Q**: Which of the elasticities should this equal?

  Marshall elasticity: Not keeping income effect constant.

Outline

Dynamic Labor Supply

- 2-period model with **saving/borrowing**
  - Perfect foresight + deterministic
  - Exogenous wages, w_1 and w_2
  - Same per-period utility as before

Dynamic Labor Supply

- 2-period model with **saving/borrowing**
  - Perfect foresight + deterministic
  - Exogenous wages, w_1 and w_2 Does not depend on eachother.
  - Same per-period utility as before
- Discounted utility is

$$U = U_1(C_1, h_1) + \rho U_2(C_2, h_2)$$

where

$$C_1 = (1-\tau)w_1 h_1 + N_1 + b$$
$$C_2 = (1-\tau)w_2 h_2 + N_2 - b(1+r)$$

and $b = -[(1-\tau)w_1 h_1 + N_1 - C_1]$ is borrowing.

Dynamic Labor Supply

- We find optimal h_1, h_2 and b by maximizing utility
- First order conditions (FOCs)

$$\frac{\partial U}{\partial h_1} = [(1-\tau)w_1 h_1 + N_1 + b]^{\eta} w_1(1-\tau) - \beta h_1^{\gamma} = 0 \tag{2}$$

$$\frac{\partial U}{\partial h_2} = [(1-\tau)w_2 h_2 + N_2 - b(1+r)]^{\eta} w_2(1-\tau) - \beta h_2^{\gamma} = 0 \tag{3}$$

$$\begin{aligned}\frac{\partial U}{\partial b} &= [(1-\tau)w_1 h_1 + N_1 + b]^{\eta} \\ &\quad - \rho[(1-\tau)w_2 h_2 + N_2 - b(1+r)]^{\eta}(1+r) = 0\end{aligned} \tag{4}$$

Dynamic Labor Supply

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- First order conditions (FOCs)

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- Again, no closed-form solution for optimal labor supply
- We can find elasticities
  - and simulate them!
  - e_H and e_M basically the same as in static case (because no human capital)

Frisch Elasticity

- We now have a *new* elasticity: How much am I willing to substitute my consumption over time.
  Elasticity of intertemporal substitution

Frisch Elasticity

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- Frisch elasticity

$$e_F \equiv \left.\frac{\partial \log h_t}{\partial \log w_t}\right|_{\lambda_t}$$

where

$$\lambda_t : \text{marginal utility of wealth}$$

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  where

$$\lambda_t : \text{marginal utility of wealth}$$

- λ_t potentially depends on all state variables, such as wages.
- In our setting (see extra slide)

$$e_F = \frac{1}{\gamma}$$

Summary: Elasticities

- **Combining**, we have

$$e_F > e_H > e_M$$

in this model.
  - **Frisch:** Can be simulated as an anticipated *transitory* increase in wage (income and wealth effects are small and assumed negligible)
  - **Marshall:** Can be simulated as a unanticipated *permanent* increase in wage
  - **Hicks:** Can be simulated as a unanticipated *permanent compensated* increase in wage

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- If we can estimate $e_F = 1/\gamma$, we can **bound the policy-relevant elasticities**!
  - We can then bound the efficiency loss from labor income taxation.
  - And bound the optimal tax rate (see table from beginning).

Summary: Elasticities

- **Combining**, we have

$$e_F > e_H > e_M$$

in this model. Frisch: If I know that my income in period 2 will be higher, I may consume/borrow more today.

  - **Frisch:** Can be simulated as an anticipated *transitory* increase in wage (income and wealth effects are small and assumed negligible)
  - **Marshall:** Can be simulated as a unanticipated *permanent* increase in wage
  - **Hicks:** Can be simulated as a unanticipated *permanent compensated* increase in wage

- If we can estimate $e_F = 1/\gamma$, we can **bound the policy-relevant elasticities**!
  - We can then bound the efficiency loss from labor income taxation.
  - And bound the optimal tax rate (see table from beginning).

- **Next time:** If there is learning by doing (human capital accumulation) this relationship/result might not hold due to downward bias in e_F!

Solving the 2-period model

- **Backwards induction.**
- **Last period:** What is the state variable?

Solving the 2-period model

- **Backwards induction.**
- **Last period:** What is the state variable? We need to know what b is.

$$V_2(b) = \max_{C_2, h_2} U_2(C_2, h_2)$$

s.t.

$$C_2 = (1-\tau) w_2 h_2 + N_2 - (1+r) b > 0$$

such that for a grid of $\overrightarrow{b}$ we solve for griid is an array

$$h_2^\star(b) = \arg\max_{h_2} U_2((1-\tau) w_2 h_2 + N_2 - (1+r) b, h_2)$$

Solving the 2-period model

- **Backwards induction.**
- **Last period:** What is the state variable?

$$V_2(b) = \max_{C_2, h_2} U_2(C_2, h_2)$$

s.t. We are choosing our consumption based on the equality below.

$$C_2 = (1-\tau)w_2 h_2 + N_2 - (1+r)b > 0$$

such that for a grid of $\overrightarrow{b}$ we solve for

$$h_2^\star(b) = \arg\max_{h_2} U_2((1-\tau)w_2 h_2 + N_2 - (1+r)b, h_2)$$

- **First period:**

$$V_1 = \max_{C_1, h_1} U_1(C_1, h_1) + \rho V_2(b)$$

s.t.

$$b = -[(1-\tau)w_1 h_1 + N_1 - C_1]$$

Life-Cycle Model

- T periods, a_t is savings. No uncertainty (for now)
- Bellman Equation

$$V_t(a_t) = \max_{C_t, h_t} U(C_t, h_t) + \rho V_{t+1}(a_{t+1})$$

a_t is the level of wealth

s.t.

$$a_{t+1} = (1+r)(a_t + (1-\tau_t) w_t h_t - C_t)$$

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- **Exogenous wages**:
  - Elasticities are as discussed throughout (Keane, 2016)

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$$a_{t+1} = (1+r)(a_t + (1-\tau_t) w_t h_t - C_t)$$

- **Exogenous wages**:
  - Elasticities are as discussed throughout (Keane, 2016)
- **Endogenous wages** (through human capital, as next time):

$$w_t = w\,(1 + \alpha k_t)$$
$$k_{t+1} = k_t + h_t$$

  - Human capital, k_t, is a new state variable
  - Elasticities are different
  - In general no simple formula
  - *Can always simulate!*

Outline

Estimating Elasticities

- **Regressions** often like (pioneered by MaCurdy, 1981)

$$\log h_{it} = \alpha + e \log(w_{it}(1 - \tau_t)) + \beta_I N_{it} + \varepsilon_{it}$$

- Controlling for non-labor income, $N \to e$ is **Marshall** elasticity.

Estimating Elasticities

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- Controlling for non-labor income, $N \to e$ is **Marshall** elasticity.
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- Our **models can illuminate** potential problems (Keane, 2011, sec 4)
  1. Endogeneity of wages: tastes for work
  2. Endogeneity of wages: simultaneity
  3. Endogeneity of taxes (non-linear)
  4. Measurement error (downward bias)
  5. Wages only observed for workers (selection)
  6. Savings and non-labor earnings
  7. Human capital and other dynamics (next time)

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  4. Measurement error (downward bias)
  5. Wages only observed for workers (selection)
  6. Savings and non-labor earnings
  7. Human capital and other dynamics (next time)
- Structural estimation can handle most of these

Estimated Elasticities, Men.

TABLE 6
SUMMARY OF ELASTICITY ESTIMATES FOR MALES

Authors of study	Year	Marshall	Hicks	Frisch
Static models				
Kosters	1969	−0.09	0.05	
Ashenfelter-Heckman	1973	−0.16	0.11	
Boskin	1973	−0.07	0.10	
Hall	1973	n/a	0.45	
Eight British studies[a]	1976–83	−0.16	0.13	
Eight NIT studies[a]	1977–84	0.03	0.13	
Burtless-Hausman	1978	0.00	0.07–0.13	
Wales-Woodland	1979	0.14	0.84	
Hausman	1981	0.00	0.74	
Blomquist	1983	0.08	0.11	
Blomquist-Hansson-Busewitz	1990	0.12	0.13	
MaCurdy-Green-Paarsch	1990	0.00	0.07	
Triest	1990	0.05	0.05	
Van Soest-Woittiez-Kapteyn	1990	0.19	0.28	
Ecklof-Sacklen	2000	0.05	0.27	
Blomquist-Ecklof-Newey	2001	0.08	0.09	
Dynamic models				
MaCurdy	1981	0.08[b]		0.15
MaCurdy	1983	0.70	1.22	6.25
Browning-Deaton-Irish	1985			0.09
Blundell-Walker	1986	−0.07	0.02	0.03
Altonji[c]	1986	−0.24	0.11	0.17
Altonji[d]	1986			0.31
Altug-Miller	1990			0.14
Angrist	1991			0.63
Ziliak-Kniesner	1999	0.12	0.13	0.16
Pistaferri	2003	0.51[b]		0.70
Imai-Keane	2004	0.40[e]	1.32[e]	0.30–2.75[f]
Ziliak-Kniesner	2005	−0.47	0.33	0.54
Aaronson-French	2009			0.16–0.61
Average		0.06	0.31	0.85

Notes: Where ranges are reported, mid-point is used to take means.

Estimated Elasticities, Women.

- Women have been viewed as more "complex"
  Literature started later
  When dynamic models was used more

TABLE 7
Summary of Elasticity Estimates for Women

Authors of study	Year	Marshall	Hicks	Frisch	Uncompensated (dynamic)	Tax response
Static, life-cycle and life-cycle consistent models						
Cogan	1981	0.89[a]				
Heckman-MaCurdy	1982			2.35		
Blundell-Walker	1986	−0.20	0.01	0.03		
Blundell-Duncan-Meghir	1998	0.17	0.20			
Kimmel-Kniesner	1998			3.05[b]		
Moffitt	1984				1.25	
Dynamic structural models						
Eckstein-Wolpin	1989				5.0	
Van der Klauuw	1996				3.6	
Francesconi	2002				5.6	
Keane-Wolpin	2010				2.8	
Difference-in-difference methods						
Eissa	1995, 1996a					0.77–1.60[b]

Notes:
[a] = Elasticity conditional on positive work hours.
[b] = Sum of elasticities on extensive and intensive margins.

Next Time

- **Next time:**
  Dynamic labor supply with learning by doing
  Human capital accumulation from working
  (Uncertainty?)

- **Literature:**
  Keane (2016): "Life-Cycle Labour Supply with Human Capital: Econometric and Behavioural Implications"
  - **Read** before lecture
  - **Reading guide:**
    Section 0: Introduction
    Section 1: Dynamic model. *Key section, main focus*.
    Section 2: Simulations of 2-period model. skim/drop.
    Section 3: Quantitative role of HC. Read fast, *focus on 3.2*.
    Section 4: Comparison with extensive margin. Read if time

References I

Keane, M. P. (2011): "Labor Supply and Taxes: A Survey," *Journal of Economic Literature*, 49(4), 961–1075.

——— (2016): "Life-cycle Labour Supply with Human Capital: Econometric and Behavioural Implications," *The Economic Journal*, 126(592), 546–577.

MaCurdy, T. E. (1981): "An Empirical Model of Labor Supply in a Life-Cycle Setting," *Journal of political Economy*, 89(6), 1059–1085.

Finding elasticities in static model

- Taking logs of eq. (1):

$$\underbrace{\log(1-\tau)+\log(w)}_{\text{LHS}} = \underbrace{\log(\beta)+\gamma\log(h)-\eta\log((1-\tau)wh+N)}_{\text{RHS}} \quad (5)$$

- Derivative wrt. $\log w$ gives (while keeping N fixed)

$$\frac{\partial \text{RHS}}{\partial \log w} = \gamma \underbrace{\frac{\partial \log h}{\partial \log w}}_{e_M} - \eta\left(\frac{\partial \log(\bullet)}{\partial w}\underbrace{\frac{\partial w}{\partial \log w}}_{w} + \frac{\partial \log(\bullet)}{\partial h}\underbrace{\frac{\partial h}{\partial \log h}}_{h}\underbrace{\frac{\partial \log h}{\partial \log w}}_{e_M}\right)$$

where $\log(\bullet) = \log((1-\tau)wh+N)$.

Marshall

- Letting $S = \frac{(1-\tau)wh}{(1-\tau)wh+N}$, we have that

$$\frac{\partial \text{RHS}}{\partial \log w} = \gamma e_M - \eta(S + Se_M)$$
$$\frac{\partial \text{LHS}}{\partial \log w} = 1$$

such that

$$1 = \gamma e_M - \eta(S + Se_M)$$
$$\Updownarrow$$
$$e_M = \frac{1+\eta S}{\gamma - \eta S}$$

Income effect

- Similarly, the income elasticity is

$$e_I = \frac{\partial \log h}{\partial \log N} = \frac{\eta(1-S)}{\gamma - \eta S}$$

- Found similarly, using that $1 - S = \frac{N}{(1-\tau)wh+N}$.
- The income effect is then

$$ie = \frac{wh(1-\tau)}{N} e_I = \frac{\eta S}{\gamma - \eta S}$$

where

$$ie < 0$$

because $\eta \leq 0$.

Hicks

- The Hicks, or "compensated" elasticity is

$$
\begin{aligned}
e_H &= \left.\frac{\partial \log h}{\partial \log w}\right|_u \\
&= e_M - ie \\
&= \frac{1+\eta S}{\gamma - \eta S} - \frac{\eta S}{\gamma - \eta S} \\
&= \frac{1}{\gamma - \eta S}
\end{aligned}
$$

using the Slutsky equation.

- We have

$$
e_H \geq e_M
$$

Frisch Elasticity

- Deriving the Frisch via. the Lagrangian

$$\max_{h_1,h_2,C_1,C_2,b} U + \lambda_1[(1-\tau)w_1h_1 + N_1 + b - C_1] + \lambda_2[(1-\tau)w_2h_2 + N_2 - b(1+r) - C_2]$$

Frisch Elasticity

- Deriving the Frisch via. the Lagrangian

$$\max_{h_1,h_2,C_1,C_2,b} U + \lambda_1[(1-\tau)w_1h_1 + N_1 + b - C_1] \\ + \lambda_2[(1-\tau)w_2h_2 + N_2 - b(1+r) - C_2]$$

- FOC for hours in period 1

$$\frac{\partial U}{\partial h_1} = -\beta h_1^{\gamma} + \lambda_1(1-\tau)w_1 = 0$$

such that

$$\log h_1 = \frac{1}{\gamma}\log(\lambda_1) + \frac{1}{\gamma}\log((1-\tau)w_1) - \frac{1}{\gamma}\log(\beta)$$

Frisch Elasticity

- Deriving the Frisch via. the Lagrangian

$$\max_{h_1,h_2,C_1,C_2,b} U + \lambda_1[(1-\tau)w_1 h_1 + N_1 + b - C_1] \\ + \lambda_2[(1-\tau)w_2 h_2 + N_2 - b(1+r) - C_2]$$

- FOC for hours in period 1

$$\frac{\partial U}{\partial h_1} = -\beta h_1^{\gamma} + \lambda_1(1-\tau)w_1 = 0$$

such that

$$\log h_1 = \frac{1}{\gamma}\log(\lambda_1) + \frac{1}{\gamma}\log((1-\tau)w_1) - \frac{1}{\gamma}\log(\beta)$$

- and the partial derivative (fixed λ_1)

$$e_F = \left.\frac{\partial \log h_1}{\partial \log w_1}\right|_{\lambda_1} = \frac{1}{\gamma}.$$

Dynamic Labor Supply: Human Capital

Thomas H. Jørgensen

2025

Plan for today

- Dynamic Labor supply w/w.o. **human capital** (HC)
  Keane (2016)
  - Reduced-form estimation, bias from HC
  - Elasticities, HC affect relation between them
  - Age effects
  - Simulate
- Related literature Imai and Keane (2004); Keane and Wasi (2016).
- "Human capital" = "learning by doing" here
  educational choices etc. not the focus *here*

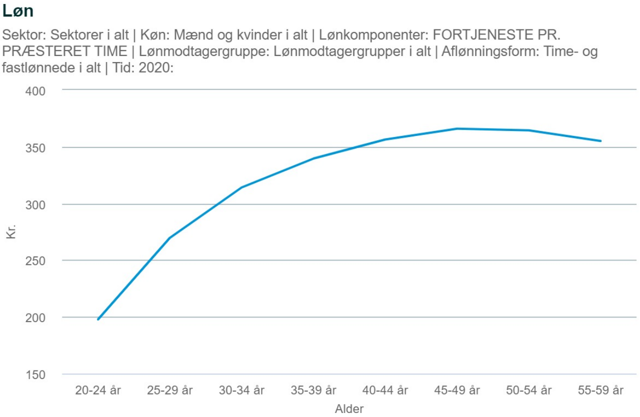
Empirical Motivation: I

- **From Arrow (1962):**
  Lundberg (1961, pp. 129-133) has given the name " Horndal effect" to a very similar phenomenon. The Horndal iron works in Sweden had no new investment (and therefore presumably no significant change in its methods of production) for a period of 15 years, yet productivity (output per manhour) rose on the average close to 2 % per annum. We find again steadily increasing performance which can only be imputed to learning from experience.

Empirical Motivation: II

- **Wages vary** over life...
  could be due to learning by doing (HC)

Figure: Hourly Wage over the Life Cycle in Denmark (LONS50).



Empirical Motivation: III (Motivation for Keane, 2016)

- **Low Frisch** elasticities estimated in large literature (Keane, 2011)

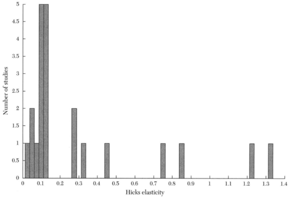


Figure 5. Distribution of Hicks Elasticity of Substitution Estimates

Note: The figure contains a frequency distribution of the twenty-two estimates of the Hicks elasticity of substitution discussed in this survey.

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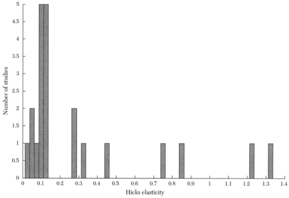

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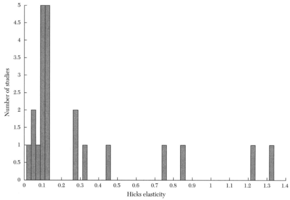

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- **Basic life-cycle model**: Frisch is *upper bound* of Hicks and Marshall
  $\rightarrow$ taxes hardly distort behavior
  $\rightarrow$ optimal tax rates are high
- **Could be erroneous conclusion**: Human capital
  $\rightarrow$ Value of time $\neq$ net-of-tax wage
  $\rightarrow$ Frisch might not be upper bound
  $\rightarrow$ Elasticities vary with age (i.e. estimation sample important)

Simple Life-Cycle Model (Keane, 2016)

- Workers chose throughout life, $t = 1, \dots, T$
  - consumption, $c_t > 0$
  - labor hours worked, $h_t \geq 0$
- To maximize the discounted value

$$\frac{c_1^{1+\eta}}{1+\eta} - \beta \frac{h_1^{1+\gamma}}{1+\gamma} + \sum_{t=2}^{T} \rho^{t-1} \left(\frac{c_t^{1+\eta}}{1+\eta} - \beta \frac{h_t^{1+\gamma}}{1+\gamma} \right)$$

  where
  - $\rho \in (0, 1)$ is the discount factor
  - $\eta \leq 0$ is the CRRA
  - $\gamma \geq 0$ is the curvature wrt. hours worked
  - $\beta > 0$ is the strength of the dis-utility of work
- No uncertainty - perfect foresight

Simple Life-Cycle Model, Keane (2016)

- **Inter-temporal budget** constraint is

$$a_{t+1} = (1+r)\big(a_t + (1-\tau_t)w_t h_t - c_t\big),\ a_0 = 0$$

  where
  - r is the real interest rate
  - τ_t is the tax rate
- **Human capital** accumulation,

$$k_{t+1} = k_t + h_t,\ k_0 = 0$$

- **Endogenous wages**, $h_t \rightarrow k_{t+1} \rightarrow w_{t+1}$,

$$\begin{aligned} w_t &= w\,(1 + \alpha k_t) \\ &= w\left(1 + \alpha \sum_{s=1}^{t-1} h_s\right) \end{aligned}$$

Optimal Consumption

- We will assume $\rho(1+r)=1$
  $\rightarrow$ consumption is perfectly smoothed across periods, $c_t = C\ \forall t$.
- C can be found from the life-time constraint that the NPV of consumption must equal the NPV of resources,

$$\sum_{t=1}^{T}\frac{c_t}{(1+r)^t} = \sum_{t=1}^{T}\frac{w_t(1-\tau_t)h_t}{(1+r)^t}$$

$$C(1+r)^T\sum_{t=1}^{T}(1+r)^{-t} = (1+r)^T\sum_{t=1}^{T}w_t(1-\tau_t)h_t(1+r)^{-t}$$

$$C = \frac{\sum_{t=1}^{T}w_t(1-\tau_t)h_t(1+r)^{T-t}}{\sum_{t=1}^{T}(1+r)^{T-t}} \tag{1}$$

MRS and Human Capital: General

- **Bellman equation** formulation

$$V_t(a_t, k_t) = \max_{c_t, h_t} \frac{c_t^{1+\eta}}{1+\eta} - \beta \frac{h_t^{1+\gamma}}{1+\gamma} + \rho V_{t+1}(a_{t+1}, k_{t+1})$$

- **First order conditions**

$$c_t^{\eta} - \rho(1+r)V_{t+1}^1(a_{t+1}, k_{t+1}) = 0$$
$$-\beta h_t^{\gamma} + \rho V_{t+1}^2(a_{t+1}, k_{t+1}) + \rho(1+r)(1-\tau_t)w_t V_{t+1}^1(a_{t+1}, k_{t+1}) = 0$$

- **The MRS** is, using $\rho(1+r) = 1$ and thus $c_t = C$,

$$\beta h_t^{\gamma} / C^{\eta} = w_t(1-\tau_t) + \rho V_{t+1}^2(a_{t+1}, k_{t+1}) / C^{\eta}$$

- Last term is related to the endogeneity of wages from human capital

Solution

- **No analytical solution** for optimal hours, $h_t^\star(a_t, k_t)$
- **Our tools can be used** to solve for this function
  Backwards induction
  Interpolation
- **See notebook.**

Elasticities

- **Without human capital** and non-labor income ($\alpha = 0$, $N = 0$), we had

$$\underbrace{e_F}_{\frac{1}{\gamma}} \geq \underbrace{e_H}_{\frac{1}{\gamma-\eta}} \geq \underbrace{e_M}_{\frac{1+\eta}{\gamma-\eta}}$$

- How does the elasticities now look like?
(The inequalities might not hold for all parameter values!)

OCT and Human Capital

- The Marginal Rate of Substitution (MRS) condition is here

$$\underbrace{\beta h_t^{\gamma} / C^{\eta}}_{\text{OCT}} = \underbrace{w_t(1-\tau_t)}_{\text{wage}} + \underbrace{\alpha w F_t}_{\text{HC}} \tag{2}$$

where

$$F_t = \sum_{s=t+1}^{T} \frac{h_s(1-\tau_s)}{(1+r)^{s-t}},\ F_T = 0$$

measures the NPV of future human capital investments.

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- Taking logs of eq. (2) gives

$$\log h_t = \frac{1}{\gamma} \log[w_t(1-\tau_t) + \alpha w F_t] + \frac{\eta}{\gamma} \log C - \frac{1}{\gamma} \log \beta \tag{3}$$

OCT and Human Capital

Figure: Opportunity Cost of Time and Human Capital, Keane (2016).

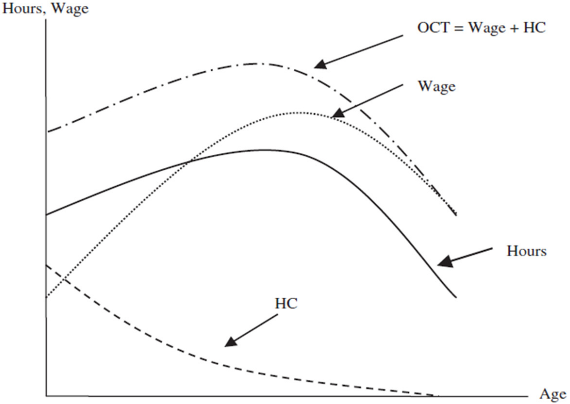

Fig. 1. *Hours, Wages and Price of Time over the Life-cycle*

Notes. This Figure plots the components of the first-order condition for labour supply generated by the life-cycle model with human capital: $\beta h_t^{\gamma}/C^{\eta} = w_t(1-\tau_t) + \alpha w F_t$. Here, $Wage \equiv w_t(1-\tau_t)$, *the 'human capital term'* $HC \equiv \alpha w F_t$, *and the 'opportunity cost of time'* $OCT \equiv Wage + HC$. Note that the term HC captures the return to an hour of work experience, in terms of increased present value of future wages.

Frisch Elasticity

- **Q:** What does the disconnect between OCT and Wage mean for the responsiveness of young workers to *transitory* tax-changes?

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- **A:** *transitory* tax-changes is a smaller part of the OCT for young $\rightarrow$ **Frisch elasticity is lower** (compared to when $\alpha = 0$)

$$e_{F,t} \equiv \left. \frac{\partial \log h_t}{\partial \log(1-\tau_t)} \right|_{dC=0} = \frac{1}{\gamma} \frac{w_t(1-\tau_t)}{w_t(1-\tau_t) + \alpha w F_t} < \frac{1}{\gamma}$$

with $F_t \rightarrow 0$ for $t \rightarrow T$.

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  with $F_t \rightarrow 0$ for $t \rightarrow T$.

- **Derivation:** taking logs of eq. (2) and *partial* derivative wrt. $(1-\tau_t)$

$$\log h_t = \frac{1}{\gamma}\log[w_t(1-\tau_t)+\alpha w F_t] + \frac{\eta}{\gamma}\log C - \frac{1}{\gamma}\log\beta$$

  such that

$$\frac{\partial \log h_t}{\partial \log(1-\tau_t)} = \frac{\partial \log h_t}{\partial(1-\tau_t)}\underbrace{\frac{\partial(1-\tau_t)}{\partial \log(1-\tau_t)}}_{=(1-\tau_t)} = \frac{1}{\gamma}\frac{w_t}{w_t(1-\tau_t)+\alpha w F_t}(1-\tau_t)$$

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Marshall Elasticity

- **Q:** What does the disconnect between OCT and Wage mean for the responsiveness of young workers to *permanent* tax-changes?

- **A:** *permanent* tax-changes now increases the return to human capital $\rightarrow$ **Marshall elasticity can be higher** (compared to when $\alpha = 0$)

$$e_{M,t} \equiv \left. \frac{\partial \log h_t}{\partial \log(1-\tau)} \right| = \frac{1 + \eta \cdot E_t / C}{\gamma - \eta \cdot C_t^* / C}$$

where

$$E_t = \sum_{s=t}^{T} \frac{w_s(1-\tau)}{(1+r)^{s-t}}$$
$$C_t^* = w_t(1-\tau)h_t + \alpha w h_t F_t = OCT_t \cdot h_t$$

is the PV of after-tax earnings and "effective earnings", respectively.

- Note $e_{M,t} \rightarrow \frac{1}{\gamma} = e_F$ for $t \rightarrow T$ since $E_T = C_T^* = 0$.

Marshall vs. Frisch Elasticity when $\alpha > 0$

- **We might have a violation** of the old inequality

$$e_F \overset{?}{\geq} e_M$$

- **Frisch is only an upper bound** if

$$\frac{1}{\gamma}\frac{w_t(1-\tau_t)}{w_t(1-\tau_t)+\alpha w F_t} \geq \frac{1+\eta \cdot E_t / C}{\gamma - \eta \cdot C_t^* / C}$$

which – if there is no income effects ($\eta = 0$) – is satisfied only if

$$\alpha = 0.$$

- **They converge** in age:
$e_{M,t} \to \frac{1}{\gamma} = e_F$ for $t \to T$ since $E_T = C_T^* = 0$.

Reduced-form Estimation

- How is "standard" regressions affected by HC?

$$\Delta \log h_t = \frac{1}{\hat{\gamma}} \Delta \log(w_t(1-\tau_t)) + \epsilon_t$$

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- Where $F_t < F_{t-1}$.
  $\rightarrow \left[\bullet \right] < \Delta \log(w_t(1-\tau_t))$, which is the included regressor
  $\rightarrow \frac{1}{\hat{\gamma}} < \frac{1}{\gamma}$, as variation $\Delta \log h_t$ is "unchanged"/"fixed" by data
  $\rightarrow \hat{e}_F < e_F$ (downwards bias in Frisch, which might not upper bound)

What to do then?

- We can do **structural estimation**, but we must specify everything... Reduced form nice since we can remain "agnostic" about wage process.

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  Re-write the "true" equation

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$$= \frac{1}{\gamma}\Delta \log(w_t(1-\tau_t)) + \underbrace{f(w_t, w_{t-1}, \tau_t, \tau_{t-1}, F_t, F_{t-1}) + u_t}_{=\epsilon_t\text{: error-term in standard regression}}$$

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  Any *relevant* instrument, Z_t, with $Cov(w_t(1-\tau_t), Z_t) \neq 0$
  will also be ***invalid***, as $Cov(\epsilon_t, Z_t) \neq 0$!

Simulated Elasticities

- Keane (2016, sec. 3) simulates elasticities
  Based on model estimated in Imai and Keane (2004)
- **Richer version** of our toy model
  but quite similar

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  Based on model estimated in Imai and Keane (2004)
- **Richer version** of our toy model
  but quite similar
- **Short run:**
  Current period response
  Like the Frisch, Marshall and Hicks
- **Long run:**
  Average effect from permanent increase throughout life
- See notebook after discussion

Simulated Elasticities: Short Run

- **Short Run**:
  Response in period t to a transitory change in period t

Table 3

Short-run Labour Supply Responses to Taxes in the Imai–Keane Model

	Transitory		Permanent (unanticipated)	
Age	Unanticipated	Anticipated (Frisch)	Uncompensated (Marshall)	Compensated (Hicks)
20	**0.30**	**0.30**	0.14	**0.64**
25	**0.36**	**0.36**	0.12	**0.54**
30	**0.44**	**0.44**	0.12	**0.48**
35	0.52	0.52	0.10	0.46
40	0.64	0.66	0.14	0.46
45	0.76	0.84	0.20	0.56
50	0.94	1.06	0.46	0.84
55	1.24	1.44	1.06	1.44
60	**1.74**	**1.96**	1.88	**2.09**

Notes. All figures are elasticities of current hours with respect to tax changes. The 'transitory' increase only applies for one year at the indicated age. In the 'anticipated' case this has no wealth effect, so it is a pure Frisch effect. The 'permanent' tax increases take effect (unexpectedly) at the indicated age and last until age 65. In the 'compensated' case the proceeds of the tax (in each year) are distributed back to agents in lump sum form. Figures in bold are cases where permanent tax effects exceed transitory tax effects.

- **Bold:** $e_H > e_F$ (unlike baseline model)

Simulated Elasticities: Long Run

- **Long Run**:
  Response in period t to a permanent change from period 20 until 65 ("regime shift". Could alternatively have been from age t to 65)

Table 4

Lifetime Effects of a Permanent Tax Increase on Labour Supply

Age	Uncompensated	Compensated
20	0.14	0.64
30	0.14	0.66
40	0.18	0.84
45	0.24	1.14
50	0.42	1.74
60	1.82	4.00
Lifetime hours (Ages 20–65)	0.40	1.32

Notes. This Table compares the baseline simulation of the Imai and Keane (2004) model with an alternative scenario where the tax rate on earnings is permanently higher. The increase is in effect from the first period (age 20) until the terminal period (age 65). The Table reports both the uncompensated case and the case where the proceeds of the tax (in each year) are distributed back to agents in lump sum form.

Next Time

- **Next time:**
  Labor supply and children.

- **Literature:**
  Adda, Dustmann and Stevens (2017): "The Career Costs of Children"
  - **Read** before lecture
  - **Reading guide:**
    Section 1: Introduction. Key
    Section 2: Data. Skim fast.
    Section 3: Model. *Key*, but complex. Get the idea.
    Section 4: Results. *Simulations in sections E, F and G are key*!

References I

Adda, J., C. Dustmann and K. Stevens (2017): "The Career Costs of Children," *Journal of Political Economy*, 125(2), 293–337.

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Career Costs of Children

Thomas H. Jørgensen

2025

Plan for today

- Dynamic Labor supply w. HC and **children**
  Adda, Dustmann and Stevens (2017): "The Career Costs of Children"

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- **Reading guide:**
  1. What are the main *research question*s?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

Plan for today

- Dynamic Labor supply w. HC and **children**
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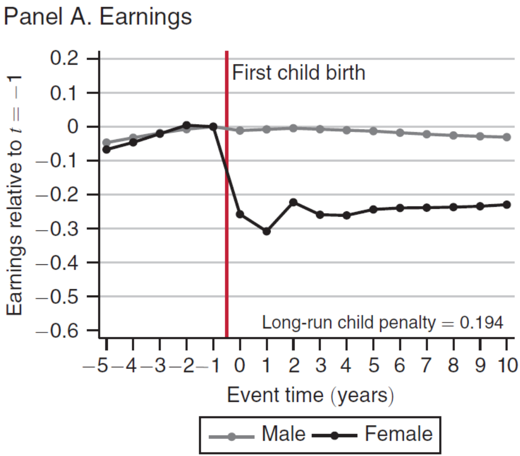
- **Reading guide:**
  1. What are the main *research questions*?
     - How **costly** are children for careers over the life cycle?
     - How does pro-fertility **reforms** affect completed fertility?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

Empirical Motivation: I

- "Child penalty" (Kleven, Landais and Søgaard, 2019)



Empirical Motivation: II

TABLE 1
DESCRIPTIVE STATISTICS, BY OCCUPATION

	Routine	Abstract	Manual	Whole Sample
Initial occupation	25.0%	44.8%	30.3%	100%
Occupation of work	25.4%	52.7%	21.9%	
A				
Annual occupational transition rates:				
If in routine last year	97.9%	1.5%	.5%	
If in abstract last year	.7%	99.0%	.2%	
If in manual last year	.9%	.8%	98.3%	
B				
Log wage at age 20	3.598	3.742	3.470	3.634
	(.297)	(.301)	(.386)	(.337)
Log wage growth, at potential experience = 5 years	.0485	.0551	.0450	.0510
	(.187)	(.156)	(.196)	(.175)
Log wage growth, at potential experience = 10 years	.0181	.0240	.0152	.0208
	(.187)	(.206)	(.223)	(.206)
Log wage growth, at potential experience = 15 years	.00995	.0147	.0127	.0133
	(.206)	(.195)	(.211)	(.200)
C				
Total work experience after 15 years	11.55	12.81	12.14	12.34
	(3.273)	(2.624)	(2.880)	(2.909)
Full-time work experience after 15 years	10.32	11.92	10.86	11.29
	(3.907)	(3.348)	(3.570)	(3.617)
Part-time work experience after 15 years	1.229	.889	1.274	1.056
	(2.187)	(1.828)	(2.125)	(1.997)
D				
Total log wage loss, after interruption = 1 year	−.0968	−.147	−.105	−.121
	(.560)	(.636)	(.633)	(.613)
Total log wage loss, after interruption = 3 years	−.152	−.253	−.223	−.216
	(.604)	(.639)	(.619)	(.625)
E				
Age at first birth	27.27	28.39	25.94	27.56
	(4.138)	(3.783)	(3.517)	(3.943)

Empirical Motivation: III

- **Selection** into family friendly occupations
  $\rightarrow$ correlation $\neq$ causation!
  $\rightarrow$ we need a model!

- **Short run** effects of pro-fertility reforms on labor supply are substantial
  Reduced form evidence

- **Long run effects**: “need” a model!

Outline

1. Model and Mechanisms
2. Simulation Results
3. Simple Model

Model Overview

- **Choices:**
  $b_t \in \{0, 1\}$: fertility, try to conceive a child
  $c_t \in (0, \overline{M}]$: consumption (household)
  $o_t \in \{1, 2, 3\}$: occupation of women (effect end of period)
  $l_t \in \{\text{OLF}, \text{U}, \text{PT}, \text{FT}\}$: labor supply of women (effect end of period)
  (and o_0 initial occupation/education)

OLF: Out of labor force,

Model Overview

- **Choices:**
  $b_t \in \{0,1\}$: fertility, try to conceive a child
  $c_t \in (0, \overline{M}]$: consumption (household)
  $o_t \in \{1,2,3\}$: occupation of women (effect end of period)
  $l_t \in \{\text{OLF}, \text{U}, \text{PT}, \text{FT}\}$: labor supply of women (effect end of period)
  (and o_0 initial occupation/education)
- **States**, Ω_t:
  l_{t-1} and o_{t-1}: Past labor market
  A_{t-1} : wealth, no borrowing
  $h_{t-1} \in \{0,1\}$: presence of husband last period
  age_t^M: age of woman
  x_t: human capital of women
  n_t: number of children
  age_t^K: age of youngest child
  Υ_t: preference and income shocks (see e.g. p.331)
  $f = (f^P, f^F, f^L, f^C)$: heterogeneity
  (productivity, fertility, taste for leisure, taste for children)

Model Overview

Key mechanisms:

1. **Early life occupational choices** (age 15) locks in on fertility effects
   - **Family friendliness** differs across occupations
     Wage level/growth
     Costs of temporary leave (atrophy)
     Offer probabilities
     Amenities (utility value)
2. **Human capital** leads to persistent effects of early life behavior
3. **Endogenous fertility** trades of timing of children and career

State Transitions, $\Omega_{t+1} \sim \Gamma(\Omega_t, b_t, c_t, o_t, l_t)$

- l_t and o_t are choices. A_t: Assets in period t, A_(t-1): Number of assets you have from the previous period

$$A_t = (1+r)A_{t-1} + net(GI_t; h_t, n_t) \\ - c_t - \kappa(age_t^K, n_t)\mathbf{1}(n_t > 0, l_t \in \{\text{PT}, \text{FT}\})$$

Cost associated with taking care of the kids

where GI_t is gross household income (next slide)

- h_t (husband) is random and function of age_t^M, x_t, f^C. Husbands earnings are a function of women's characteristic
- $age_{t+1}^M = \frac{1}{2} + age_t^M$

$$x_{t+1} = \begin{cases} x_t + 1 & \text{if } l_t = \text{FT} \\ x_t + \frac{1}{2} & \text{if } l_t = \text{PT} \\ x_t \rho(x_t, o_t) & \text{else} \end{cases}$$

Loss of human capital if woman is not working

where $\rho(x_t, o_t)$ is deprecation rate.

- $n_{t+1} = n_t + 1$ with prob. $\pi(age_t^M, f^C)\mathbf{1}(b_t = 1)$. Else $n_{t+1} = n_t$.
- $age_{t+1}^K = \frac{1}{2} + age_t^K$ if $n_{t+1} = n_t$ else $age_{t+1}^K = 0$

State Transitions, Gross Income

- Never write up the gross income. This is my take.
- Gross household income is (note timing)

Her wage w_t multiplied by labor supply last period l_(t-1)

$$GI_t = w_t l_{t-1} + \mathbf{1}(h_t = 1) earn_t^h + \text{benefits...}$$

If husband is present we get his earnings

- Husbands earnings are

$$earn_t^h = \alpha_0^h + \alpha_1^h age_t^M + \alpha_2^h (age_t^M)^2 + \alpha_o(o_t) + \alpha_P^h f^P + \eta_t^h$$

- Wages of women are Mincer-type

$$\log w_t = f^P + a_0(o_t) + a_X(o_t) x_t + a_{XX}(o_t) x_t^2 + \eta_t$$

Recursive Formulation

- **Bellman equation** is

$$V_t(\Omega_t) = \max_{b_t, c_t, o_t, l_t} u(\bullet) + \beta \mathbb{E}_t[V_{t+1}(\Omega_{t+1})]$$

s.t.

$$\Omega_{t+1} \sim \Gamma(\Omega_t, b_t, c_t, o_t, l_t)$$

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  Split up the different discrete choices. See their appendix.
  I will illustrate how the fertility choice is made (conditional on o_t, l_t)

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- **Implemented "sequentially"**
Split up the different discrete choices. See their appendix.
I will illustrate how the fertility choice is made (conditional on o_t, l_t)
- **Discuss:** What is good/bad about this model for the research purpose?

Recursive Formulation, working + conceiving

- Value of **W**orking ($l_t \in \{\mathrm{PT}, \mathrm{FT}\}$) and **C**onceiving ($b_t = 1$)

$$
\begin{aligned}
V^{W,C}(\Omega_t) = \max_{c_t} u(c_t, \bullet) &+ \pi(age_t^M, f^C)\beta\mathbb{E}_t[V^{L_w}(\Omega_{t+1}^P)] \\
&+ \delta(1 - \pi(age_t^M, f^C))\beta\mathbb{E}_t[V^U(\Omega_{t+1})] \\
&+ (1-\delta)(1 - \pi(age_t^M, f^C))(1 - \phi_0(o_t, l_t))\beta Emax_t \\
&+ (1-\delta)(1 - \pi(age_t^M, f^C))\phi_0(o_t, l_t)\beta \widetilde{Emax}_t
\end{aligned}
$$

Here he goes through all the expected values of the different combinations

where
$V^{L_w}(\Omega_{t+1}^P)$: value of parental leave
$V^U(\Omega_{t+1})$: value of unemployment (w. prob δ)
$\phi_0(o_t, l_t)$: job-offer prob.
$Emax_t = \mathbb{E}_t[\max\{V_{t+1}^W + \eta_{t+1}^W, V_{t+1}^U + \eta_{t+1}^U, V_{t+1}^O + \eta_{t+1}^O\}]$
$\widetilde{Emax}_t = \ldots$ Also choose between leaving and staying (see p. 331).

Recursive Formulation, working + not conceiving

- Value of **W**orking ($l_t \in \{PT, FT\}$) and **N**ot **C**onceiving ($b_t = 0$)

$$\begin{aligned} V^{W,NC}(\Omega_t) = & \max_{c_t} u(c_t, \bullet) \\ & + \delta\beta\mathbb{E}_t[V^U(\Omega_{t+1})] \\ & + (1-\delta)(1-\phi_0(o_t, l_t))\beta Emax_t \\ & + (1-\delta)\phi_0(o_t, l_t)\beta \widetilde{Emax_t} \end{aligned}$$

- **Fertility choice** is then (conditional on k/working)

$$b_t^\star(k) = \arg\max\{V^{k,C}(\Omega_t), V^{k,NC}(\Omega_t)\}$$

[but also extreme value taste-shocks in utility wrt. conceiving]

Estimation Results

- **Simulated method of moments (SMM)**
  Weighting matrix: diagonal, inverse of variance of empirical moments.
  763 moments (Table 2)
- **Estimate 88 parameters** (allowing for unobserved types)

$$
\begin{aligned}
u_{it} = {} & \frac{(c_{it}/\bar{c})^{(1-\gamma c)}-1}{1-\gamma c} \exp\Big[\gamma^1_{PT} I_{l_{it}=PT} + (\gamma^1_U + f^L_i) I_{l_{it}=U} \\
& + (\gamma^1_{\mathrm{OLF}} + f^L_i) I_{l_{it}=\mathrm{OLF}}\Big] \exp\Big(\gamma_{NC} I_{n_{it}>0}\Big) \\
& + \Big[\gamma^1_N (f^C_i) I_{n_{it}=1} + \gamma^2_N (f^C_i) I_{n_{it}=2}\Big] \cdot \exp\Big(\gamma_{NH} I_{n_{it}>0 \& h_{it-1}}\Big) \\
& \cdot \exp(\gamma_U)^{I_{l_{it}=U}} \cdot \exp\Big(\gamma_{\mathrm{OLF}} + \gamma^1_{A,\mathrm{OLF}} I_{\mathrm{age}^{\kappa}_{it} \in [0,3]} \\
& + \gamma^2_{A,\mathrm{OLF}} I_{\mathrm{age}^{\kappa}_{it} \in [4,6]} + \gamma^3_{A,\mathrm{OLF}} I_{\mathrm{age}^{\kappa}_{it} \in [7,9]}\Big)^{I_{l_{it}=\mathrm{OLF}}} \\
& \cdot \exp\Bigg(\sum_{i_o=1}^{3} \gamma_{i_o,PT} I_{o_{it}=i_o} + \gamma^1_{A,PT} I_{\mathrm{age}^{\kappa}_{it} \in [0,3]} \\
& + \gamma^2_{A,PT} I_{\mathrm{age}^{\kappa}_{it} \in [4,6]} + \gamma^3_{A,PT} I_{\mathrm{age}^{\kappa}_{it} \in [7,9]}\Bigg)^{I_{l_{it}=PT}} \\
& \cdot \exp\Bigg(\sum_{i_o=1}^{3} \gamma_{i_o,W} I_{o_{it}=i_o}\Bigg)^{I_{l_{it}=PT,FT}} + \eta^C_{it} b_{it} + \eta^{NC}_{it}(1-b_{it}).
\end{aligned}
$$

Estimation Results

TABLE 3
OCCUPATION-SPECIFIC PARAMETERS

Parameter	Routine	Abstract	Manual
	A. Atrophy Rates Parameters (Annual Depreciation Rates)		
At 3 years of uninterrupted work experience	−.06%	−.11%	−.03%
	(1e−5%)	(2e−5%)	(2e−5%)
At 6 years of uninterrupted work experience	−.50%	−6.90%	−3.45%
	(.11%)	(.17%)	(.24%)
At 10 years of uninterrupted work experience	−.61%	−2.65%	−3.08%
	(14.2%)	(.01%)	(.18%)
	B. Wage Equation Parameters		
Log wage constant	3.39	3.6	3.32
	(.0038)	(.0054)	(.0059)
Years of uninterrupted work experience	.1	.09	.123
	(3.3e−05)	(3.6e−05)	(.0001)
Years of uninterrupted work experience, squared	−.00382	−.0021	−.00463
	(3e−06)	(4.1e−06)	(6.4e−06)
	C. Amenity Value of Occupations		
Utility of work if children	0	−.056	−.014
		(.001)	(.0005)
Utility of part-time work if children	0	−.42	−.08
		(.003)	(.007)

Outline

1. Model and Mechanisms
2. Simulation Results
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Simulation Results: Career Costs of Children

- **Career costs of children**
  NPV difference in income of women
  between baseline and alternative model
- **Alternative model:** no children (known by all)

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TABLE 6
CAREER COST OF CHILDREN: PERCENTAGE LOSS IN NET PRESENT VALUE OF INCOME AT AGE 15, WITH AND WITHOUT FERTILITY

	Percentage Loss Compared to Baseline
Total cost	−35.3%
	A. Oaxaca Decomposition of Total Cost
Labor supply contribution	−27%
Wage contribution	−8.5%
	B. Oaxaca Decomposition of Wage Contributions
Contribution of atrophy	−1.8%
Contribution of other factors	−6.7%
Contribution of occupation	−1.6%
Contribution of other factors	−7%

NOTE.—The career costs are evaluated using simulations and comparing the estimated model with a scenario in which the woman knows ex ante that she cannot have children. The costs are computed as the net present value of female incomes, including all wages, unemployment benefits, and maternity benefits in the calculations. The discount factor is set to 0.95 annually. Initial occupation is the one in the no-fertility scenario.

Simulation Results: Career Costs of Children

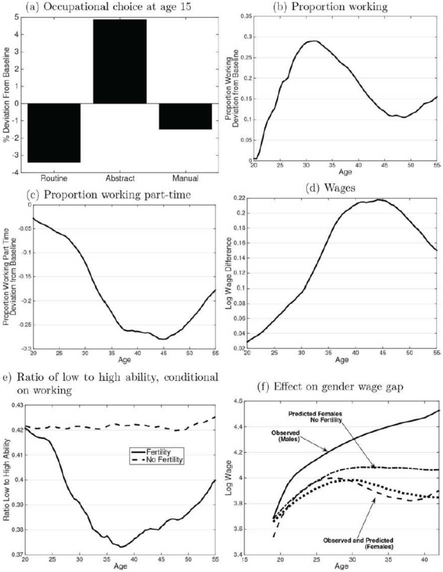

FIG. 3.—Effect of no fertility. The different panels display the difference in outcomes between a baseline scenario and one in which a woman knows that she is infertile.

Counterfactual reform: Pro-fertility

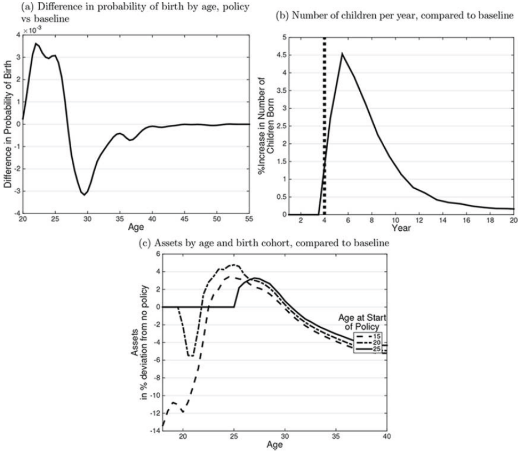

FIG. 4.—Effect of child premium. Panel a shows the effect of the policy (cash transfer of €6,000 at birth) by age on the probability of giving birth, comparing the policy to the baseline. In the policy scenario, women learn at age 15 about the policy. Panel b depicts the aggregate effect of the policy, by year, in an overlapping generation economy. The graph aggregates each year the behavior of women aged 15–60. Each year a new cohort of 15-year-olds enters the economy and the cohort who is 60 exits. The policy starts in year 4. Panel c displays the percentage change in assets as a function of age, compared to a baseline without transfer. The birth cohort who is 15 at the start of the policy can adjust right away their behavior. The cohorts who are 20 or 25 when the policy starts do not anticipate the policy.

Outline

Extending our simple model

- **We can extend** the simple dynamic model of Keane (2016)
  Random arrival of a child, $n_t \in \{0, 1\}$
  Dis-utility from work depend on children

Extending our simple model

- **We can extend** the simple dynamic model of Keane (2016)
  Random arrival of a child, $n_t \in \{0, 1\}$
  Dis-utility from work depend on children
- **Bellman equation**

$$V_t(n_t, a_t, k_t) = \max_{c_t, h_t} \frac{c_t^{1+\eta}}{1+\eta} - \beta(n_t)\frac{h_t^{1+\gamma}}{1+\gamma} + \rho \mathbb{E}_t[V_{t+1}(n_{t+1}, a_{t+1}, k_{t+1})]$$

s.t.

$$n_{t+1} = \begin{cases} n_t + 1 & \text{with prob. } p(n_t) \\ n_t & \text{with prob. } 1 - p(n_t) \end{cases}$$

$$a_{t+1} = (1+r)(a_t + (1-\tau_t)w_t h_t - c_t)$$

$$k_{t+1} = k_t + h_t$$

Extending our simple model

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$$w_t = w\,(1 + \alpha k_t)$$

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$$p(n_t) = \begin{cases} p_n & \text{if } n_t = 0 \\ 0 & \text{if } n_t = 1 \end{cases}$$

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such that *original model is nested* if $\beta_1 = 0$.

- **Expected value** is

$$\begin{aligned} \mathbb{E}_t[V_{t+1}(n_{t+1}, a_{t+1}, k_{t+1})] &= p(n_t) V_{t+1}(n_t + 1, a_{t+1}, k_{t+1}) \\ &+ (1 - p(n_t)) V_{t+1}(n_t, a_{t+1}, k_{t+1}) \end{aligned}$$

- See notebook...

Next Time

- **Next time:** Labor supply of couples.
  Remember: Assignment!

- **Literature:**
  Borella, De Nardi and Yang (2023): "Are Marriage-Related Taxes and Social Security Benefits Holding Back Female Labor Supply?"
  - **Read** before lecture.
    Focus on "working-stage of couples" and removal of joint taxation
  - **Reading guide:**
    Section 1: Introduction. Read
    Section 2+3: Taxation of Couples in the US (short). *Motivation, key*.
    Section 4: Model. *Key*, but complex. Get the idea. Focus on "working-stage of couples". Think about how children enter.
    Section 5: Estimation. Skim.
    Section 6: "Validation", short. Labor supply elasticities, read.
    Section 7: Counterfactual simulations. Key - Read with focus on 7.1.
    Section 8: Sensitivity/robustness. Can drop.

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Household Labor Supply and Taxes

Thomas H. Jørgensen

2025

Plan for today

- Dynamic labor supply of **couples**
  Borella, De Nardi and Yang (2023): "Are Marriage-Related Taxes and Social Security Benefits Holding Back Female Labor Supply?"

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- **Reading guide:**
  1. What are the main *research questions*?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

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- **Reading guide:**
  1. What are the main *research question*s?
     - How does household-level taxes and transfers affect labor supply?
     - Could individual taxes/transfers increase welfare?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

Empirical Motivation: I

- High marginal tax rates for secondary earner (often women historically)
  $\rightarrow$ labor supply discouraged
  $\rightarrow$ specialization
  $\rightarrow$ intra-household inequality

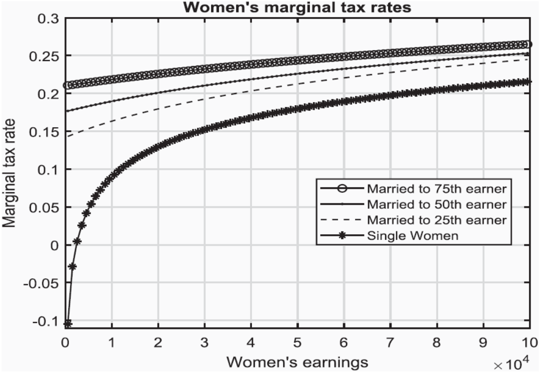
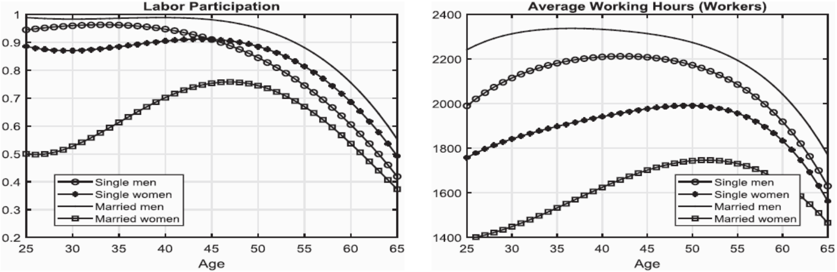


FIGURE 1

Women's marginal tax rates as a function of earnings (2016 dollars). Single (starred) and married to men at different

Empirical Motivation: II



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Model Overview

- **Three stages**
  1. Working (25–61)
  2. Early retirement (62–65)
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    Consumption/Savings, c_t, a_{t+1}
  - **States:**
    Savings, a_t
    Income shocks of both, ϵ_t^i
    Human capital of both, $\overline{y}_t^i$

Preferences

- **Individual preferences** are [my notation]

$$u(c_t, l_t, i, j) = \frac{[(c_t/\eta^{i,j})^{\omega} l_t^{1-\omega}]^{1-\gamma} - 1}{1-\gamma}$$

where
$l_t^{i,j} = L^{i,j} - n_t^i - \Phi_t^{i,j}\mathbf{1}(n_t^i > 0)$ is leisure.
(4 parameters estimated for each gender/marital status)
$\eta^{i,j}$ is equivalence scales
ω is the Cobb-Douglas input elasticity
γ is the CRRA coefficient

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ω is the Cobb-Douglas input elasticity
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- **Utility of a single** man and woman is $u(c_t, l_t, 1, 1)$ and $u(c_t, l_t, 2, 1)$.
- **Utility of a couple** is

$$U(c_t, l_t^1, l_t^2) = u(c_t, l_t^1, 1, 2) + u(c_t, l_t^2, 2, 2)$$

Human Capital and Wages

- **Human capital** is previous avg. earnings, approximated as

$$\overline{y}^i_{t+1} = \frac{\overline{y}^i_t(t - t_0) + \min(Y^i_t, \tilde{y}_t)}{t + 1 - t_0} \tag{1}$$

where
$Y^i_t = w^i_t n^i_t$ is labor earnings
$\tilde{y}_t$ is Social Security cap
$t_0 = 25$.

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  where
  $Y^i_t = w^i_t n^i_t$ is labor earnings
  $\tilde{y}_t$ is Social Security cap
  $t_0 = 25$.

- **Wages** are

$$w^i_t = e^i_t(\overline{y}^i_t)\epsilon^i_t$$

  where

y\bar is a state variable

$$e^i_t(\overline{y}^i_t) : \text{age, gender and HC. Table 1 in Appendix}$$

$$\log \epsilon^i_{t+1} = \rho^i_\epsilon \log \epsilon^i_t + v^i_{t+1},\ v^i_{t+1} \sim \mathcal{N}(0, (\sigma^i_v)^2) \tag{2}$$

Government: Taxes and Transfers

- Labor income taxes are approximated as

$$T(Y, i, j, t) = (1 - \lambda_t^{i,j} Y^{-\tau_t^{i,j}}) \cdot Y$$

  where
  $Y = ra_t + Y_t^1 + Y_t^2$ is total *household* income
  $\lambda_t^{i,j}$ and $\tau_t^{i,j}$ are gender/marital specific tax-parameters (not reported).
- Payroll tax: $\min(Y, \tilde{y}_t)\tau_t^{SS}$
- Consumption floor, $c(j)$. See table 10 in Appendix.

Figure 1 is based on different parameters for lambda and tau

Children

- **Exogenous/Perfect foresight** and continuous. Only women + couples.

Children

- **Exogenous/Perfect foresight** and continuous. Only women + couples.
- $f^{0,5}(i,j,t)$: number of children in age-group 0-5
  $\tau_c^{0,5}$: child care cost, pct of income (estimated)
- $f^{6,11}(i,j,t)$: number of children in age-group 6-11
  $\tau_c^{6,11}$: child care cost, pct of income (estimated)
- $f(1,1,t) = 0$ (single men)

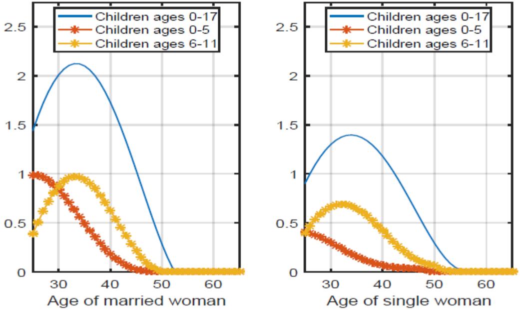

Figure: Figure 5 in Online Appendix. 1945 cohort.

Marriage and Divorce

- **Marriage** probability depends on wage-shock

What is the probability that there is a spouse. Allow to be function of current income shock

$$v_{t+1}(i, \epsilon_t^i) = \Pr(j_{t+1} = 2 | j_t = 1, t, i, \epsilon_t^i)$$

- Probability of *matching* a partner with states $(a_{t+1}^p, \overline{y}_{t+1}^p, \epsilon_{t+1}^p)$:

$$\Pr(a_{t+1}^p, \overline{y}_{t+1}^p, \epsilon_{t+1}^p | \epsilon_t^i, i) = \theta_{t+1}(a_{t+1}^p, \overline{y}_{t+1}^p | \epsilon_{t+1}^p) \cdot \xi_{t+1}(\epsilon_{t+1}^p | \epsilon_t^i, i)$$

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Condition on what my own income shock is. Could I also condition on own human capital?

- **Divorce** probability depends on both members wage shocks

$$\zeta_{t+1}(\epsilon_t^1, \epsilon_t^2) = \Pr(j_{t+1} = 1 | j_t = 2, t, \epsilon_t^1, \epsilon_t^2)$$

- Wealth *equally split* + no alimony.

Recursive Formulation: Working-Stage Couple

- **Bellman Equation** for couple is (subject to (1) and (2))

$$
\begin{aligned}
W_t^c(a_t, \epsilon_t^1, \epsilon_t^2, \overline{y}_t^1, \overline{y}_t^2) = \max_{c_t, n_t^1, n_t^2} & \; U(c_t, l_t^1, l_t^2) \\
& + (1-\zeta_{t+1})\beta \mathbb{E}_t[W_{t+1}^c(a_{t+1}, \epsilon_{t+1}^1, \epsilon_{t+1}^2, \overline{y}_{t+1}^1, \overline{y}_{t+1}^2)] \\
& + \zeta_{t+1}\beta \sum_{i=1}^{2} \mathbb{E}_t[W_{t+1}^s(i, a_{t+1}/2, \epsilon_{t+1}^i, \overline{y}_{t+1}^i)]
\end{aligned}
$$

s.t.

$$
\begin{aligned}
a_{t+1} = & \; (1+r)a_t + Y_t^1 + Y_t^2(1-\tau_c(2,2,t)) - c_t \\
& - \tau_t^{SS} \sum_{i=1}^{2} \min(Y_t^i, \tilde{y}_t) - T(ra_t + Y_t^1 + Y_t^2, 2, t)
\end{aligned}
$$

where (see e.q 2)

$W_{t+1}^s(\bullet)$ is value of being single

$\mathbb{E}_t[W_{t+1}^c(a_{t+1}, \epsilon_{t+1}^1, \epsilon_{t+1}^2, \overline{y}_{t+1}^1, \overline{y}_{t+1}^2)] =$

$$
\int\int W_{t+1}^c(\bullet, \underbrace{\exp(\rho_\epsilon^1 \log \epsilon_t^1 + v_{t+1}^1)}_{\epsilon_{t+1}^1}, \underbrace{\exp(\rho_\epsilon^2 \log \epsilon_t^2 + v_{t+1}^2)}_{\epsilon_{t+1}^2}, \bullet)\phi(dv_{t+1}^1)\phi(dv_{t+1}^2)
$$

Outline

1. Model and Mechanisms
2. Simulation Results
3. Simple Model

Labor Supply Elasticities

- **Frisch:** Anticipated transitory income changes

TABLE 4
Model-implied elasticities of labour supply

	Participation				Hours among workers			
	Married		Single		Married		Single	
	W	M	W	M	W	M	W	M
30	1.0	0.0	0.5	0.2	0.2	0.3	0.4	0.4
40	0.7	0.1	0.4	0.2	0.4	0.5	0.4	0.5
50	0.6	0.2	0.4	0.5	0.4	0.5	0.8	0.5
60	1.1	0.8	1.8	1.4	0.3	0.3	0.5	0.4

- Highest for women
- Extensive margin important

Labor Supply Elasticities

- **Marshall:** permanent increase in *wages of women* from age 25 (t_0), I think, i.e. "regime shift".

- Large for married women
- U-shaped
- Small negative cross-elasticity for men.

Counterfactual Policy Simulations

- **Remove the Joint taxation.**
  Unclear exactly how, but I think it is like

$$a_{t+1} = (1+r)a_t + Y_t^1 + Y_t^2(1 - \tau_c(2,2,t)) - c_t - \tau_t^{SS}\sum_{i=1}^{2}\min(Y_t^i, \tilde{y}_t)$$
$$- T(ra_t/2 + Y_t^1, 1, 1, t) - T(ra_t/2 + Y_t^2, 2, 1, t)$$

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$$- T(ra_t/2 + Y_t^1, 1, 1, t) - T(ra_t/2 + Y_t^2, 2, 1, t)$$

- **Balance government budget** by changing $\lambda_t^{i,j}$ in

$$T(Y, i, j, t) = (1 - \lambda_t^{i,j} Y^{-\tau_t^{i,j}}) \cdot Y$$

Counterfactual Policy Simulations

- **Remove the Joint taxation**.
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$$T(Y, i, j, t) = (1 - \lambda_t^{i,j} Y^{-\tau_t^{i,j}}) \cdot Y$$

- **Also: Remove spousal dependence** on social and survivor benefits
  Only affects in later life stages (ignore a bit here)

Simulation Results: Participation

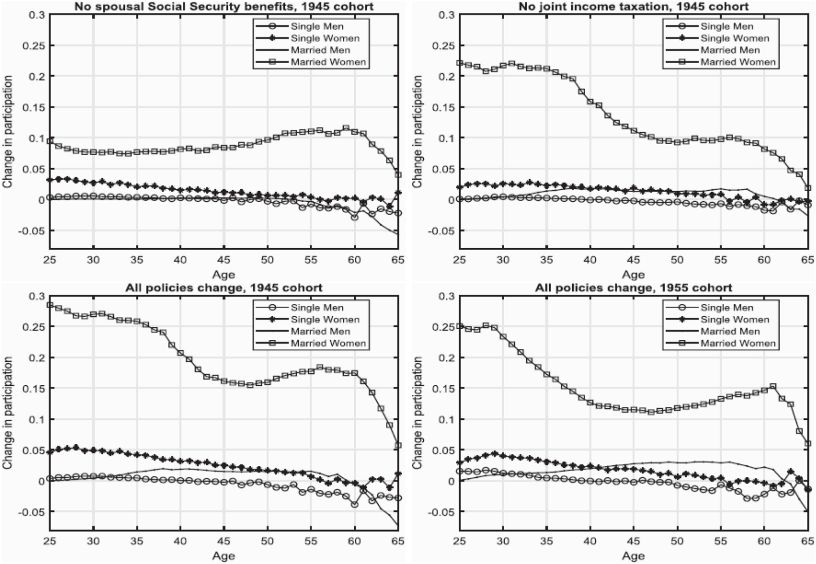

FIGURE 9

Simulation Results: Welfare

- **Welfare effects:** Level of wealth at age 25 (t_0) in the baseline model that makes individuals indifferent between the baseline and the new

Simulation Results: Welfare

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- Let the alternative value be (all couples here)

$$V^{reform} = \frac{1}{s} \sum_{s=1}^{S} V_{t_0}(a_{s,t_0}, \epsilon^1_{s,t_0}, \epsilon^2_{s,t_0}, \overline{y}^1_{s,t_0}, \overline{y}^2_{s,t_0}; reform)$$

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- Let baseline value be

$$V^{base}(a) = \frac{1}{s}\sum_{s=1}^{S} V_{t_0}(a_{s,t_0} + a, \epsilon^1_{s,t_0}, \epsilon^2_{s,t_0}, \overline{y}^1_{s,t_0}, \overline{y}^2_{s,t_0})$$

as a function of additional initial wealth, a.

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  as a function of additional initial wealth, a.
- Asset compensation required

$$a^\star = \{a : V^{reform} - V^{base}(a) = 0\}$$

  (normalized by avg income: $\frac{1}{2}a^\star/(\overline{y}^1_{s,t_0} + \overline{y}^2_{s,t_0})$ (?))

Simulation Results: Welfare

TABLE 7

Asset compensation required for staying in the benchmark economy, normalized as a fraction of average income

	All			Winners			Losers		
	Couples	SW	SM	Couples	SW	SM	Couples	SW	SM
1945 cohort									
(1) Remove Social Security spousal benefits, unbalanced budget									
Average	−0.24	−0.20	0.25	0.00	0.00	0.25	−0.24	−0.20	−0.02
Percentage				0.0	0.0	100.0	100.0	100.0	0.0
(2) Remove Social Security spousal benefits, balanced budget									
Average	0.66	0.19	1.15	0.66	0.20	1.15	0.00	−0.03	0.00
Percentage				100.0	92.5	100.0	0.0	7.5	0.0
(3) Remove joint income taxation, unbalanced budget									
Average	0.06	−0.18	0.81	0.29	0.06	0.81	−0.19	−0.19	0.00
Percentage				52.8	4.9	100.0	47.2	95.1	0.0
(4) Remove joint income taxation, balanced budget									
Average	0.31	−0.08	1.06	0.42	0.12	1.06	−0.09	−0.13	0.00
Percentage				78.8	20.6	100.0	21.2	79.4	0.0
(5) Remove all marital-related policies, balanced budget									
Average	0.80	0.05	1.97	0.80	0.33	1.97	−0.03	−0.12	0.00
Percentage				98.8	37.4	100.0	1.2	62.6	0.0
1955 cohort									
(6) Remove all marital-related policies, balanced budget									
Average	0.73	0.21	1.14	0.74	0.30	1.14	−0.04	−0.04	−0.03
Percentage				98.2	74.1	100.0	1.8	25.9	0.0

Notes: Top line for each experiment: average welfare gain or loss. Bottom line for each experiment: fraction in that group gaining or losing welfare. SM, single men; SW, single women.

Outline

1. Model and Mechanisms

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3. Simple Model

Our simple model

- **Dual-earner model**
- **Simplifications:**
  No savings
  Couple cannot divorce (no singlehood)
  Deterministic (no shocks)
- **Taxes:**
  On household level
- **Reform** of interest:
  Individual taxation

Our simple model

- **Recursive formulation**

$$V_t(K_{1,t}, K_{2,t}) = \max_{h_{1,t}, h_{2,t}} U(c_t, h_{1,t}, h_{2,t}) + \beta V_{t+1}(K_{1,t+1}, K_{2,t+1})$$

$$c_t = \sum_{j=1}^{2} w_{j,t} h_{j,t} - T(w_{1,t} h_{1,t}, w_{2,t} h_{2,t})$$

$$\log w_{j,t} = \alpha_{j,0} + \alpha_{j,1} K_{j,t},\ j \in \{1, 2\}$$

$$K_{j,t+1} = (1 - \delta) K_{j,t} + h_{j,t},\ j \in \{1, 2\}$$

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- **Preferences** are sum of individual

$$U(c_t, h_{1,t}, h_{2,t}) = 2\frac{(c_t/2)^{1+\eta}}{1+\eta} - \rho_1 \frac{h_{1,t}^{1+\gamma}}{1+\gamma} - \rho_2 \frac{h_{2,t}^{1+\gamma}}{1+\gamma}$$

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- **Taxes** are

$$T(Y_1, Y_2) = (1 - \lambda (Y_1 + Y_2)^{-\tau}) \cdot (Y_1 + Y_2)$$

Next Time

- **Next time:**
  Labor supply and child-related transfers.

- **Literature:**
  Guner, Kaygusuz and Ventura (2020): "Child-Related Transfers, Household Labor Supply and Welfare"
  - **Read** before lecture
  - **Reading guide:**
    (focus on types child-related transfers + policy experiments)
    Section 1: Introduction + topic. Super important - Read.
    Section 2: Background, US. Read.
    Section 3: Model. Complex. Get the idea. Focus on married couples and childcare costs.
    Section 4: Calibrations. Skim.
    Section 5: Understanding childcare subsidies. Key - read.
    Section 6: Counterfactual policies. Key - read.

References I

Borella, M., M. De Nardi and F. Yang (2023): "Are Marriage-Related Taxes and Social Security Benefits Holding Back Female Labor Supply?," *Review of Economic Studies*, 90(1), 102–131.

Guner, N., R. Kaygusuz and G. Ventura (2020): "Child-Related Transfers, Household Labor Supply and Welfare," *Review of Economic Studies*, 87(5), 2290–2321.

Household Labor Supply and Child-Related Transfers

Thomas H. Jørgensen

2025

Plan for today

- Dynamic labor supply of **couples** + children
  Guner, Kaygusuz and Ventura (2020): "Child-Related Transfers, Household Labor Supply and Welfare"
  Related paper: Wang (2022, R&R, ReStud)

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- **Reading guide:**
  1. What are the main *research questions*?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

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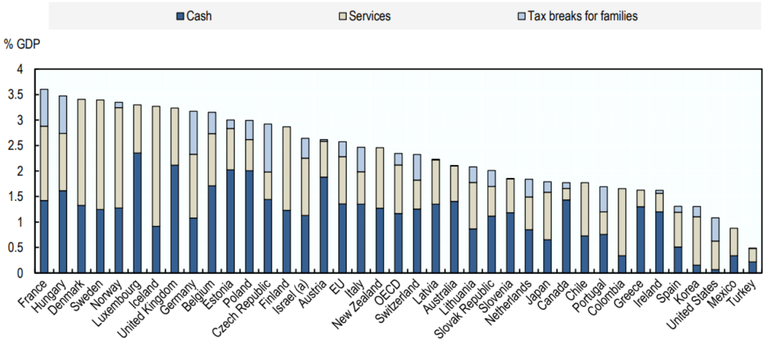
- **Reading guide:**
  1. What are the main *research questions*?
     - How do different child-related transfers affect labor supply?
     - Could alternative combinations increase welfare?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

Empirical Motivation: I

- Large differences in governmental spending on child-related policies
  https://www.oecd.org/els/soc/PF1_1_Public_spending_on_family_benefits.pdf

Chart PF1.1.A. Public spending on family benefits

Public expenditure on family benefits by type of expenditure, in per cent of GDP, 2017 and latest available



Note: Public spending accounted for here concerns public support that is exclusively for families (e.g. child payments and allowances, parental leave benefits and childcare support), only. Spending in other social policy areas such as health and housing support also assists families, but not exclusively, and is not included here. Coverage of spending on family and community services in the OECD Social Expenditure data may be limited as such services are often provided and/or co-financed by local governments. The latter may receive general block grants to finance their activities, and reporting requirements may not be sufficient for central statistical agencies to have a detailed view of the nature of local spending. In Nordic countries (where local government is heavily involved in service

Empirical Motivation: II

- **Childcare subsidies can increase labor supply of mothers**
  See e.g. Lefebvre and Merrigan (2008) for RF evidence
  $\rightarrow$ Such policies should be analyzed!

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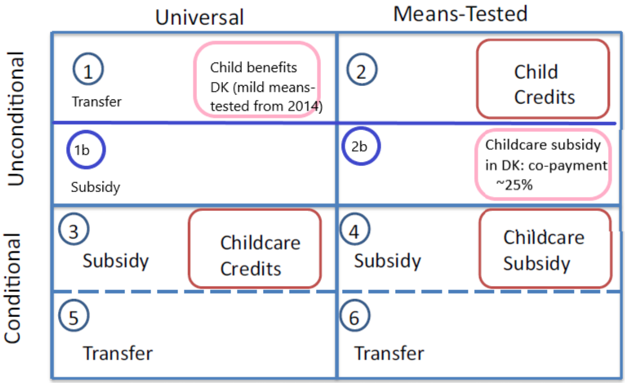
  $\rightarrow$ Such policies should be analyzed!

- **Many ways** to subsidize families, however
  - Conditional on work ("Unconditional"/"Conditional")
    (incentivize work)
  - Means-tested on income ("Universal"/"Means-tested")
    (re-distribution but dis-incentivize work a bit)
  - Conditional on childcare usage ("Transfer"/"Subsidy")
    (incentivize work through co-payment)

$\rightarrow$ Need a **unified framework** (model) to evaluate them!

Taxonomy (my modifications)



Universal: Benefits are given to all families, regardless of income.
Means-tested: Benefits are targeted based on income or need (only lower-income families qualify, or they receive more).

FIGURE 1

Taxonomy of child-related transfers.

Unconditional: Families receive support without needing to meet specific behavioral requirements.
Conditional: Support is given only if certain conditions are met, such as working, using childcare, etc.

- Guner, Kaygusuz and Ventura (2020) do not entertain 1b and 2b.

US Background

- **Childcare subsidies (4):** Child Care Development Fund (CCDF)
  $\sim$75% childcare subsidy to low-income employed households
  (useful for working people with childcare expenses)

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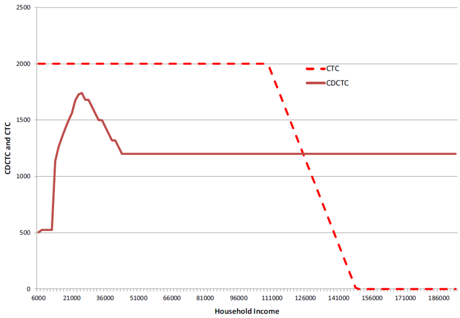

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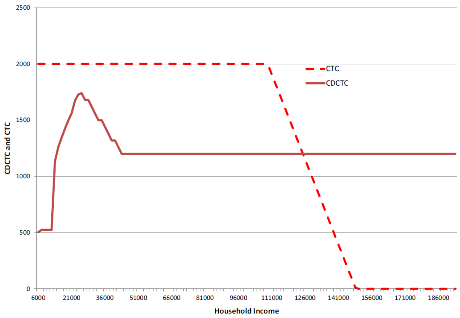
- **Childcare Credits (3):** Child and Dependent Care Tax Credit (CDCTC) ∼\$1,200-\$2,100 (dep. on income) tax deduction of childcare expenses. (useful for working people with childcare expenses)
- **Child Credits (2):** Child Tax Credit (CTC) + Additional CTC (ACTC) ∼\$1,000 per child but is a tax-credit (reason for ACTC).



Outline

1. Model Overview
2. Simulation Results
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Model Overview

- **"Macro" model**
  General equilibrium (w and r)
  Overlapping generations (OLG)
  Period is 5 years, j is period

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- **Choices:**
  Labor supply (w. human capital accumulation of women)
  Consumption/saving

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  Children can arrive "early" or "late" ($k(x, z)$)
- **Choices:**
  Labor supply (w. human capital accumulation of women)
  Consumption/saving
- **Childcare:**
  If a woman works, the household has to pay childcare costs.
  Access to informal childcare ($g = 1$) reduces cost

Model Overview

- **Types:** x, z is educational type of men and women
- **Income:**

scalar (equilibrium)·labor supply · shock · deterministic wage profile

$$\text{men} : w\varpi_m(z,j)\varepsilon_z l_m$$

$$\text{women} : wh\varepsilon_x l_f$$ fixed scalar · human capital · shock · labor supply?

$$\log h' = \log h + \alpha_j^x \chi(l_f) - \delta_x(1 - \chi(l_f))$$

\Chi is equal to 1 if the person works.

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- **Income:**

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$$\text{women}: wh\varepsilon_x l_f$$
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- **Childcare costs:** Number of children (couple) is $k(z,x)$
  If women work, they incur costs of

$$D = wk(z,x)d(s,x,z,g)$$

  where s is age of child, g is informal childcare availability.

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  If women work, they incur costs of

$$D = wk(z,x)d(s,x,z,g)$$

  where s is age of child, g is informal childcare availability. x, z is educational level
- **Childcare subsidies:** Parents pay $(1-\theta)$ of childcare costs if working:

$$(1-\theta)D\chi(l_f)$$

  if household income less that $\hat{I}$ (determined endo. to match take-up).

Model Overview: Preferences

- **Individual** preferences [my notation]

$$U(c, l, k_y) = \log c - \phi(l + k_y\eta)^{1+\frac{1}{\gamma}}$$

where $k_y = 1$ if there is a young child present

Model Overview: Preferences

- **Individual** preferences [my notation]

$$U(c, l, k_y) = \log c - \phi(l + k_y\eta)^{1+\frac{1}{\gamma}}$$

  where $k_y = 1$ if there is a young child present

- **Household** utility is sum

$$U(c, l_f, l_m, k_y, q) = U(c, l_f, k_y) + U(c, l_m, 0) - q\chi(l_f)$$

  where q is a (random) household-specific cost of "joint work" (but should it then be $q\chi(l_f)\chi(l_m)$?)

- **Imposes** that women are less likely to work when children.

Bellman Equation, Working Couple.

3.7.2. The problem of married households. Like singles, married couples decide how much to consume, how much to save, and how much to work. They also decide whether the female member of the household should work. Their problem is given by

$$V^M(a,h,\mathbf{s}^M,j) = \max_{a',\, l_f,\, l_m} \{[U_f^M(c,l_f,q,k_y) + U_m^M(c,l_m,l_f,q)] + \beta V^M(a',h',\mathbf{s}^M,j+1)\},$$

Something close to the Bellman equation

subject to

(i) *With kids: if* $b=\{1,2\}$, $j\in\{b,b+1,b+2\}$, *then* the household has $k(x,z)$ children and

Split between whether your income is above a certain threshold I hat.

$$c+a' = \begin{cases} a(1+r(1-\tau_k)) + w(\varpi_m(z,j)\varepsilon_z l_m + h\varepsilon_x l_f)(1-\tau_p) \\ -T^M(I,k(x,z)) + TR^M(I,D(1-\theta),k(x,z)) \\ -D(1-\theta)\chi(l_f), \text{ if } I \leq \widehat{I} \\ \text{assets + combined labor income} - \text{taxes + transfers} - \text{childcare} \\ a(1+r(1-\tau_k)) + w(\varpi_m(z,j)\varepsilon_z l_m + h\varepsilon_x l_f)(1-\tau_p) \\ -T^M(I,k(x,z)) + TR^M(I,D,k(x,z)) \\ -D\chi(l_f), \text{ otherwise} \end{cases}$$

where $I = w\varpi_m(z,j)\varepsilon_z l_m + wh\varepsilon_x l_f + ra$ and $D = wd(j+1-b,x,z,g)k(x,z)$. Furthermore, if $b=j$, then $k_y=1$.

- $\mathbf{s}^M = (x, z, \varepsilon_x, \varepsilon_z, q, b, g)$. $b \in \{1, 2\}$, where 2 is child-bearing
- TR: transfers

Outline

1. Model Overview
2. Simulation Results
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Simulation Results, Guner, Kaygusuz and Ventura (2020)

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- **keeps** government budget fixed (I think).
  Meant to illustrate the incentives
- **Implies that the current system is not optimal** (if fixed budget)
  Could still be optimal to have a mix (not analyzed)
  What about the 1b and 2b alternatives? Co-payments encourage work.

Counterfactual Reforms

Q2: **What happens if we** expand the current system by alternative reforms?

Budget neutral: For each reform, a flat-rate marginal tax rate is found to keep government budget fixed at baseline.

Counterfactual Reforms

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1. **Universal subsidies (childcare):**
   Remove means-testing in 75% childcare subsidy (CCDF and 1.2% tax) [by setting income threshold $\hat{I} = \infty$]
2. **Child credit expansion:**
   Increase CTC from $1,000 to $1,800 per child, unchanged income threshold (and 1.2% tax)
3. **Childcare credit expansion:**
   Increase the CDCTC by factor ~2
   (non-usable tax credit payed out and 1.2% tax)
4. **New child credit:**
   Increase CTC from $1,000 to $2,000 per child, increased income threshold (and 1.35% tax)

Counterfactual Reforms, Labor Supply

TABLE 4
Expansion of child-related transfers (% changes relative to benchmark)

	Universal subsidies (75%)	Child credit expansion	Childcare credit expansion	New child credit
Participation of married females	$\boxed{10.2}$	−2.4	$\boxed{10.6}$	−2.6
Total hours	$\boxed{1.8}$	−1.4	$\boxed{1.5}$	−1.5
Total hours (married females)	$\boxed{8.6}$	−3.1	$\boxed{8.6}$	−3.3
Hours per worker (all females)	−1.1	−1.1	−1.6	−1.3
Hours per worker (married females)	−1.8	−0.7	−2.2	−0.9
Hours per worker (single females)	0.2	−1.5	−0.3	−1.9
Hours per worker (all males)	−1.5	−0.7	−1.7	−0.7
Human capital (married females)	2.8	−0.8	2.5	−0.8
Output	0.5	−1.7	0.7	−1.5
Tax rate (%)	1.2	1.2	1.2	1.35
Participation of married females:				
By education				
<HS	25.4	−6.4	32.0	−7.2
HS	13.3	−4.4	16.9	−4.8
SC	9.1	−2.5	10.4	−2.8
COL	9.4	−1.2	7.0	−1.3
COL+	5.2	−0.7	2.8	−0.3
By child bearing status				
Early	14.9	−4.0	17.0	−4.4
Late	8.2	−1.5	6.9	−1.4

Counterfactual Reforms, Labor Supply

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Counterfactual Reforms, Welfare

TABLE 5
Expansion of child-related transfers: welfare effects (newborns, %)

	Childcare subsidy (75%)	Child credit	Childcare credit	New child credit
Single F				
No children	−1.41	−1.40	−1.46	−1.62
Early	4.25	5.99	10.06	6.71
Late	3.40	3.58	7.40	4.25
Informal care	4.15	5.44	9.62	6.03
No informal care	3.69	5.23	8.84	6.15
<HS	1.85	8.43	6.95	9.55
HS	2.54	4.93	6.66	5.62
SC	2.41	2.39	6.40	2.65
COL	1.08	0.33	2.43	0.37
COL+	0.56	−0.54	1.19	−0.56
Married				
No children	−3.16	−3.14	−3.29	−3.61
Early	2.90	3.59	5.80	4.76
Late	0.50	0.85	1.51	1.41
Informal care	2.02	2.09	3.84	3.96
No informal care	1.18	2.95	3.74	2.93
All newborns	0.84	1.28	2.51	1.73
(%) winners	48.0	54.3	50.9	57.7
All newborns (weighted welfare)	0.04	0.04	0.14	∼0

Non-mothers always lose they just have to pay more in taxes

Counterfactual Reforms, Welfare

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but majority of voters and the government might prefer it

Outline

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Our simple model

- **Dual-earner model from last time**
- **Modification:**
  A child can arrive with probability

$$p(n_t) = \begin{cases} p_n & \text{if } n_t = 0 \\ 0 & \text{if } n_t = 1 \end{cases}$$

- **Taxes and transfers:**
  Taxes on household level
  Childcare costs if both work
  Child credits
- **Reform** of interest:
  Child-related transfers

Our simple model: **Recursive formulation**

$$V_t(n_t, K_{1,t}, K_{2,t}) = \max_{h_{1,t}, h_{2,t}} U(c_t, h_{1,t}, h_{2,t}, n_t) + \beta \mathbb{E}_t [V_{t+1}(n_{t+1}, K_{1,t+1}, K_{2,t+1})]$$

$$c_t = Y_t - T(Y_t + \mathcal{C}) + \mathcal{C}$$

$$Y_t = \sum_{j=1}^{2} w_{j,t} h_{j,t}$$

$$n_{t+1} = \begin{cases} n_t + 1 & \text{with prob. } p(n_t) \\ n_t & \text{with prob. } 1 - p(n_t) \end{cases}$$

$$\log w_{j,t} = \alpha_{j,0} + \alpha_{j,1} K_{j,t}, \ j \in \{1, 2\}$$

$$K_{j,t+1} = (1 - \delta) K_{j,t} + h_{j,t}, \ j \in \{1, 2\}$$

- **Child-related transfers** not means tested and mean tested

$$\mathcal{C}(n_t, h_{1,t}, h_{2,t}, w_{1,t}, w_{2,t}) = \mathcal{C}_1(n_t) + \mathcal{C}_2(n_t, Y_t) + [\mathcal{C}_3(n_t) + \mathcal{C}_4(n_t, Y_t)] \cdot \mathbf{1}(h_{1,t} \cdot h_{2,t} > 0)$$

Our simple model

- **We assume that if both work, they have to buy childcare.**
  Means that conditional transfers are always a subsidy (cannot do 5 or 6).

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- **Preferences** are sum of individual

$$U(c_t, h_{1,t}, h_{2,t}) = 2\frac{(c_t/2)^{1+\eta}}{1+\eta} - \rho_1(n_t)\frac{h_{1,t}^{1+\gamma}}{1+\gamma} - \rho_2(n_t)\frac{h_{2,t}^{1+\gamma}}{1+\gamma}$$

with

$$\rho_j(n_t) = \rho_{0,j} + \rho_{1,j} n_t$$

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with

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- **Expected value** is

$$\begin{aligned}\mathbb{E}_t[V_{t+1}(n_{t+1}, K_{1,t+1}, K_{2,t+1})] = {} & p(n_t)V_{t+1}(n_t+1, K_{1,t+1}, K_{2,t+1}) \\ & + (1-p(n_t))V_{t+1}(n_t, K_{1,t+1}, K_{2,t+1})\end{aligned}$$

Next Time

- **Next time:**
  Models of Household Behavior.

- **Literature:**
  Chiappori and Mazzocco (2017): "Static and Intertemporal Household Decisions"
  + Guide on limited commitment models
  - **Read** before lecture
  - **Reading guide:**
    Section 1: Introduction + overview. Read.
    Section 2: Static models. Read. Read details fast.
    Section 3: Dynamic models. Get the idea of limited commitment. Do not get stuck.
    Section 4: Tests. We won't cover this, you can skip.
    Section 5: Policies. Short, might be worth a read.

References I

CHIAPPORI, P.-A. AND M. MAZZOCCO (2017): "Static and Intertemporal Household Decisions," *Journal of Economic Literature*, 55(3), 985–1045.

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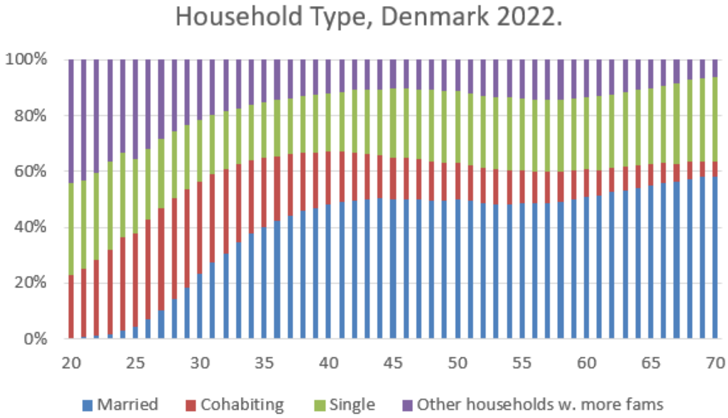
Models of Household Behavior

Thomas H. Jørgensen

2025

Empirical Motivation

- **Many people live in couples**



Introduction

- **Unitary model** until now
  The couple acted as one unit

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- There are ∞ many ways of modeling household decisions
  Some large and some small differences
- I will focus on some main types of models
  Give an idea of the main similarities and differences

Introduction

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  The couple acted as one unit

- There are ∞ many ways of modeling household decisions
  Some large and some small differences
- I will focus on some main types of models
  Give an idea of the main similarities and differences

- Remember: Models are ***abstractions***, hopefully useful...

- Notation on dynamic models differ from Chiappori and Mazzocco (2017)
  Based on guide: Hallengreen, Jørgensen and Olesen (2024)

Outline

1. Static Models
   - Setting
   - Unitary Model
   - Non-cooperative
   - Cooperative: Collective

2. Dynamic Models
   - Full Commitment
   - Limited Commitment

Production Technology

- **Superscript:** individual (1,2), **subscript:** element
- **Private** goods ($h = 1, \ldots, n$) produced as

$$q_h = q_h^1 + q_h^2 = f_h(x_h, d_h) \tag{1}$$

q_h^1 is a private good for individual 1

where

x_h: market goods inputs

$d_h = (d_h^1, d_h^2)$: time inputs

Production Technology

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private goods small letters
public goods capital letters

- **Public** goods ($k = 1, \ldots, N$) produced as

$$Q_k = F_k(X_k, D_k) \tag{2}$$

where
X_k: market goods inputs
$D_k = (D_k^1, D_k^2)$: time inputs

Preferences: Utility and Felicity Function

- Individual *utility* function

$$U^i(Q, q^1, q^2, l^1, l^2)$$

Public goods, private goods for 1 and 2 and leisure for 1 anf 2

where
l^i is leisure time
$T^i = h^i + l^i + \sum_{k=1}^{N} D_k^i + \sum_{h=1}^{n} d_h^i$ is available time
h^i is hours worked

Preferences: Utility and Felicity Function

- **Caring preferences:**
  Care not about the allocation of partner but only their welfare:

$$U^i(Q, q^1, q^2, l^1, l^2) = W^i(u^1(Q, q^1, l^1), u^2(Q, q^2, l^2))$$

where
$u^i(Q, q^i, l^i)$ is called the *felicity* function

Preferences: Utility and Felicity Function

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  where
  $u^i(Q, q^i, l^i)$ is called the *felicity* function
- **Egotistic preferences:**
  Care not about the partner:

$$U^i(Q, q^1, q^2, l^1, l^2) = F^i(u^i(Q, q^i, l^i), a)$$

  where a can contain marital status etc.

Budget Constraint

- **Budget constraint**

$$p'\left(\sum_{k=1}^{N} X_k + \sum_{h=1}^{n} x_n\right) = \sum_{i=1}^{2}(y^i + w^i h_i) \tag{3}$$

where
p is vector of market prices
y^i is non-market income.
w^i is wage rate

Budget Constraint

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- Can be written as in Chiappori and Mazzocco (2017)

$$p'\left(\sum_{k=1}^{N} X_k + \sum_{h=1}^{n} x_n\right) + \sum_{i=1}^{2} w^i (l^i + \sum_{k=1}^{N} D_k^i + \sum_{h=1}^{n} d_h^i) = \underbrace{\sum_{i=1}^{2}(y^i + w^i T^i)}_{Y \text{ (pot. inc.)}}$$

(note that $T^i = h^i + l^i + \sum_{k=1}^{N} D_k^i + \sum_{h=1}^{n} d_n^i$, they miss l^i on p. 989)

- **Income pooling:** non-labor income, y^i, enters identically for both

Unitary Model

- **Unitary model**, households solve
  (conditional on *potential* income, $Y = \sum_{i=1}^{2}(y^i + w^i T^i)$)

$$\max_{X,x,l^1,l^2,d^1,d^2,D^1,D^2} U^H(Q, q, l^1, l^2)$$

s.t.

$$Q_k = F_k(X_k, D_k),\ k = 1, \ldots, N$$

$$q_h = q_h^1 + q_h^2 = f_h(x_h, d_h),\ h = 1, \ldots, n$$

$$Y = p'\left(\sum_{k=1}^{N} X_k + \sum_{h=1}^{n} x_h\right) + \sum_{i=1}^{2} w^i(l^i + \sum_{k=1}^{N} D_k^i + \sum_{h=1}^{n} d_h^i)$$

where $U^H(Q, q, l^1, l^2)$ is *some* household-level utility function

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- **Rationalized** via
  Samuelson's welfare index
  Becker's rotten kid (we skip)
  Transferable Utility (TU)

Unitary Motivation: Samuelson's welfare index

- **Samuelson's welfare index**

$$U^H(Q, q, I^1, I^2) = \max_{q_1, q_2} W(u^1(Q, q^1, I^1), u^2(Q, q^2, I^2))$$

$$\text{s.t.}$$

$$q = q_1 + q_2$$

Unitary Motivation: Samuelson's welfare index

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$$\text{s.t.}$$

$$q = q_1 + q_2$$

- **Example** could be

$$W(u^1(Q, q^1, I^1), u^2(Q, q^2, I^2)) = \lambda u^1(Q, q^1, I^1) + (1 - \lambda) u^2(Q, q^2, I^2)$$

where
λ is a constant weight on each member's utility; "power"

Unitary Motivation: Samuelson's welfare index

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  where
  λ is a constant weight on each member's utility; "power"

- **Arbitrary** that households should have some $W()$... but this example is a special form of the "collective model" below [nice]

Unitary Motivation: Transferable Utility

- **TU: If there exists a Pareto frontier**, such that a *cardinal transformation,* $k()$ gives
$$k(u^1(Q, q^1, l^1)) + k(u^2(Q, q^2, l^2)) = K(p, w, Y)$$
$\rightarrow$ utility possibility frontier has a slope of -1,
$$k(u^1(Q, q^1, l^1)) = K(p, w, Y) - k(u^2(Q, q^2, l^2))$$
- **Then** we can describe the optimization problem using $U^H(Q, q, l^1, l^2)$

Unitary Motivation: Transferable Utility

- **TU: If there exists a Pareto frontier**, transfer of utility to partner
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- **Then** we can describe the optimization problem using $U^H(Q, q, l^1, l^2)$
- **Example:** Conditional on Y,
  $$U^{m\star}(\overline{u}^f) = \max_{c^m} U^m(c^m) = \sqrt{c^m}$$
  $$\text{s.t.}$$
  $$Y = c^m + c^f$$
  $$\overline{u}^f = U^f(c^f) = \sqrt{c^f}$$
  gives $U^{m\star}(\overline{u}^f)^2 = \text{constant}(Y) - (\overline{u}^f)^2$.

Unitary Model: Not Consistent with Data

- **Two testable implications**
  1. Income pooling (source of non-labor income does not matter for behavior)
  2. Slutsky symmetry (commodity prices affect members' demand similarly)

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- **Two testable implications**
  1. Income pooling (source of non-labor income does not matter for behavior)
  2. Slutsky symmetry (commodity prices affect members' demand similarly)

- **Almost always rejected** see Chiappori and Mazzocco (2017, p. 1022)

- **Alternatives** have been proposed
  Non-cooperative
  Cooperative (collective)

Non-Cooperative

- **Non-cooperative models:**
  A game with two players
  Nash equilibrium

$$\max_{Q^1, q^1, l^1} u^1(Q^1 + Q^2, q^1, q^2, l^1, l^2)$$

$$\text{s. t.}$$

$$PQ^1 + p'q^1 = Y^1$$

and

$$\max_{Q^2, q^2, l^2} u^2(Q^1 + Q^2, q^1, q^2, l^1, l^2)$$

$$\text{s. t.}$$

$$PQ^2 + p'q^2 = Y^2$$

- **Generally not efficient:** Partner's gains not internalized.

Cooperative: Collective

- **Collective model:** Pareto efficient allocations (def)

Cooperative: Collective

Assumes that couples can allocate all of their decisions such that the solution is Pareto efficient

- **Collective model:** Pareto efficient allocations (def)
  Typically formulated as

$$\max_{X,x,l^1,l^2,d^1,d^2,D^1,D^2} \lambda(z)u^1(Q,q^1,l^1) + (1-\lambda(z))u^2(Q,q^2,l^2)$$

s.t.

$$Q_k = F_k(X_k, D_k),\ k = 1, \ldots, N$$

$$q_h = q_h^1 + q_h^2 = f_h(x_h, d_h),\ h = 1, \ldots, n$$

$$Y = p'\left(\sum_{k=1}^{N} X_k + \sum_{h=1}^{n} x_h\right) + \sum_{i=1}^{2} w^i(l^i + \sum_{k=1}^{N} D_k^i + \sum_{h=1}^{n} d_h^i)$$

- **Distribution factors:** z
  Anything that affect power, such as p, w, y.
  Cannot be endogenous: Over-investment in power $\rightarrow$ inefficient.

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- **Distribution factors:** z
  Anything that affect power, such as p, w, y.
  Cannot be endogenous: Over-investment in power $\rightarrow$ inefficient.
- **Unitary model** is nested, $\lambda(z) = \text{constant} \rightarrow$ unitary model

Outline

1. Static Models
   - Setting
   - Unitary Model
   - Non-cooperative
   - Cooperative: Collective

how do we define the value function in a dual member household in a static world?

2. Dynamic Models
   - Full Commitment
   - Limited Commitment

Dynamic Models

- All comes down to how the bargaining weight is updated.
- My slides combine Chiappori and Mazzocco (2017) with Theloudis, Velilla, Chiappori, Giménez-Nadal and Molina (2022) and Hallengreen, Jørgensen and Olesen (2024)

General Setup: Choices

- Period $t = 0$: Individuals A and B become a couple
- Periods $t > 0$: As a couple, they decide on
  - private consumption, c_t^A and c_t^B (and thus savings, a_t)
  - labor supply, $l_t^A, l_t^B \in \{0, 0.75, 1\}$
  - whether to split up (no re-partnering for simplicity)
- Period T: both die with certainty

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- **Inter-temporal budget** constraint of couple

$$a_t + c_t^A + c_t^B = Ra_{t-1} + w^A l_t^A + w^B l_t^B$$

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- **Example throughout**, but apply more generally.

General Setup: Utility

- **Individual utility** is for $j \in \{A, B\}$

$$u^j(c_t^j, l_t^j)$$

- **Household utility** is weighted sum

$$U(c_t^A, c_t^B, l_t^A, l_t^B; \psi_t, \mu_t) = \mu_t u^A(c_t^A, l_t^A) + (1 - \mu_t) u^B(c_t^B, l_t^B) + \psi_t$$

where match quality/"love" is

$$\psi_t = \psi_{t-1} + \varepsilon_t, \ \varepsilon \sim iid\mathcal{N}(0, \sigma_\varepsilon^2)$$

and μ_t is the **bargaining power** of agent A

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- How μ_t is determined defines the different types of models:
  Unitary: $\mu_t = \mu$ is a **constant number**
  Full commitment: $\mu_t = \mu_0(Z)$ is a **function** (of states known i $t = 0$)
  No commitment: μ_t is updated in each period
  Limited commitment: $\mu_t = \mu_t(\bullet, \mu_{t-1})$ is a function of past power

General Setup: Recursive Formulation

- **Outside option:** Value of being single

$$V^s_{j,t}(a_{t-1}) = \max_{c^j_t, l^j_t} u^j(c^j_t, l^j_t) + \beta V^s_{j,t+1}(a_t)$$

$$\text{s.t.}$$

$$a_t = Ra_{t-1} + w^j l^j_t - c^j_t$$

where I do not allow for re-partnering ($V^{m\to s}_{j,t} = V^s_{j,t}$).

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- **Partnership dissolution:**
  Share κ_1 of wealth is transferred to agent A and $\kappa_2 = 1 - \kappa_1$ to agent B.
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- **Value of being in a couple**
  depends on what we assume about the *bargaining process*.

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- **Value of a couple** is

$$W_t(a_{t-1}, \psi_t) = \max_{c_t^A, c_t^B, l_t^A, l_t^B} U(c_t^A, c_t^B, l_t^A, l_t^B, \psi_t; \mu) + \beta \tilde{W}_{t+1}(a_t, \psi_t)$$

the *expected* (w.r.t. ψ_{t+1}) continuation value is

$$\tilde{W}_{t+1}(a_t, \psi_t) \\ = \mathbb{E}_t[\max\{W_{t+1}(a_t, \psi_{t+1}) \ ; \ \underbrace{\mu V_{A,t+1}^s(a_t^A) + (1-\mu) V_{B,t+1}^s(a_t^B)}_{\text{weighted value of singlehood}}\}]$$

Commitment Models

- **Endogenously determined μ_t**

  FC: Full commitment, μ_t is a **constant function**
  We will see in Bruze, Svarer and Weiss (2015)

  NC: No commitment, μ_t updated **every period**
  We will just discuss today

  LC: Limited commitment, μ_t updated **sometimes** $\rightarrow$ function of past power
  We will see in several papers + code

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- Bargaining power function is determined and agreed upon **at beginning of partnership**
- Bargaining power is thus a *constant function*

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  - If time-varying elements in Z_t: Assuming perfect foresight

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- **How is this function determined?**
  We will come back to this in a few slides

Full Commitment

- **Value of being a couple** is then

$$W_t(a_{t-1}, \psi_t) = \max_{c_t^A, c_t^B, l_t^A, l_t^B} U(c_t^A, c_t^B, l_t^A, l_t^B; \psi_t, \mu_0(Z_t)) + \beta \tilde{W}_{t+1}(a_t, \psi_t)$$

where *expected continuation* value is

$$\tilde{W}_{t+1}(a_t, \psi_t) = \mathbb{E}_t[\max\{W_{t+1}(a_t, \psi_{t+1}) \\ ; \underbrace{\mu_0(Z_{t+1}) V^s_{A,t+1}(a_t^A) + (1 - \mu_0(Z_{t+1})) V^s_{B,t+1}(a_t^B)}_{\text{weighted value of singlehood}}\}]$$

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- **Transferable utility:** Household jointly decide to divorce if

possible to transfer some utility to wife/spouse in order to stay in relationship

$$W_t(a_t) < \mu_0(Z_t) V^s_{A,t}(a_t^A) + (1 - \mu_0(Z_t)) V^s_{B,t}(a_t^B)$$

- No constraints on individual members' utilities

Full Commitment: Determining Bargaining Power

- How could $\mu_0(Z_t)$ be determined?

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- **Idea 1: Nash-bargaining** at the point of partnership formation

$$\mu_0(Z) = \arg\max_{\mu\in[0,1]} \left(\mu W_0(a_{-1}) - V^s_{A,0}(\lambda a_{-1})\right)^{0.5} \times \left((1-\mu)W_0(a_{-1}) - V^s_{B,0}((1-\lambda)a_{-1})\right)^{0.5}$$

  - μ_0 "non-parametric" constant function of e.g. $Z = (a_{-1}, w_0^A, w_0^B)$

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- **Idea 2: Assume a functional form** (Bruze, Svarer and Weiss, 2015)

$$\mu_0(w_t^A / w_t^B) = \frac{\exp(\alpha_0 + \alpha_1 w_t^A / w_t^B)}{1 + \exp(\alpha_0 + \alpha_1 w_t^A / w_t^B)}$$

and estimate parameters α_0 and α_1 using data.
[perfect foresight assumption on wages]

  - If $\alpha_1 = 0$: Similar to the unitary model.

No- and Limited Commitment

- My definition of "No commitment" is different from that of Mazzocco (2007)
  - I will call his setup "Limited commitment" (as is standard now)
- They are closely related: Both **do not assume transferable utility**
  - Only differ in how the bargaining power is updated dynamically

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- They are closely related: Both **do not assume transferable utility**
  - Only differ in how the bargaining power is updated dynamically
- We thus need to check **individual "participation" constraints**: Is it optimal for each agent to be part of the couple *without* receiving any utility from the other partner
- **We need to define new objects** for this purpose: The value of agent j from being in the couple if μ_{t-1} is the bargaining power coming into period t

Notation: Partnership Status and Transitions

- All values are **individual**
- **Four** transition possibilities

from\to	married	single
married	$V_{j,t}^{m\to m}(\mathcal{S}_t, \mu_{t-1})$	$V_{j,t}^{m\to s}(\mathcal{S}_t, \mu_{t-1})$
single	$V_{j,t}^{s\to m}(\mathcal{S}_t)$	$V_{j,t}^{s\to s}(\mathcal{S}_t)$

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- Value of **starting** in a state

Value of married to singlehood

$$V_{j,t}^{m}(\mathcal{S}_t, \mu_{t-1}) = D_t^\star V_{j,t}^{m\to s}(\mathcal{S}_t, \mu_{t-1}) + (1 - D_t^\star) V_{j,t}^{m\to m}(\mathcal{S}_t, \mu_{t-1})$$
$$V_{j,t}^{s}(\mathcal{S}_t) = M_t^\star V_{j,t}^{s\to m}(\mathcal{S}_t) + (1 - M_t^\star) V_{j,t}^{s\to s}(\mathcal{S}_t)$$

where $D_t^\star \in \{0,1\}$ is divorce and $M_t^\star \in \{0,1\}$ is marriage

We don't allow remarriage

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- When singlehood is absorbing (and no div. costs): $V_{j,t}^{s} = V_{j,t}^{m\to s}$.

Recursive Formulation

- **Individual value of choice** c_t^j, l_t^j while remaining in couple is

$$v_{j,t}^{m\to m}(c_t^j, l_t^j; a_{t-1}, \psi_t, \mu) = u^j(c_t^j, l_t^j) + \psi_t + \beta \mathbb{E}_t[V_{j,t}^m(a_t^j, \psi_{t+1}, \mu)]$$

  where μ is *some* bargaining power, we will discuss in great detail.
  - μ_{t-1} is the specific value when entering period t (the state)

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1. **Conditional on remaining together**, optimal choices are

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2. **Marital surplus** for agent j is

mu is between 0 and 1, which is the bargaining power

$$S_t^j(\mu) = \underbrace{v_{j,t}^{m\to m}(\tilde{c}_t^j(\mu), \tilde{l}_t^j(\mu); a_{t-1}, \psi_t, \mu)}_{V_{j,t}^{m\to m}(a_{t-1}, \psi_t, \mu)} - V_{j,t}^{m\to s}(a_{t-1}^j)$$

Limited Commitment: Bargaining Process

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3.3 If e.g. $S_t^A(\mu_{t-1}) < 0$ (member A not happy with current division) They re-bargain to find a new potential distribution both can accept (point 4 on next slide)

Limited Commitment: Updating Bargaining Weight

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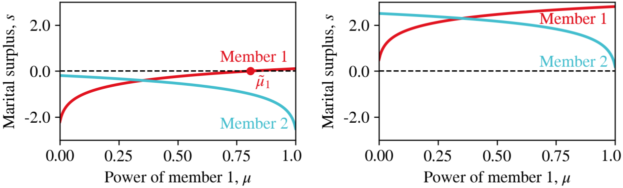
   - If $S_t^B(\tilde{\mu}^A) < 0$, they divorce ($D_t^\star = 1$)

- Combined the co-state evolves as

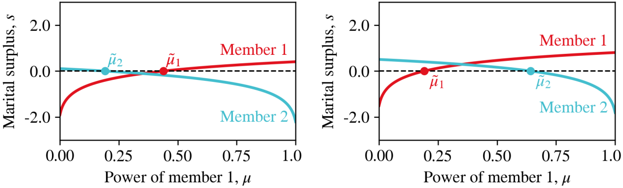
$$\mu_t = \mu_t^\star(\mathcal{S}_t, \mu_{t-1}) = \begin{cases} \mu_{t-1} & \text{if } S_t^A(\mathcal{S}_t, \mu_{t-1}) \geq 0 \text{ and } S_t^B(\mathcal{S}_t, \mu_{t-1}) \geq 0 \\ \tilde{\mu}^A & \text{if } S_t^A(\mathcal{S}_t, \mu_{t-1}) < 0 \text{ and } S_t^B(\mathcal{S}_t, \tilde{\mu}^A) \geq 0 \\ \tilde{\mu}^B & \text{if } S_t^A(\mathcal{S}_t, \tilde{\mu}^B) \geq 0 \text{ and } S_t^B(\mathcal{S}_t, \mu_{t-1}) < 0 \\ \emptyset & \text{else} \end{cases} \tag{4}$$

Limited Commitment: Updating Bargaining Weight

marital surplus (s) measures the net gain a household member receives from staying married compared to their outside option (typically divorce or separation)



(a) 2 will never remain. (b) Both will always remain.

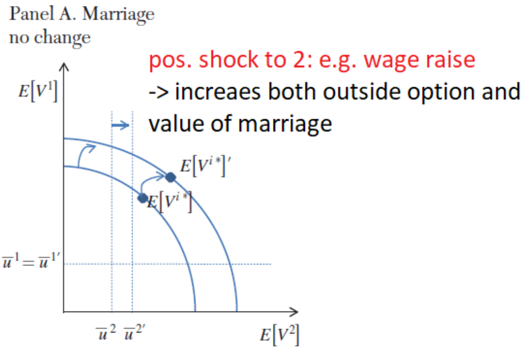


(c) No room for bargaining. (d) Room for bargaining.

Source: Hallengreen, Jørgensen and Olesen (2024).

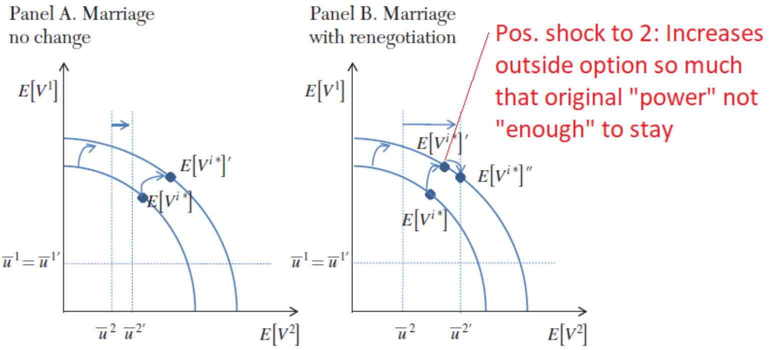
Limited Commitment: Updating Bargaining Weight

- Shock to agent 2's outside option: **small**.



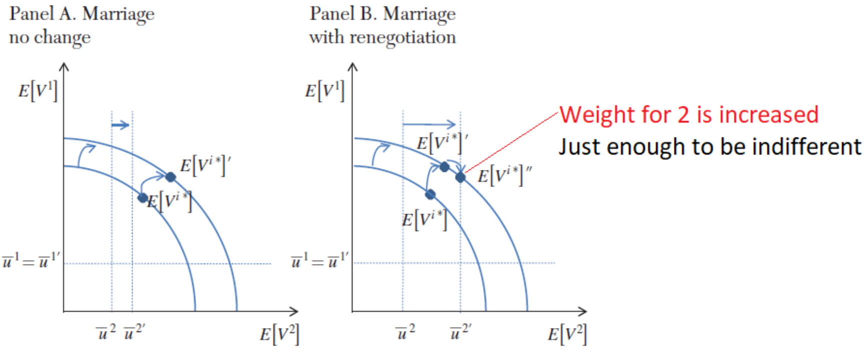
Limited Commitment: Updating Bargaining Weight

- Shock to agent 2's outside option: **medium**.



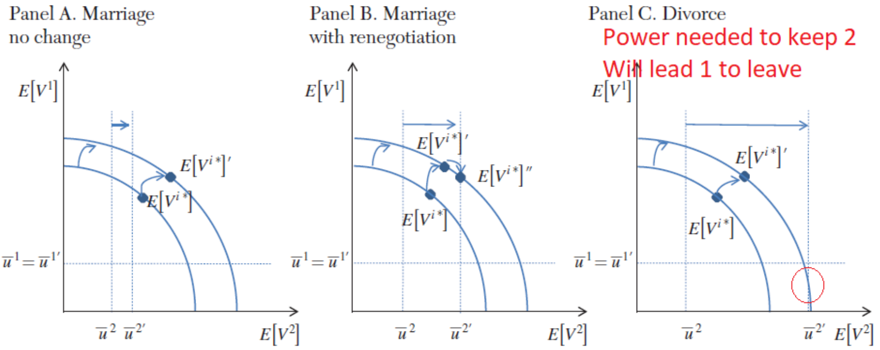
Limited Commitment: Updating Bargaining Weight

- Shock to agent 2's outside option: **medium**.

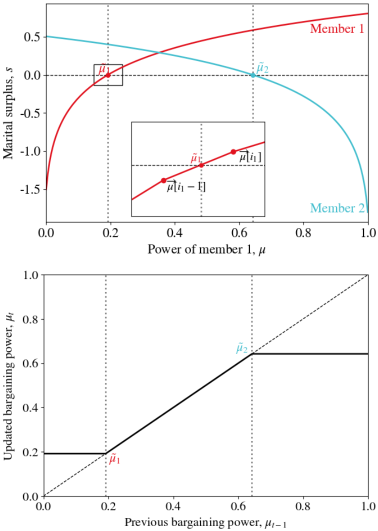


Limited Commitment: Updating Bargaining Weight

- Shock to agent 2's outside option: **large**.



Limited Commitment: Updating Bargaining Weight



Source: Hallengreen, Jørgensen and Olesen (2024).

No Commitment

- **Limited commitment:** Bargaining power updated if individual participation constraints violated at current bargaining position, μ_{t-1},

$$\mu_t = \mu_t^{\star}(a_{t-1}, \psi_t, \mu_{t-1})$$

No Commitment

- **Limited commitment:** Bargaining power updated if individual participation constraints violated at current bargaining position, μ_{t-1},

$$\mu_t = \mu_t^{\star}(a_{t-1}, \psi_t, \mu_{t-1})$$

- **No commitment:** Bargaining power updated in *all periods*,

$$\mu_t = \mu_t^{\star}(a_{t-1}, \psi_t)$$

  replace step 3 with e.g. [instead of the discussion before]

$$\mu_t = \arg\max_{\tilde{\mu}} S_t^A(\tilde{\mu})^{0.5} S_t^B(\tilde{\mu})^{0.5}$$

- **We focus on limited commitment** in the code
  What about initial bargaining power, μ_0, then?
  Could be found through Nash bargaining :)

Next Time

- **Next time:**
  Divorce Laws, Savings and Labor Supply.

- **Literature:**
  Voena (2015): "Yours, Mine, and Ours: Do Divorce Laws Affect the Intertemporal Behavior of Married Couples?"
  - **Read** before lecture
  - **Reading guide:**
    Section 0: Introduction. Key
    Section 1: US divorce law. Key.
    Section 2: Model. *Key*, but complex. Get the idea.
    Under unilateral divorce: limited commitment model.
    Section 3: Data and RF motivation. Get the overall results/motivation.
    Section 4: Structural Estimation: Read fast.
    Section 5: Counterfactual simulations. Key.

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Bruze, G., M. Svarer and Y. Weiss (2015): "The Dynamics of Marriage and Divorce," *Journal of Labor Economics*, 33(1), 123–170.

Chiappori, P.-A. and M. Mazzocco (2017): "Static and Intertemporal Household Decisions," *Journal of Economic Literature*, 55(3), 985–1045.

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VOENA, A. (2015): "Yours, Mine, and Ours: Do Divorce Laws Affect the Intertemporal Behavior of Married Couples?," *American Economic Review*, 105(8), 2295–2332.

Divorce Laws and Intra-Household Bargaining

Thomas H. Jørgensen

2025

Plan for today

- Divorce law and intra-household bargaining
  Voena (2015): "Yours, Mine, and Ours: Do Divorce Laws Affect the Intertemporal Behavior of Married Couples?"
  - Limited commitment model as last time
    different notation $\rightarrow$ good to see again but different!

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- **Reading guide:**
  1. What are the main *research questions*?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

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  - Limited commitment model as last time
    different notation $\rightarrow$ good to see again but different!

- **Reading guide:**
  1. What are the main *research questions*?
     - How does divorce laws affect saving and female labor supply in marriage?
     - What are the welfare consequences of unilateral divorce?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

Empirical Motivation: I

- **Reduced Form** evidence from the US
  Using *time- and state variation* in adoption in unilateral divorce

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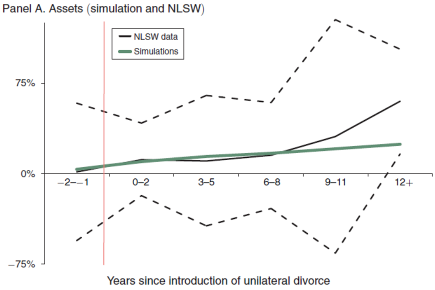

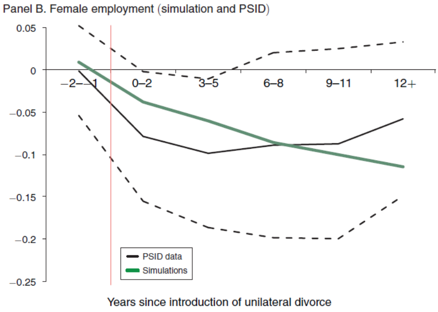

Empirical Motivation: I

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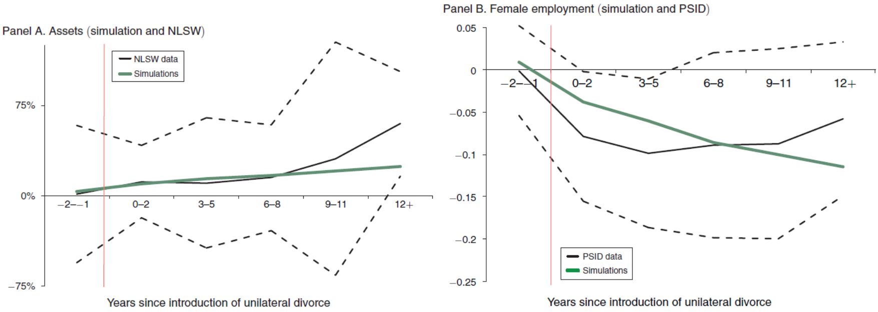

- **Interpretation:** women with *low bargaining power pre-reform:* unilateral → threat to leave → increase bargaining power → work less.

Empirical Motivation: II

1. Unilateral vs. mutual consent divorce
   [One can decide vs. both has to agree]
2. Community vs. title-based division of property
   [50-50 vs. individual ownership]

Table: Mutual → Unilateral (rows 1+2, Tab. 2).

	Savings	Employment	
Community	↑	↓	increased power of women (last slide)
Title-based	–	–	no sign. effect (everything is private)

Outline

Model Overview

- **Choices:**
  c_t^j: consumption of member $j \in \{H, W\}$
  P_t^W: labor market participation, wife (men always work)
  A_{t+1}^j: assets of member $j \in \{H, W\}$
  D_t: divorce
- **States** (ω_t)**:**
  A_t^j: assets of member $j \in \{H, W\}$
  z_t^j: income shock (perm)
  ε_t^j: match quality shock (love)
  h_t^W: human capital, wife only.
  Ω_t: divorce laws.
  $(\tilde{\theta}_t^W, \tilde{\theta}_t^H)$: bargaining weights (in unilateral/limited commitment).
  (Childbirth occurs at predetermined ages, perfect foresight)

As usual we have the Bellman equation - how do all of the states evolve?

State Transitions: Income and Human Capital

- **Income** is

human capital + permanent shock

$$\log(y_t^j) = \ln(h_t^j) + z_t^j$$
$$z_t^j = z_{t-1}^j + \zeta_t^j, \quad \zeta_t^j \sim iid\mathcal{N}(0, \sigma^2_{\zeta^j})$$

- **Human capital** is

$$\log(h_t^j) = \log(h_{t-1}^j) - \delta(1 - P_{t-1}^j) + (\lambda_0^j + \lambda_1^j t)P_{t-1}^j$$

- **Why** only need to keep track of h_t^W?

State Transitions: Income and Human Capital

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$$\log(y_t^j) = \ln(h_t^j) + z_t^j$$
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$$\log(h_t^j) = \log(h_{t-1}^j) - \delta(1 - P_{t-1}^j) + (\lambda_0^j + \lambda_1^j t)P_{t-1}^j$$

- **Why** only need to keep track of h_t^W?
  Because since men always work, $P_t^H = 1$, we have

$$\begin{aligned}\log(h_t^H) &= \log(h_{t-1}^H) + (\lambda_0^H + \lambda_1^H t) \\ &= \log(h_{t-2}^H) + (\lambda_0^H + \lambda_1^H (t-1)) + (\lambda_0^H + \lambda_1^H t) \\ &= \underbrace{\overset{\text{initial human capital}}{\log(h_0^H)}}_{\text{estimated as intercept}} + \sum_{s=1}^{t} (\lambda_0^H + \lambda_1^H s)\end{aligned}$$

If heterogeneity in initial condition, we would solve for a grid of h_0^H.

State Transitions: Love

- **Match quality (love)** is an AR(1) process

$$\zeta_t^j = \zeta_{t-1}^j + \epsilon_t^j, \quad \epsilon_t^j \sim iid\mathcal{N}(0, \sigma^2)$$

unit root process

State Transitions: Assets (Inter-temporal Budget)

- $e(k) \geq 1$ is equiv. scale as function of children, k.
- d_t^k is child-care costs

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- **Budget constraint** depends on status
  Singles (share childcare costs):

$$A_{t+1}^j = (1+r)A_t^j + (y_t^j - d_t^k/2) \cdot P_t^j - c_t^j \cdot e(k) \tag{1}$$

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Couples ($A_t = A_t^H + A_t^W$):

$$A_{t+1} = (1+r)A_t + y_t^H + (y_t^w - d_t^k)P_t^W - x_t \tag{2}$$

where *expenditures* are (couples have econ. of scale, $\rho \geq 1$)

$$x_t = \left[(c_t^H)^\rho + (c_t^W)^\rho\right]^{\frac{1}{\rho}} e(k)$$

Preferences

- **Individual preferences** are [my notation]

$$u(c_t^i, P_t^i, D_t^i) = \frac{(c_t^i)^{1-\gamma}}{1-\gamma} - \psi P_t^i + \xi_t^i(1 - D_t^i)$$

where
γ is the CRRA coefficient
ψ is the dis-utility of working
ξ_t^i is a marital match shock ("love")

Value of a Divorcee

- **Re-marriage** with prob. $\pi_t^{j\Omega_t}$
  Match with someone similar to j (I think).

Value of a Divorcee

- **Re-marriage** with prob. $\pi_t^{j\Omega_t}$ remarriage probability depends on match prob?
  Match with someone similar to j (I think).
- **(expected) Value of entering** period t as divorced ($\approx V^s$)
  D = divorce, R= remarry

$$V_t^{jDR}(\omega_t) = \pi_t^{j\Omega_t} V_t^{jR}(\omega_t) + (1 - \pi_t^{j\Omega_t}) V_t^{jD}(\omega_t)$$

where V_t^{jR} is *value of re-marriage* (defined next) and

$$V_t^{jD}(\omega_t) = \max_{c_t^j, P_t^j} u(c_t^j, P_t^j, 1) + \beta \underbrace{\{\pi_{t+1}^{j\Omega_t} \mathbb{E}_t[V_{t+1}^{jR}(\omega_{t+1})] + (1 - \pi_{t+1}^{j\Omega_t}) \mathbb{E}_t[V_{t+1}^{jD}(\omega_{t+1})]\}}_{\mathbb{E}_t[V_{t+1}^{jDR}(\omega_{t+1})]}$$

is the *value of remaining divorced* ($V^{s \to s}$ in my notation).

Value of a Divorcee

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  is the *value of remaining divorced* ($V^{s \to s}$ in my notation).
- **Outside option** is (my notation)

$$V_{j,t}^{m \to s}(\omega_t; \kappa) = V_t^{jD}(\omega_t, (A_{t-1} - CD)\kappa_j)$$

  where CD is divorce costs (tab 3), $\kappa_j = 0.5$ in community property.

Value of a Remarried

- **Re-marriage** is absorbing. See footnote 7.
- In turn,
$$V_t^{jR}(\omega_t) = u(c^{j*R}, P^{j*R}) + \beta \mathbb{E}_t[V_{t+1}^{jR}(\omega_{t+1})]$$
where
$$\begin{aligned} c^{W*R}, c^{H*R}, P^{W*R} = \arg \max_{c^W, c^H, P^W} & \theta u(c^H, 1, 0) + (1-\theta) u(c^W, P^W, 0) \\ & + \beta \mathbb{E}_t[\theta V_{t+1}^{HR}(\omega_{t+1}) + (1-\theta) V_{t+1}^{WR}(\omega_{t+1})] \end{aligned}$$

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- This means that, in the model, divorce can only happen once.

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- This means that, in the model, divorce can only happen once.
- Reason is computational: Assets brought into the marriage is private Keeping track of assets from all previous marriages would be unfeasible. No second divorce $\rightarrow$ does not need to keep track of individual assets.

Household Planning

- **Two cases:**
  1. *Mutual Consent:* Both must prefer divorce for it to happen.
     Committed by law (there are exceptions).
  2. *Unilateral divorce:* If one prefers divorce, they can divorce.
     Limited commitment.
     See lecture note for my notation, I follow Voena (2015).

Household Planning

- **Two cases:**
  1. *Mutual Consent:* Both must prefer divorce for it to happen. Committed by law (there are exceptions).
  2. *Unilateral divorce:* If one prefers divorce, they can divorce. Limited commitment. See lecture note for my notation, I follow Voena (2015).

- **Timing-issue**: The bargaining weight is updated in current period. (See guide)

Household Planning: Mutual Consent

- **Couples** $(D_{t-1} = 0)$ in *mutual consent* regime solve

$$V_t(\omega_t) = \max_{c_t^H, c_t^W P_t^W, A_{t+1}^H, A_{t+1}^W, D_t}$$

$$(1 - D_t)\Big(\theta u(c_t^H, 1, 0) + (1-\theta)u(c_t^W, P_t^W, 0) + \beta \mathbb{E}_t[V_{t+1}(\omega_{t+1}$$

$$+ D_t\Big(\theta\{u(c_t^H, 1, 1) + \beta \mathbb{E}_t[V_{t+1}^{HDR}(\omega_{t+1})]\}$$

$$+ (1-\theta)\{u(c_t^W, P_t^W, 1) + \beta \mathbb{E}_t[V_{t+1}^{WDR}(\omega_{t+1})]\}\Big)$$

with *constant bargaining weights* θ and $1 - \theta$.

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with *constant bargaining weights* θ and $1 - \theta$.

- **Subject to** *non*-participation constraints, when $D_t = 1$,

$$V_{H,t}^{m \to s}(\omega_t; \kappa) = u(c_t^H, 1, 1) + \beta \mathbb{E}_t[V_{t+1}^{HDR}(\omega_{t+1})] > V_t^{HM}(\omega_t)$$

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Household Planning: Mutual Consent

- **Divorce only if both want a divorce.**

Household Planning: Mutual Consent

- **Divorce only if both want a divorce.**
- **If one is unhappy** in the marriage, say the wife
  θ remains unchanged
  asset-split in divorce, κ_m, is changed in his favor until he is indifferent
  $\rightarrow$ she transfers wealth in divorce to convince him to accept divorce.

Household Planning: Unilateral Divorce

- **Couples** $(D_{t-1} = 0)$ in *unilateral* regime solve

$$V_t(\omega_t) = \max_{c_t^H, c_t^W P_t^W, A_{t+1}^H, A_{t+1}^W, D_t} \tag{3}$$

$$(1 - D_t)\Big(\tilde{\theta}_{t+1}^H u(c_t^H, 1, 0) + \tilde{\theta}_{t+1}^W u(c_t^W, P_t^W, 0) + \beta \mathbb{E}_t[V_{t+1}(\omega_{t+1}$$

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$$+ \tilde{\theta}_{t+1}^W \{u(c_t^W, P_t^W, 1) + \beta \mathbb{E}_t[V_{t+1}^{WDR}(\omega_{t+1})]\}\Big)$$

where $\tilde{\theta}_{t+1}^j = \tilde{\theta}_t^j + \mu_t^j$ and μ_t^j are Lagrange multipliers on

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where $\tilde{\theta}_{t+1}^j = \tilde{\theta}_t^j + \mu_t^j$ and μ_t^j are Lagrange multipliers on **participation constraints**, when $D_t = 0$,

$$V_{H,t}^{m \to s}(\omega_t; \kappa = \frac{1}{2}) \leq V_t^{HM}(\omega_t)$$

$$V_{W,t}^{m \to s}(\omega_t; \kappa = \frac{1}{2}) \leq V_t^{WM}(\omega_t)$$

Household Planning: Unilateral Divorce

- **Individual value of *remaining in marriage*** (RHS of constraint) is

$$V_t^{jM}(\omega_t) = u(c_t^{j*}, P_t^{j*}, 0) + \beta \mathbb{E}_t[V_{t+1}^j(\omega_{t+1})]$$

where $c_t^{j*}, P_t^{j*}, A_{t+1}^{j*}$ are optimal choices from eq. (3) and

$$V_{t+1}^j(\omega_{t+1}) = (1 - D_{t+1}^*) V_{t+1}^{jM} + D_{t+1}^* V_{t+1}^{jD}$$

is individual **value of *entering as married*** in $t+1$.

Household Planning: Unilateral Divorce

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$$V_{t+1}^j(\omega_{t+1}) = (1 - D_{t+1}^*) V_{t+1}^{jM} + D_{t+1}^* V_{t+1}^{jD}$$

is individual **value of *entering as married*** in $t+1$.

- **Choices are made as a household** (with weights on individual utility) **individual values** are only based on own utility (and future).

Household Planning: Unilateral Divorce

- **Beginning of period** bargaining weights, $\tilde{\theta}^j_t$, are in ω_t.
- **If both participation constraints are not violated** at $\tilde{\theta}^H_t$ and $\tilde{\theta}^W_t$, the Lagrange multipliers are zero and $\tilde{\theta}^j_{t+1} = \tilde{\theta}^j_t$ is not updated.

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- **To solve** this model (last time + note)
  1. solve the model for couples assuming they remain together, for a grid of bargaining weights.
  2. If, for a given weight, one spouse is not satisfied ($V^{jD}_t > V^{jM}_t$), update the weight on that spouse until indifferent ($V^{jD}_t = V^{jM}_t$). If the other spouse wants to remain in marriage at this new weight, then update weight and carry on!
     Otherwise, divorce.

Outline

1. Model and Mechanisms
2. Estimation and Counterfactuals
3. Simple Model

Estimation

- **2-step estimation:**
  1. calibrate (preset) parameters in Table 3+4
  2. estimate by SMM 3 parameters in Table 5

  using policy variation from mutual to unilateral

TABLE 5—ESTIMATED STRUCTURAL PARAMETERS AND MATCH OF THE AUXILIARY MODEL

Parameter	Symbol	Estimate	Standard error
Standard deviation of preference shocks	σ	0.0008	0.0004
Disutility from labor market participation	ψ	0.0107	0.0025
Husbands' Pareto weight	θ	0.7	0.0155
Auxiliary model parameter	**Symbol**	**Target**	**Simulated**
Effect of uni. divorce on savings in CP	ϕ_1	13.54 percent	13.43 percent
Effect of uni. divorce on participation in CP	ϕ_2	−6.93 pcpt	−6.86 pcpt
Baseline participation rate in CP	ϕ_3	55.97 percent	56.03 percent
Baseline divorce probability in CP	ϕ_4	19.44 percent	19.44 percent

Notes: Parameters of the dynamic model $\{\sigma, \psi, \theta\}$ estimated by indirect inference. The parameters of the auxiliary model are $\{\phi 1\ ,\ \phi 2\ ,\ \phi_3, \phi_4\}$.

Simulation: A

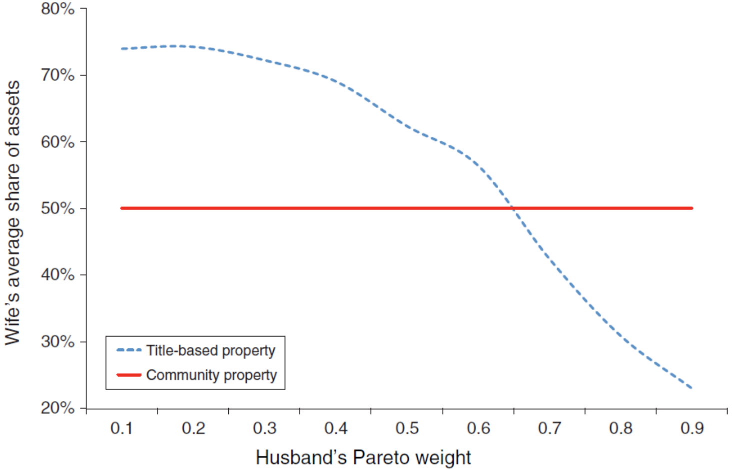
- **Effects from mutual to unilateral**
  in *community* (50-50) regime

Simulation: A

- **Effects from mutual to unilateral**
  in *community* (50-50) regime
- **Mutual:**
  $\theta = 0.7$
  consumption share of women: 39%
- **Unilateral:**
  19% re-bargained their power
  consumption share of women: 41%
  labor supply: $\downarrow$ 6.86pp.

Simulation: B

- **Effects of property division regimes**



Simulation: C

- **Divorce laws and consumption insurance**

TABLE 6—DIVORCE LAWS AND CONSUMPTION INSURANCE AGAINST INCOME SHOCKS

	Married couples			
	Men		Women	
Regimes	Mutual consent	Unilateral divorce	Mutual consent	Unilateral divorce
Title-based	0.372	0.410	0.233	0.207
Community property	0.371	0.390	0.235	0.192
Equitable distribution	0.375	0.384	0.238	0.197

Notes: The table reports the estimates of coefficients μ^j obtained from the regressions

$$\Delta\log(c_{it}^H) = \kappa^H + \mu^H \Delta\log(y_{it}^H) + \boldsymbol{\nu}'^H \mathbf{X}_{it}^j + e_{it}^H \quad \text{and}$$

$$\Delta\log(c_{it}^W) = \kappa^W + \mu^W \Delta\log(y_{it}^W) + \boldsymbol{\nu}'^W \mathbf{X}_{it}^j + e_{it}^W$$

in each legal regime, where X_{it}^j are spouse j's age and age squared. The coefficients are estimated on data obtained from simulating the model using the preset parameters and the estimated parameters for a sample of simulated households. I account for the differential selection of couples out of marriage because of divorce laws by simulating income and consumption profiles using only the policy functions of married couples.

1. Men have more consumption insurance under mutual (lower pass-through of income shocks; col 1 $<$ col 2)
2. Property division does not matter in mutual (col 1 $+$ 3 constant across rows)

Outline

Our simple model

- **Same model as last time** (see notebook)
- We cannot model the same counterfactuals as Alessandra Voena in our simple model.
  But we can **change wealth distribution upon divorce**.

Our simple model

- **Same model as last time** (see notebook)
- We cannot model the same counterfactuals as Alessandra Voena in our simple model.
  But we can **change wealth distribution upon divorce**.
- Now, κ_j denotes the share of wealth to member j, $\kappa_1 + \kappa_2 = 1$.

$$
\begin{aligned}
V^m_{j,t}(a_{t-1}, \psi_t, \mu_{t-1}) = {} & D^\star_t V^{m\to s}_{j,t}(\kappa_j a_{t-1}, \psi_t, \mu_{t-1}) \\
& + (1 - D^\star_t) V^{m\to m}_{j,t}(a_{t-1}, \psi_t, \mu_{t-1})
\end{aligned}
$$

Next Time

- **Next time:**
  Marriage and Divorce (in Denmark).

- **Literature:**
  Bruze, Svarer and Weiss (2015): "The Dynamics of Marriage and Divorce"
  [full commitment]
  - **Read** before lecture
  - **Reading guide:**
    Section 1: Introduction + overview. Read.
    Section 2: Data. Skim.
    Section 3: Marriage patterns. Read (many figures).
    Section 4: Model. Key, get the idea.
    Section 5: Estimation. Skim.
    Section 6: Results. Read.

References I

BRUZE, G., M. SVARER AND Y. WEISS (2015): "The Dynamics of Marriage and Divorce," *Journal of Labor Economics*, 33(1), 123–170.

VOENA, A. (2015): "Yours, Mine, and Ours: Do Divorce Laws Affect the Intertemporal Behavior of Married Couples?," *American Economic Review*, 105(8), 2295–2332.

Marriage and Divorce Dynamics in Denmark

Thomas H. Jørgensen

2025

Plan for today

- Bruze, Svarer and Weiss (2015): "The Dynamics of Marriage and Divorce"
    - Full commitment!
      Danish data for cohorts 1960 (men), 1962 (women).

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- **Reading guide:**
    1. What are the main *research questions*?
    2. What is the *(empirical) motivation*?
    3. What are the central *mechanisms in the model*?
    4. What is the *simplest model* in which we could capture these?

Plan for today

- Bruze, Svarer and Weiss (2015): "The Dynamics of Marriage and Divorce"
  - Full commitment!
    Danish data for cohorts 1960 (men), 1962 (women).
- **Reading guide:**
  1. What are the main *research questions*?
     - How does marriage and divorce behavior vary across age and educational groups?
     - How does educational differences influence intra-household inequality?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

Empirical Motivation: I

- **Marriage and divorce**
  Age of female cohort = age of male cohort-2

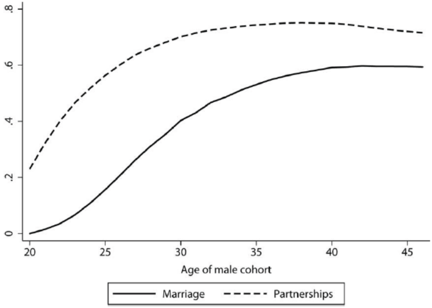

FIG. 1.—Fraction married or in partnerships (marriage plus cohabitation) by age. A color version of this figure is available online.

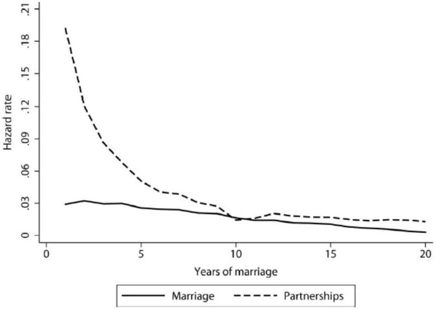

FIG. 3.—Divorce hazard for first marriage or partnership (marriage plus cohabitation). A color version of this figure is available online.

Empirical Motivation: II

- **Highly educated** people partner later but more "stable". Especially if both highly educated

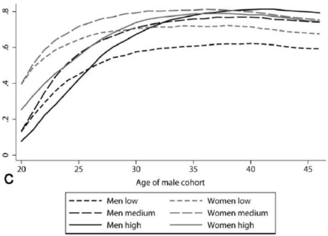

FIG. 4.—A, Fraction married men and women by age and education; B, fraction cohabiting men and women by age and education; C, fraction men and women in partnerships by age and education. A color version of this figure is available online.

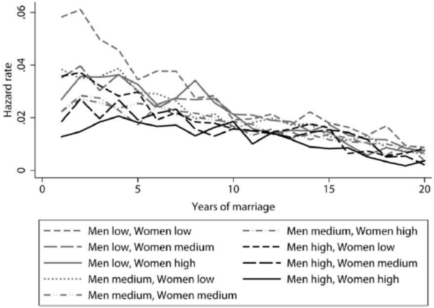

FIG. 6.—Divorce hazards for first marriages by education of the husband and wife. A color version of this figure is available online.

Empirical Motivation: III

- **Highly educated** people re-marry faster.
  And stay in second marriage longer.

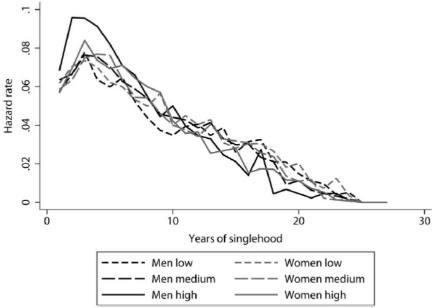

FIG. 7.—Hazard rate into second marriage for men and women by education. A color version of this figure is available online.

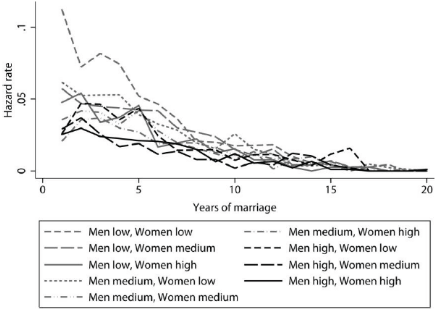

FIG. 8.—Divorce hazards when at least one spouse is in second marriage, by education of the husband and wife. A color version of this figure is available online.

Empirical Motivation: IV

- **Assortative matching:**
  people more likely to marry one with same education

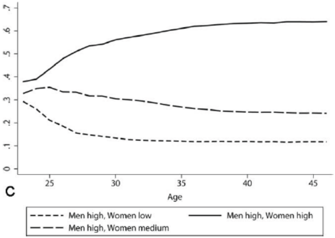

FIG. 9.—Distribution of marriages for men with low (top), medium (middle), or high (bottom) education. A color version of this figure is available online.

(a) $P(educ_w | educ_m = high)$

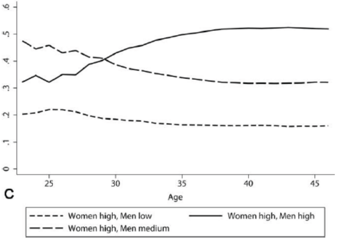

FIG. 10.—Distribution of marriages for women with low (top), medium (middle), or high (bottom) education. A color version of this figure is available online.

(b) $P(educ_m | educ_w = high)$

Empirical Motivation: V

- **What is the point of the model?**

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  - "Explain" the data
  - Structure $\rightarrow$ identify bargaining power $\rightarrow$ intra-household inequality
    Fundamentally unobserved
    Important for understanding gender-inequality

Empirical Motivation: V

- **What is the point of the model?**
  - "Explain" the data
  - Structure $\rightarrow$ identify bargaining power $\rightarrow$ intra-household inequality
    Fundamentally unobserved
    Important for understanding gender-inequality

- **From Abstract:**
  Education raises the share of the marital surplus for men but not for women. As men and women get older, husbands receive a larger share of the marital surplus

Outline

1. Model and Mechanisms

2. Estimation

Model Overview

- **Full commitment:**
  Transferable utility
  Perfect foresight wrt bargaining power.
  Particular timing/expectation assumptions (get back)
- **Choices:**
  Marriage: which type IJ
  Divorce
- **States:**
  d_t: duration of marriage
  "TYPES":
  $e \in E = \{l, m, h\}$: Educational type of both members
  $p_t \in P = \{nm, pm\}$: never/previously married
  $u \in U = \{1, 2\}$ (unobserved type)
  $\rightarrow I \in E \times P \times U$ (men) and $J \in E \times P \times U$ (women)
  (love-shock, $\theta_t \sim iid\mathcal{N}(0, 1)$)

Bellman Equation: Married

- **Bellman equation** for type IJ (remaining) couple is

$$\underbrace{W_t^{IJ}(d_t)+\theta_t}_{V_t^{m\to m}} = \underbrace{\zeta^{IJ}+\theta_t}_{U^{IJ}} + R\mathbb{E}_t[\underbrace{\max\{\underbrace{W_{t+1}^{IJ}(d_{t+1})+\theta_{t+1}}_{V_{t+1}^{m\to m}}, \underbrace{V_{t+1}^{I}+V_{t+1}^{J}-s(d_{t+1})}_{V_{t+1}^{m\to s}}\}}_{V_{t+1}^{m}}]$$

where
ζ^{IJ}: type-specific utility
R: discount factor
$s(d_{t+1})$: divorce cost
$V_{t+1}^{I}+V_{t+1}^{J}$: sum of value of singlehood (TU)
(I would think that $d_{t+1}=d_t+1$, but they never write)
$\mathbb{E}_t[]$ is wrt. θ_{t+1}

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  (I would think that $d_{t+1}=d_t+1$, but they never write)
  $\mathbb{E}_t[]$ is wrt. θ_{t+1}

- **Probability** of observing divorce, $P_D(I,J,t,d_t)$:

$$\Pr(W_t^{IJ}(d_t)+\theta_t < V_t^I+V_t^J-s(d_t)) = \Phi(V_t^I+V_t^J-s(d_t)-W_t^{IJ}(d_t))$$

iid theta which tells us the likelihood of divorce. Normal distributed. probit or logit model

Bellman Equation: Single

- **Bellman equation** for type I man being single is

$$V_t^I = \varphi^I + R\mathbb{E}_t[V_{t+1}^I + \max_{J \in E \times P \times U}\{\varepsilon_{t+1}^0, \gamma_{t+1}^{IJ}[W_{t+1}(\underbrace{1}_{d_{t+1}}) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\}]$$

  where
  ε_{t+1}^0: EV taste-shock wrt value of singlehood
  ε_{t+1}^J: EV taste-shock wrt value of marriage with type J
  γ_{t+1}^{IJ}: share of (new) marital surplus to man. Focus in a bit.
  $\mathbb{E}_t[]$ is wrt. Extreme Value taste shocks over type of female match, J.
  (See discussion on following slides.)

- **Value of marriage next period** is thus the value of being single + the share of the marital surplus he gets.
- **Symmetric** for women with share $1 - \gamma_{t+1}^{IJ}$.

Bellman Equation: Single, Marital Surplus!

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$$V_t^I = \varphi_t^I + R\mathbb{E}_t[\underbrace{V_{t+1}^I + \max_{J\in E\times P\times U}\{\varepsilon_{t+1}^0, \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\}}_{V_{t+1}^s \text{ (but might re-partner)}}]$$

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where

$$\begin{aligned} V_{t+1}^s &= V_{t+1}^I + \max_{J\in E\times P\times U}\{\varepsilon_{t+1}^0, \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\} \\ &= \max_{J\in E\times P\times U}\{V_{t+1}^I + \varepsilon_{t+1}^0, V_{t+1}^I + \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\} \end{aligned}$$

such that value of marriage is "value of singlehood + his share of marital surplus"

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such that value of marriage is "value of singlehood + his share of marital surplus"

- **Marital surplus** is "special" since future love-shock does not enter...:

$$W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J = \mathbb{E}_t[W_{t+1}(1) + \underbrace{\theta_{t+1}}_{\text{theta = match quality}} - V_{t+1}^I - V_{t+1}^J]$$

since $\mathbb{E}_t[\theta_{t+1}] = 0$.

Bellman Equation: Single, Marital Surplus!

- They interpret this as a timing-thing (p. 140):
  "The quality of match ... is revealed to the partners only at the end of each period. ...
  In particular, single agents who marry at time t do not know the quality of their math θ_t and expect it to equal the mean, which is set to zero."

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  In particular, single agents who marry at time t do not know the quality of their math θ_t and expect it to equal the mean, which is set to zero."

  Thomas is pretty sure they did this to make calculations easier.

- **"Wrong":** This means that the expected value is inserted in the max, rather than taking the expected value of the max...:

$$\max_{J \in E \times P \times U} \{\varepsilon^0_{t+1}, \gamma^{IJ}_{t+1} \mathbb{E}_t [W_{t+1}(1) + \theta_{t+1} - V^I_{t+1} - V^J_{t+1}] + \varepsilon^J_{t+1}\}$$

  vs "correct"

$$\mathbb{E}_t [\max_{J \in E \times P \times U} \{\varepsilon^0_{t+1}, \gamma^{IJ}_{t+1} [W_{t+1}(1) + \theta_{t+1} - V^I_{t+1} - V^J_{t+1}] + \varepsilon^J_{t+1}\}]$$

- **Their formulation removes a numerical integral wrt.** θ_{t+1}
  $\rightarrow$ Speeds up the solution...

Bellman Equation: Single, Marital Surplus!

- The expectation in

$$V_t^I = \varphi_t^I + R\mathbb{E}_t[V_{t+1}^I + \max_{J \in E \times P \times U}\{\varepsilon_{t+1}^0, \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\}]$$

is thus only over EV-shocks (and not θ_{t+1})!

Bellman Equation: Single, Marital Surplus!

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  is thus only over EV-shocks (and not θ_{t+1})!
- **Known in closed-form:** The log-sum!

$$\mathbb{E}_t[V_{t+1}^I + \max_{J \in E \times P \times U}\{\varepsilon_{t+1}^0, \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\}]$$

$$=$$

$$\log\left[\exp(V_{t+1}^I) + \sum_{J \in E \times P \times U} \exp(V_{t+1}^I + \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J])\right]$$

Bellman Equation: Single, Marital Surplus!

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- **Probability** (logit) of entering marriage with type j, ($j = 0 \rightarrow$single)

$$P_M^I(j,t) = \frac{\exp(V_t^I + \gamma_t^{Ij}[W_t(1) - V_t^I - V_t^j])}{\exp(V_t^I) + \sum_{J \in E \times P \times U} \exp(V_t^I + \gamma_t^{IJ}[W_t(1) - V_t^I - V_t^J])}$$

Commitment

- **Full commitment:**

$$\gamma_t^{IJ}$$

is known throughout.

Commitment

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$$\gamma_t^{IJ}$$

is known throughout.

- **Assume** the functional form

$$\gamma_t^{IJ} = \frac{\exp\{\rho^{IJ} + \kappa^{IJ} t + \lambda^{IJ} t^2\}}{1 + \exp\{\rho^{IJ} + \kappa^{IJ} t + \lambda^{IJ} t^2\}}$$

which has 108 estimated parameters.
(not reported)

Outline

1 Model and Mechanisms

2 Estimation

Remaining Parameters

- **Cost of divorce** "non-parametric" (10)

$$s(d_t) = \sum_{k=1}^{9} \beta_k \mathbf{1}(d_t = k) + \beta_{10} \mathbf{1}(d_t \geq 10)$$

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$$s(d_t) = \sum_{k=1}^{9} \beta_k \mathbf{1}(d_t = k) + \beta_{10}\mathbf{1}(d_t \geq 10)$$

- **Utility** for singles are $u \in \{1, 2\}$

$$\varphi_t^I = \mu_t^I + \eta_u^I$$
$$\varphi_t^J = \mu_t^J + \eta_u^J$$

and estimated parameters are

$$\zeta^{IJ} :13 \text{ (education mix (9) or marital order mix (4))}$$
$$\mu_t^I, \mu_t^J :2 \times 18 \text{ (gender, age, educ)}$$

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$$\mu_t^I, \mu_t^J : 2 \times 18 \text{ (gender, age, educ)}$$

- **Unobserved types** (4):
  estimate u_2^I, u_2^J (relative to type 1) and the share of type 2.

Estimation

- **Maximum likelihood**
  Dynamic logit due to EV taste-shocks wrt discrete types.

Estimation

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  Dynamic logit due to EV taste-shocks wrt discrete types.
- Let $O_i = (O_{i,1}, \ldots, O_{i,T})$ and $O_j = (O_{j,1}, \ldots, O_{j,T})$ be observed choices of men and women
- Let $S_{i,0}$ and $S_{j,0}$ be initial states. These states are taken as given.

Estimation

- **Maximum likelihood**
  Dynamic logit due to EV taste-shocks wrt discrete types.
- string of events for men
  Let $O_i = (O_{i,1}, \ldots, O_{i,T})$ and $O_j = (O_{j,1}, \ldots, O_{j,T})$ be observed choices of men and women
- Let $S_{i,0}$ and $S_{j,0}$ be initial states. These states are taken as given.
- The (conditional) likelihood function of the observed data is

$$L = \prod_{i=1}^{N^m} \Pr(O_i | S_{i,0}) \times \prod_{j=1}^{N^f} \Pr(O_j | S_{j,0})$$

assuming independence.

Estimation

- **Maximum likelihood**
  Dynamic logit due to EV taste-shocks wrt discrete types.
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  assuming independence.
- The EV assumption makes $\Pr(\bullet)$ conditional multinomial logit (MNL) Can be found in closed form.

Estimation

- **Likelihood of *sequence*** of choices given $S_{i,0}$, u_i

$$\Pr(O_i|S_{i,0}, u_i) = \prod_{t=2}^{T} \Pr(O_{i,t}|O_{i,t-1}, u_i) \Pr(O_{i,1}|S_{i,0}, u_i)$$

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- **Do not observe** u; we "integrate that out":

$$\begin{aligned} \Pr(O_i|S_{i,0}) &= \mathbb{E}[\Pr(O_i|S_{i,0}, u_i)] \\ &= q^m \Pr(O_i|S_{i,0}, u_i = 1) + (1 - q^m) \Pr(O_i|S_{i,0}, u_i = 2) \end{aligned}$$

where q^m and q^f are the shares of type 1 ($u = 1$)

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  where q^m and q^f are the shares of type 1 ($u = 1$)
- **The likelihood** of observing the outcomes is then

$$\begin{aligned}L = &\prod_{i=1}^{N^m} [q^m \Pr(O_i|S_{i,0}, u_i = 1) + (1 - q^m) \Pr(O_i|S_{i,0}, u = 2)] \\ \times &\prod_{j=1}^{N^f} [q^f \Pr(O_j|S_{j,0}, u_j = 1) + (1 - q^f) \Pr(O_j|S_{j,0}, u_j = 1)]\end{aligned}$$

Identification (idea)

- **Identification** arguments in paper
  Only without unobserved types

- **Talk about some here**
  To give idea of arguments
  Ignores unobserved types, $u \in \{1, 2\}$ we will ignore this unobserved types for now.
  $\rightarrow I, J \in E \times P$ (educ and pre-marital status)

Identification: Weights

- **From probability of marriage** of I with J, relative to remaining single

$$\log\left(\frac{P_M^I(J,t)}{P_M^I(0,t)}\right) = \gamma_t^{IJ}[W_t^{IJ}(1) - V_t^I - V_t^J]$$

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- such that taking ratios **identifies weights**, γ_t^{IJ}:

by inserting data we would get the estimate for gamma

$$\frac{\gamma_t^{IJ}}{1-\gamma_t^{IJ}} = \underbrace{\log\left(\frac{P_M^I(J,t)}{P_M^I(0,t)}\right) / \log\left(\frac{P_M^J(I,t)}{P_M^J(0,t)}\right)}_{\text{data}}$$

Identification: Divorce costs

- **From probability of divorce**:

$$V_t^I + V_t^J - s(d_t) - W_t^{IJ}(d_t) = \underbrace{\Phi^{-1}(P_D(I, J, t, d_t))}_{\text{data}}$$

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We can then insert into last slide to get $s(1)$ (first-yr divorce cost):

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$$s(1) = \underbrace{\log\left(\frac{P_M^I(J,t)}{P_M^I(0,t)}\right)}_{\text{data}} / \underbrace{\gamma_t^{IJ}}_{\text{"known"}} + \underbrace{\Phi^{-1}(P_D(I,J,t,1))}_{\text{data}}$$

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- Remaining $s(d)$: Noting that $W_t^{IJ}(d_t)$ depends on d_t through $s(d)$ and

$$\underbrace{\Phi^{-1}(P_D(I,J,t,d_t)) - \Phi^{-1}(P_D(I,J,t,d_t'))}_{\text{data}} = s(d_t') - s(d_t) + W_t^{IJ}(d_t') - W_t^{IJ}(d_t)$$

Identification: Utility Flow

- **Assume** that flow-utility in couple are constant
- **"Normalize"** value of singlehood for men over age 40 to zero (but more than normalization since several periods, $T = 71$?)
- **"Normalize"** value of singlehood for women over age 38 to zero (but more than normalization since several periods, $T = 71$?)

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- **Couples:** For $t > 39$:

$$V_t^I + V_t^J - s(d_t) - W_t^{IJ}(d_t) - \theta_t = s(d_t) - W_t^{IJ}(d_t) - \theta_t$$

and thus gets ζ^{IJ} from

$$P_D(I, J, t, d_t) = \Pr(\theta_t \leq W_t^{IJ}(d_t) - s(d_t)) = \Phi(s(d_t) - W_t^{IJ}(d_t))$$

$$\Updownarrow$$

$$W_t^{IJ}(d_t) = \underbrace{s(d_t)}_{\text{"known"}} - \underbrace{\Phi^{-1}(P_D(I, J, t, d_t))}_{\text{data}}$$

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- **Singles:** time-differences in likelihood gives φ_t^I and φ_t^J.

Results: Marriage Order

- **Estimates** suggest that second marriages are less "costly" for men

Table 3
Effects of Marriage Order on the Marital Output Flow

	Wife's First Marriage	Wife's Second Marriage
Husband's first marriage	.5166	.3891
Husband's second marriage	.4709	.5364

husband tends to get a higher martital surplus in the second marriage compared to the other way around

Results: Divorce Costs

- **Estimates** suggest that divorce costs are U-shaped
  Authors are surprised, but this could still be due to children.

Table 4
Costs of Divorce by Duration of Marriage

Marital Duration	Cost of Divorce
1 year	14.3
2 years	14.1
3 years	12.4
4 years	11.5
5 years	11.6
6 years	11.6
7 years	11.5
8 years	12.7
9 years	12.7

Results: Marital Surplus Shares

- Do not want to look at γ_t^{IJ} due to selection.

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is the *expected value of being forced to remain single*.

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is the *expected value of being forced to remain single*.

- **If allowed on the marriage market**, the expected value is (log-sum):

$$C_{t+1}^I = \mathbb{E}_t[V_{t+1}^I + \max_{J \in E \times P \times U} \{\varepsilon_{t+1}^0, \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\}$$

- **Expected gains** from entering the marriage market:

$$S_t^I = C_{t+1}^I - \mathbb{E}_t[V_{t+1}^I + \varepsilon_{t+1}^0]$$

and *similarly for women.*

Results: Marital Surplus Shares

- They define the total share of surplus of the husband be

$$\Gamma_t^{IJ} = \frac{S_t^I}{S_t^I + S_t^J}$$

expected gain for the husband over the total expected gain

Table 9
Estimated Average Total Surplus Share Γ for Husband by Education of Husband and Wife

	Wife's Education		
Husband's Education	Low	Medium	High
Low	.417	.387	.402
Medium	.496	.463	.490
High	.530	.493	.498

Results: Marital Surplus Shares

Table 11
Estimated Average Total Surplus Share Γ for Husband by Marital History of Husband and Wife

	Wife's First Marriage	Wife's Second Marriage
Husband's first marriage	.464	.353
Husband's second marriage	.563	.455

Table 12
Estimated Average Total Surplus Share Γ for Husband by Age of Husband

Age of Husband	Share of Gains to Marriage
25	.425
30	.465
35	.486
40	.505
45	.541

Next Time

- **Next time:**
  Fertility and Labor Supply.

- **Literature:**
  Jakobsen, Jørgensen and Low (2022): "Fertility and Family Labor Supply"
  [unitary]
  - **Read** before lecture
  - **Reading guide:**
    Section 1: Introduction – Read.
    Section 2: Data. Skim.
    Section 3: Empirical motivation. Get idea.
    Section 4: Model. Key. Get the idea.
    Section 5: Estimation results. skim/read.
    Section 6: Simulation results. Key - read.
    Section 7: Sensitivity/robustness. Can drop.

References I

Bruze, G., M. Svarer and Y. Weiss (2015): "The Dynamics of Marriage and Divorce," *Journal of Labor Economics*, 33(1), 123–170.

Jakobsen, K., T. H. Jørgensen and H. Low (2022): "Fertility and Family Labor Supply," Working paper, Centre for Economic Behavior and Inequality.

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Fertility and Family Labor Supply

Katrine Jakobsen[1,2] Thomas H. Jørgensen[1] Hamish Low[2]

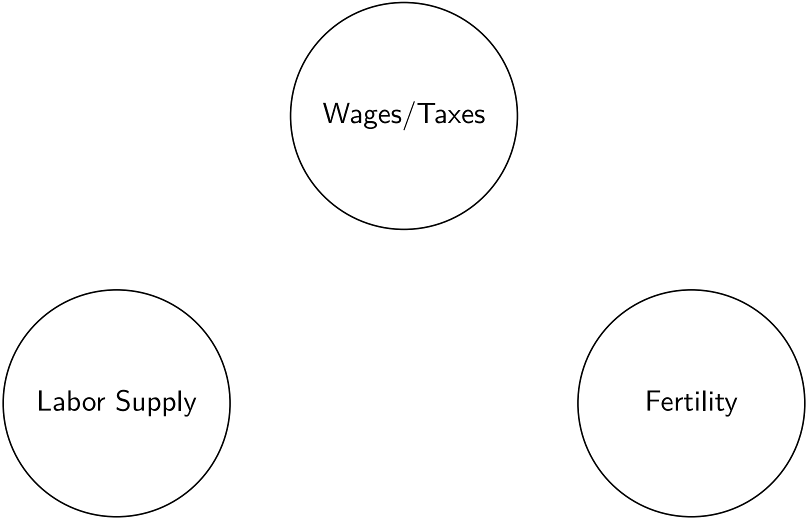
[1]CEBI, University of Copenhagen

[2]Oxford University

2026

Motivation:

- Understanding labor supply is key for policy design.



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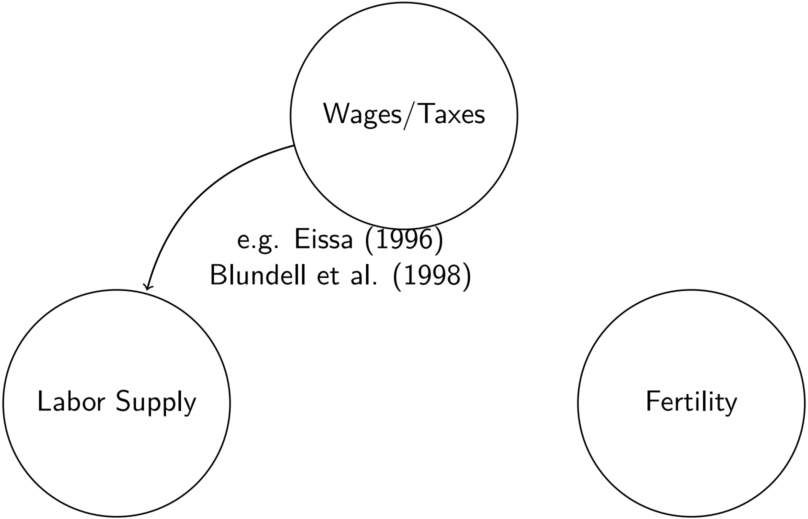

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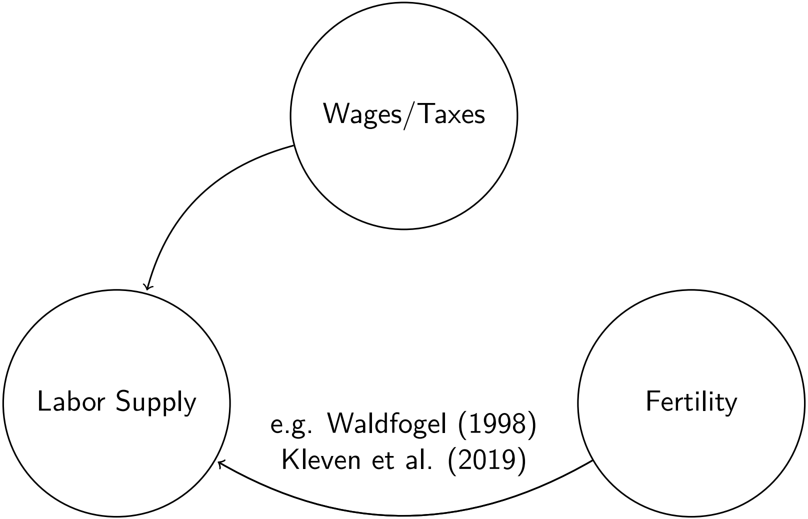

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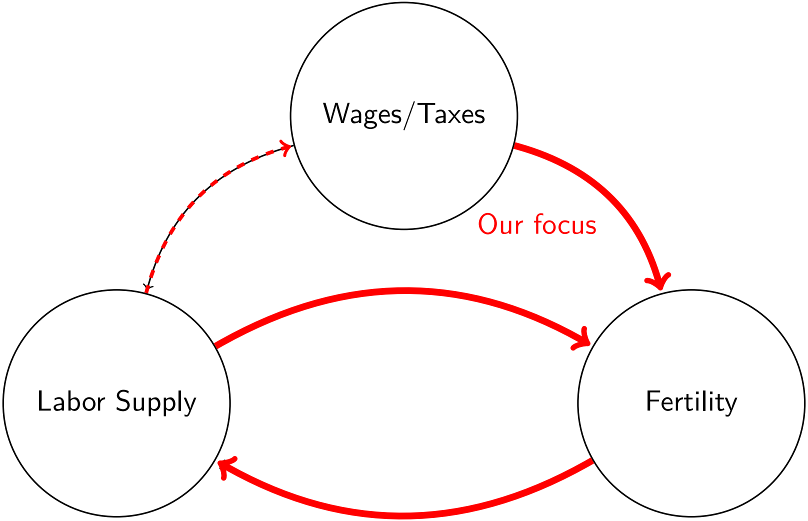

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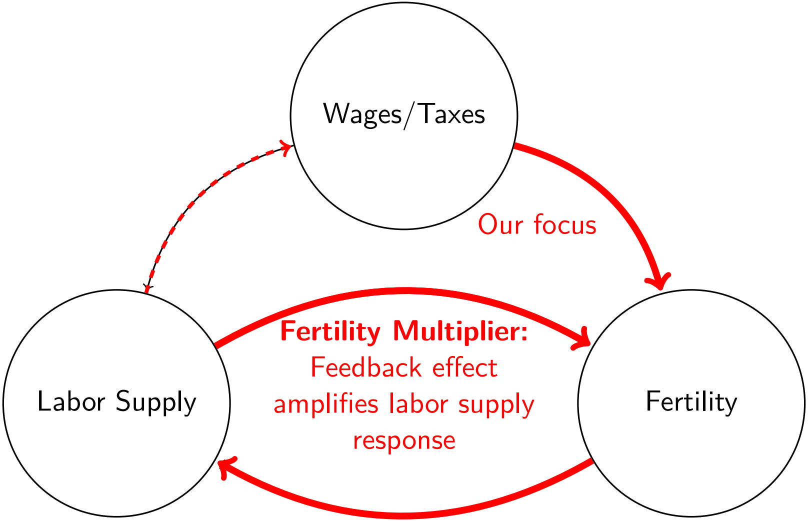

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2. **Quantify the importance of fertility** for labor market reforms through an estimated life-cycle model:
   - labor supply and human capital accumulation of both household members
   - Fertility (endogenous number and timing)
   - Wealth accumulation

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   - marginal net-of-tax wage of **men** $\uparrow \implies$ fertility $\uparrow$
   - marginal net-of-tax wage of **women** $\uparrow \implies$ fertility $\downarrow$
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   - marginal net-of-tax wage of **women** $\uparrow \Longrightarrow$ fertility $\downarrow$

2. **Quantify the importance of fertility** for labor market reforms through an estimated life-cycle model:
   - labor supply and human capital accumulation of both household members
   - Fertility (endogenous number and timing)
   - Wealth accumulation
   - Replicates empirical findings above
   - Fertility and human capital depreciation can exacerbate long run gender inequality
   - $>$10% higher Marshallian labor supply elasticity of women when fertility can respond

Related Literature

- **Fertility responses to financial incentives:**
  - **Child subsidies and tax reliefs** (see e.g. Rosenzweig, 1999; Milligan, 2005; Brewer, Ratcliffe and Smith, 2012; Cohen, Dehejia and Romanov, 2013; Laroque and Salanié, 2014)
  - **Child care costs** (Blau and Robins, 1989; Del Boca, 2002; Mörk, Sjögren and Svaleryd, 2013)
  - **Wealth** (housing) (Lovenheim and Mumford, 2013; Dettling and Kearney, 2014; Mizutani, 2015; Atalay, Li and Whelan, 2017; Clark and Ferrer, 2019; Daysal, Lovenheim, Siersbæk and Wasser, 2021).
- **Female labor supply and fertility:** Hotz and Miller (1988); Francesconi (2002); Keane and Wolpin (2010); Adda, Dustmann and Stevens (2017); Eckstein, Keane and Lifshitz (2019)
- **Long-run labor supply elasticities:** see e.g. Attanasio, Levell, Low and Sánchez-Marcos (2018) and Keane (2011, forthcoming)
- **Gender gaps and child penalties:** Goldin (2014), Goldin and Katz (2002a), Kleven et al. (2019).

Outline

1. Empirical Motivation
   - Data
   - Identification Strategy
   - Results

2. Life-Cycle Model
   - Model framework
   - Estimation
   - Simulations
   - Quantifying the Importance of Fertility

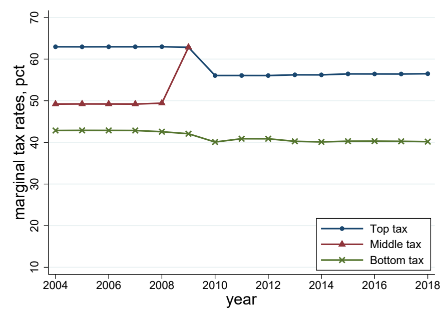
Data and Sample Selection

- **Use several Danish registers for 2004–2018**
  - Linking household members (married and cohabitating) details
  - Information on income, fertility, wealth etc.
  - Monthly pay-slip information (BFL, from 2010)
    - Aggregate to annual freq.
    - Center around calendar year or childbirth
- **Common sample selection:**
  - Aged 25–60
  - Has a partner (of opposite sex)
  - Discard people who are mainly
    self-employed, student, retired or on disability insurance
- **Two samples:**
  1. tax sample (women aged 25–40)
  2. estimation sample (2010–2018, max. 5 years age difference)
     Split by educational attainment of woman

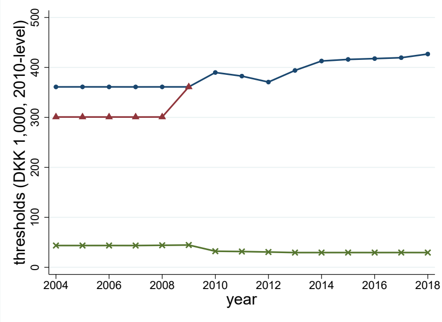
Identification Strategy: Tax Variation

Identification Strategy: Tax Variation

Figure: Danish Tax Variation, 2004–2018 (avg.).



(a) Marginal tax rates.



(b) Thresholds.

Notes: This Figure illustrates the main tax variation in the tax thresholds and marginal tax rates from 2004 through 2018 (averages). Source: Jakobsen and Søgaard (2019).

Elasticity of taxable income (ETI)
1) Household level
2) Change outcome from TI to number of children

income tax system is primarily individual in Denmark whereas in the US they have a joint tax system

Identification Strategy: Household Regressions

- Estimate equations of the form (ETI, Gruber and Saez, 2002)

$$\Delta_4 N_{i,t} = \eta_w \Delta_4 \log(1-\tau_{i,t}) + \eta_m \Delta_4 \log(1-\tau_{partner(i,t)}) + \beta X_{i,t} + g(z_{i,t}) + \varepsilon_{i,t}$$

where

- $N_{i,t}$: number of children of woman i at time t
- $\Delta_4 x_{i,t}$: four-year forward differences
- $\tau_{i,t}$: marginal tax rate not exogenous as tax varies with income
- $X_{i,t}$: year- and age dummies and human capital
- $g(z_{i,t})$: detailed income controls for both partners

- η_w: Elasticity w.r.t **women's** marginal net-of-tax wage
- η_m: Elasticity w.r.t **men's** marginal net-of-tax wage

Identification Strategy: 2SLS

- **Endogenous** marginal tax rates
- **Instrument** $\Delta_4 \log(1-\tau_{i,t})$ and $\Delta_4 \log(1-\tau_{partner(i,t)})$ with 4-year *mechanical* net-of-tax wage changes of each partner

$$\log(1-\tau_{i,t}^{t+4}) - \log(1-\tau_{i,t})$$
$$\log(1-\tau_{partner(i,t)}^{t+4}) - \log(1-\tau_{partner(i,t)})$$

details

2SLS Estimation Results

	all	less skilled	high skilled
	(1)	(2)	(3)
$\Delta_4 \log(1-\tau_{i,t})$, women	-0.018*	-0.047***	-0.015
higher income for women -> lower fertility	(0.010)	(0.014)	(0.013)
$\Delta_4 \log(1-\tau_{i,t})$, men	0.010	0.038***	-0.020
higher income for men -> higher fertility	(0.009)	(0.012)	(0.013)
Income dummies	Yes	Yes	Yes
Children dummies	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes
Hum. cap. controls	Yes	Yes	Yes
Male partner controls	Yes	Yes	Yes
Avg. dep. var. (y, level)	1.522	1.664	1.372
Obs.	2,531,181	1,299,908	1,231,273
First stage F-stat.	47,359.7	17,621.2	28,805.6

- Elasticities (less skilled):
  $-0.047/1.664 = -0.028$ wrt. wages of women
  $0.038/1.664 = 0.023$ wrt. wages of men

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Model Overview

- Households maximize the expected discounted sum of future utility
- Choose
  - C_t: Consumption
  - $l_{w,t}$: Labor supply, woman
  - $l_{m,t}$: Labor supply, men
  - e_t : Fertility effort
- Given states
  - $K_{w,t}$: Human capital, woman
  - $K_{m,t}$: Human capital, men
  - A_{t-1}: Wealth (no net-borrowing)
  - n_t: Number of children
  - o_t : Age of youngest child
  - f_t: Infertility state
  - edu: Educational attainment of woman (estimation by two groups)

Labor Supply

- Endogenous labor supply of men and women, $j \in \{m, w\}$:
  - Not working, $l_{j,t} = 0$
  - Part time, $l_{j,t} = 0.75$
  - Full time, $l_{j,t} = 1$

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- Human capital accumulation

$$K_{j,t+1} = [(1-\delta)K_{j,t} + l_{j,t}]\epsilon_{j,t+1}$$

where $\epsilon_{j,t+1}$ is an *iid* log-normal mean-one shock.

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where $\epsilon_{j,t+1}$ is an *iid* log-normal mean-one shock.

- Labor income is

$$Y_{j,t} = w_{j,t} l_{j,t}$$

where wages are

$$\log w_{j,t} = \gamma_{j,0} + \gamma_{j,1} K_{j,t}$$

details

Fertility

- If fertile, $f_t = 0$,
  - Couples chose **fertility effort**, $e_t \in \{0, 1\}$ each period
  - Imperfect fertility control

Fertility

- If fertile, $f_t = 0$,
  - Couples chose **fertility effort**, $e_t \in \{0, 1\}$ each period
  - Imperfect fertility control
  - Childbirth next period with probability

$$\wp_t(e_t) = \begin{cases} \overline{\wp}_t & \text{if } e_t = 1 \\ \overline{\wp}_t \underline{\wp} & \text{if } e_t = 0 \end{cases}$$

not trying to get pregnant but still get pregnant

$\overline{\wp}_t < 1$: biological fecundity (declining in age)
$\underline{\wp} > 0$: unintended pregnancies

- Permanent infertility is random and absorbing, $P(f_{t+1} = 1 | f_t = 0) = p_f(t)$. details

- The age of the youngest, o_t, evolves deterministically details
- Children move out stochastically details

Preferences

- Household preferences are

$$U(C_t, n_t, o_t, l_{w,t}, l_{m,t}, e_t) = \lambda u_w(\cdot) + (1-\lambda) u_m(\cdot)$$

- Individual preferences are

$$\begin{aligned} u_j(C_t, n_t, o_t, l_{w,t}, l_{m,t}, e_t) = & \frac{(C_t/\nu(n_t))^{1-\rho}}{1-\rho} \\ & + \sum_{i=1}^{3} \omega_i \mathbf{1}(n_t \geq i) \\ & + \sum_{i=0}^{2} \eta_i e_t \mathbf{1}(o_t = i) \\ & + g_j(l_{j,t}, age_{j,t}) \\ & + q_j(l_{w,t}, l_{m,t}, n_t, o_t) \mathbf{1}(n_t > 0) \end{aligned}$$

- **Flexible interaction between labor supply and children** in $q_j()$.

details

Institutional environment

- Partnership dissolution is random and absorbing details
- Retirement is exogenous and absorbing
- Involuntary unemployment risk each year
  - With probability $p_{j,0}$ member j is unemployed
    With probability $p_{j,1}$ member j can work part-time or not.

Institutional environment

- Partnership dissolution is random and absorbing details
- Retirement is exogenous and absorbing
- Involuntary unemployment risk each year
  - With probability $p_{j,0}$ member j is unemployed
    With probability $p_{j,1}$ member j can work part-time or not.
- Parsimonious versions of the Danish institutions (2010 rules)
  - Labor income tax system
  - Unemployment transfers [fixed amount in model]
  - Parental leave details
  - Child care costs
  - Child benefits details

Estimation: Two steps

1. **Calibrate** a set of parameters, $\boldsymbol{\phi}$.
   E.g. $\lambda = 0.5$.
   - Investigate the **sensitivity** to calibrated parameters (Jørgensen, 2023) details

Estimation: Two steps

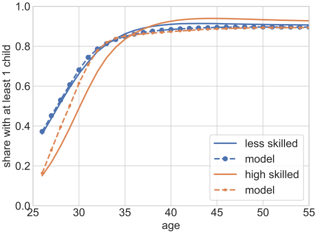
1. **Calibrate** a set of parameters, $\boldsymbol{\phi}$.
   E.g. $\lambda = 0.5$.
   - Investigate the **sensitivity** to calibrated parameters (Jørgensen, 2023) details

2. **Estimate** the remaining 2×48 parameters, $\boldsymbol{\theta}$.
   E.g. value of children, $\omega_1, \omega_2, \omega_3$ and dis-utility of work, $q(\cdot)$
   - **Simulated Method of Moments**

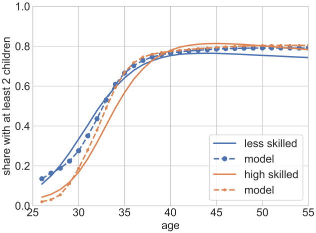
$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} g(\boldsymbol{\theta}|\boldsymbol{\phi})' W g(\boldsymbol{\theta}|\boldsymbol{\phi})$$

   - Using estimation sample from 2010 (post-reform)
   - Investigate the "**informativeness**" of estimation moments (Honoré, Jørgensen and de Paula, 2020) details

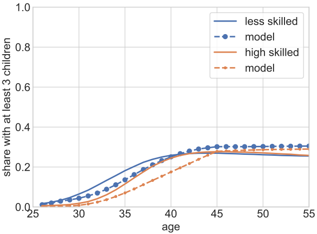
Moments and Model Fit: Fertility

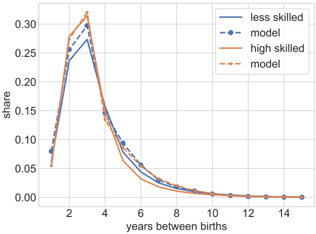


(a) Share with at least one child.

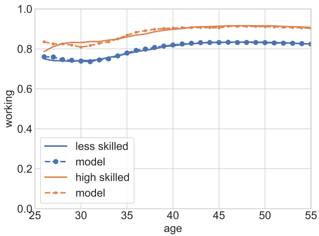


(b) Share with at least two children.

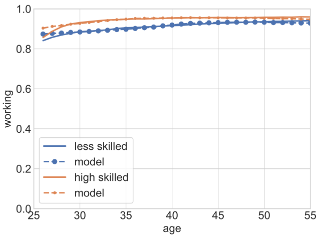




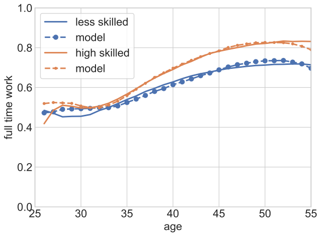
Moments and Model Fit: Selected age profiles



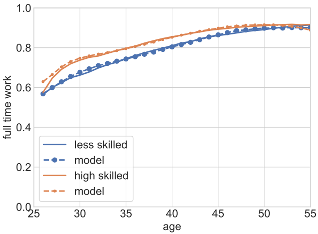
(a) Share Working, Women.



(b) Share Working, Men.

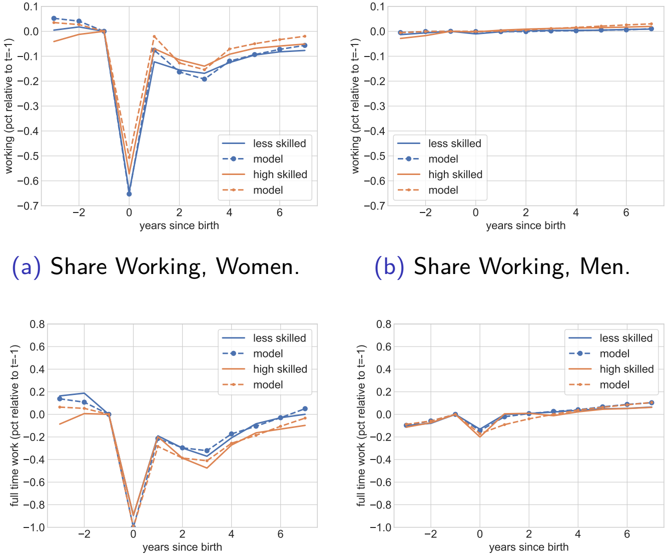


(c) Full time when working,



(d) Full time when working, Men.

Moments and Model Fit: 1. Child Arrival



(a) Share Working, Women. (b) Share Working, Men.

(c) Full time, Women. (d) Full time, Men.

Simulations: Wage Elasticities (Validation)

	Less skilled		High skilled	
	data	model	data	model
Women's wages	-0.028 [-0.045,-0.012]	-0.021	-0.011 [-0.030, 0.008]	-0.012
Men's wages	0.023 [0.009, 0.037]	0.005	-0.015 [-0.033, 0.004]	0.002

- **Data:** Based on reduced-form regressions
- **Model:** 5% unanticipated permanent wage increase at age 35, measured 4 periods later

Model-Implications

1. How much do fertility responses amplify labor supply responses?

2. What are the family-related policy responses in our framework?

3. What are the long run implications of fertility and human capital accumulation for gender inequality?

1. Quantifying the Importance of Fertility for Labor Supply

- How important are **fertility adjustments for labor supply responses**?

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- How important are **fertility adjustments for labor supply responses**?

- We quantify this through counterfactual simulations
  - How different are labor supply elasticities **if fertility cannot adjust**?

1. Quantifying the Importance of Fertility for Labor Supply

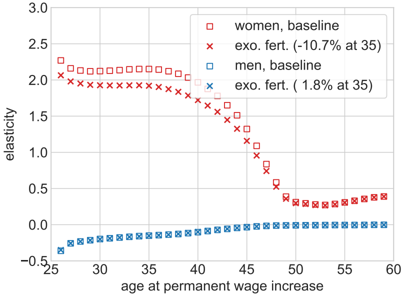
We simulate effect of wage increase from 2 models:

1. **baseline model**, with endogenous fertility
2. **exogenous fertility**, where couples cannot choose fertility
    - **Expect children to arrive *probabilistically***
      based on realized fertility from the baseline model details

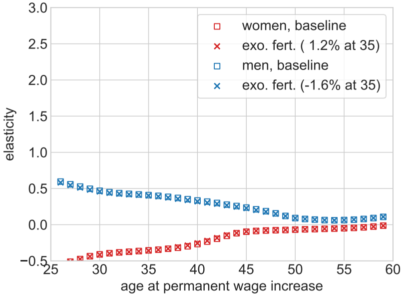
- **5% permanent (unanticipated) increase in wage rate**
    - life-cycle Marshallian elasticity

1. Quantifying the Importance of Fertility for Labor Supply

Figure: Quantifying the Role of Fertility Responses. Less Skilled.



(a) Increased wage of women.



(b) Increased wage of men.

- **Women:** Wages $\uparrow \implies$ Fertility $\downarrow \implies$ labor supply responsiveness $\uparrow$
  >10% larger long-run Marshall elasticity when fertility can adjust
- **Men:** No difference...

2. Counterfactual Policy Simulations

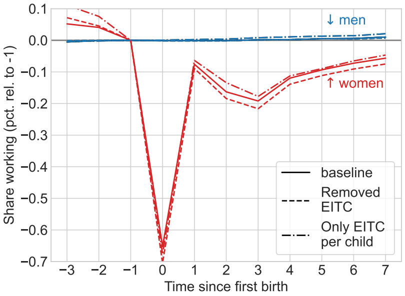
We evaluate several policies:

1. Removal of the EITC
2. Max EITC = baseline$\times$number of children
3. Free childcare (remove $\approx$ 25% co-payment)
4. No maternity leave

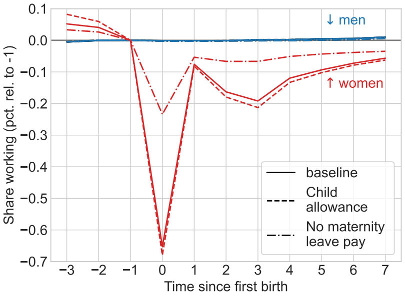
$\rightarrow$ **Clear trade-off** between labor supply (of women) and fertility.

2. Counterfactual Policy Simulations (less skilled)

Figure: Event Studies: EITC and Child-Related Policies. Less Skilled.



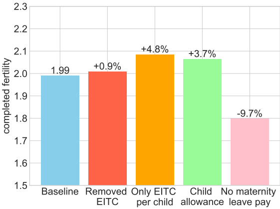
(a) EITC.



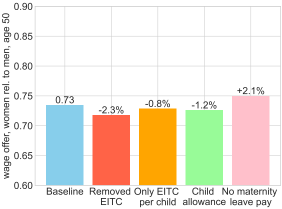
(b) Child-related.

- **Short run:** large effects of removing maternity leave

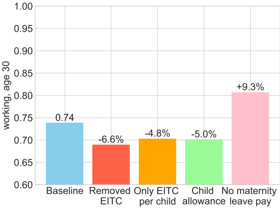
2. Counterfactual Policy Simulations (less skilled)



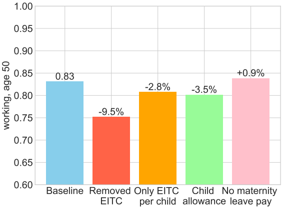
(c) Completed fertility.



(d) Offer wage gap, age 50.



(e) Share working, age 30.



(f) Share working, age 50.

- **Long run:** Only minor effects on the offer wage gap at age 50

3. Gender Inequality: Mechanisms

What are the long run implications of fertility and human capital accumulation for gender inequality?

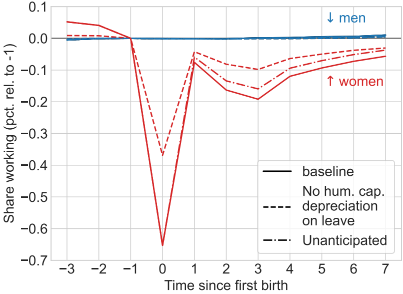
3. Gender Inequality: Mechanisms

What are the long run implications of fertility and human capital accumulation for gender inequality?

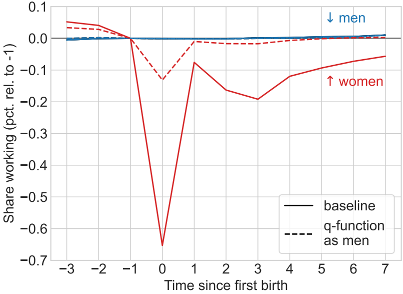
- Investigate these mechanisms through alternative models:

  1. No human capital depreciation when on maternity leave ($\delta = 0$ for women when $o_t = 0$)
  2. No human capital depreciation when on maternity leave but unanticipated
  3. Remove estimated gender differences in the the dis-utility from work when children are present (q-function for women set to that of men).

3. Gender Inequality: Mechanisms (less skilled)



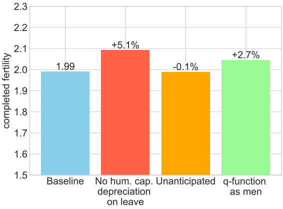
(g) Human capital.



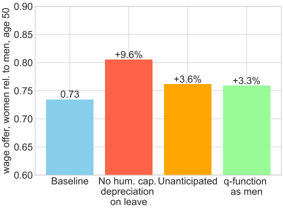
(h) Dis-utility from work.

- **Short run:** Significant short run effect from dis-utility differences

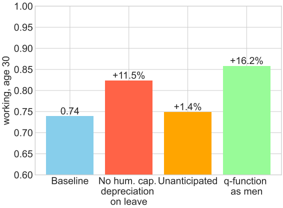
3. Gender Inequality: Mechanisms (less skilled)



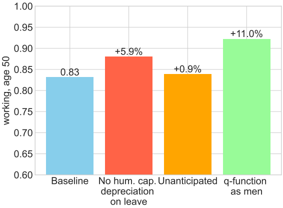
(i) Completed fertility.



(j) Offer wage gap, age 50.



(k) Share working, age 30.



(l) Share working, age 50.

- **Long run:** Substantial effect from human capital depreciation

Conclusions

- **Fertility reacts to financial incentives**
  - Asymmetric response to wages of women and men

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  - Asymmetric response to wages of women and men
- **Labor Supply Responses**
  - Family labor supply important
  - Labor supply for women responds more to wage changes when fertility can also adjust: $>$10% higher
- **Welfare reforms** have permanent effects through fertility even if wage shocks are transitory
  - "Fertility Multiplier"
- **Human capital depreciation** of women around childbirth seems important for long run gender inequality

Conclusions

- **Fertility reacts to financial incentives**
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- **Welfare reforms** have permanent effects through fertility even if wage shocks are transitory
  - "Fertility Multiplier"
- **Human capital depreciation** of women around childbirth seems important for long run gender inequality
- **Our future research:** Take the household even more seriously
  - Limited commitment (Mazzocco, 2007)
  - Likely important for asymmetric fertility effects between women and men

Extra Slides

Definition of partnership

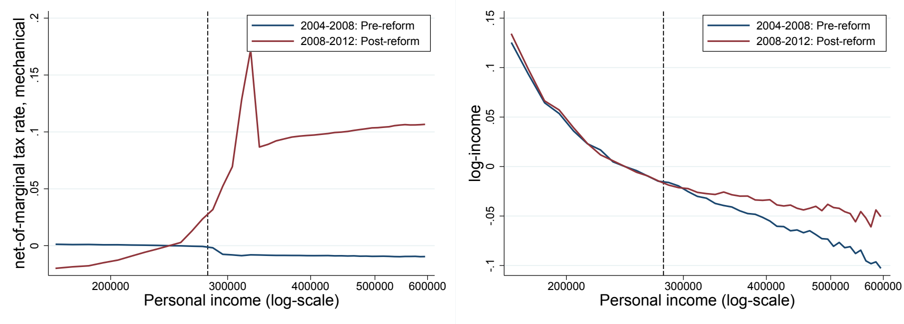
- Official definition of Statistics Denmark.
  https://www.dst.dk/da/Statistik/dokumentation/Times/cpr-oplysninger/familier-og-husstande/familie-type

- Either
  1. Legally married
  2. Living with a person with shared custody over a child
     (share legal address)
  3. Living with one other person of opposite sex with an age difference less than 15.
     (share legal address and both at least 16 years old)

back

Details on Instrument

Figure: Verification: 4-year differences across the income distribution.



(a) Mechanical tax change.

(b) Log income.

Notes: This figure illustrates the tax variation and the plausibility of the variation in generating exogeneous variation.

back

First-stage Results, $\Delta_4 \log(1-\tau_{i,t})$, Women

	(1)	(2)	(3)
$\Delta_4 \tau^m_{i,t}$, women	0.428***	0.426***	0.426***
	(0.002)	(0.002)	(0.002)
$\Delta_4 \log(y^m_{i,t})$, women	0.010***	0.010***	0.010***
	(0.000)	(0.000)	(0.000)
$\Delta_4 \tau^m_{i,t}$, men	0.019***	0.019***	0.019***
	(0.001)	(0.001)	(0.001)
$\Delta_4 \log(y^m_{i,t})$, men	0.028***	0.027***	0.027***
	(0.001)	(0.001)	(0.001)
Income dummies	Yes	Yes	Yes
Children dummies	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes
Hum. cap. controls	No	Yes	Yes
Male partner controls	No	No	Yes
Avg. dep. var. (y, level)			
Obs.	2531181	2531181	2531181
First stage F-stat.			

2SLS Results by Income back

	income $\in$ $[50, 350]$ (1)	income $\in$ $(350, 600]$ (2)	less skilled (3)	high skilled (4)
$\Delta_4 \log(1-\tau_{i,t})$, women	-0.030***	-0.048	-0.048***	-0.019
	(0.010)	(0.038)	(0.015)	(0.013)
$\Delta_4 \log(y_{i,t})$, women	0.005*	0.009	0.002	0.003
	(0.003)	(0.016)	(0.003)	(0.004)
$\Delta_4 \log(1-\tau_{i,t})$, men	0.007	0.004	0.038***	-0.026*
	(0.010)	(0.027)	(0.012)	(0.014)
$\Delta_4 \log(y_{i,t})$, men	0.048***	0.040***	0.000	0.025**
	(0.016)	(0.010)	(0.013)	(0.011)
Income dummies	Yes	Yes	Yes	Yes
Children dummies	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes	Yes
Hum. cap. controls	Yes	Yes	Yes	Yes
Male partner controls	Yes	Yes	Yes	Yes
Avg. dep. var. (y, level)	1.526	1.496	1.664	1.372
Obs.	2205258	325923	1299908	1231273
First stage F-stat.	19869.3	1996.9	11197.1	15910.2

back

2SLS Results: Labor Supply back

	Women (1)	Men (2)
$\Delta_4 \log(1-\tau_{i,t})$, women	0.213***	0.111***
	(0.015)	(0.013)
$\Delta_4 \log(y_{i,t})$, women	-0.016***	0.003
	(0.005)	(0.003)
$\Delta_4 \log(1-\tau_{i,t})$, men	-0.004	0.200***
	(0.015)	(0.014)
$\Delta_4 \log(y_{i,t})$, men	0.006	-0.019
	(0.011)	(0.016)
Income dummies	Yes	Yes
Children dummies	Yes	Yes
Year dummies	Yes	Yes
Age dummies	Yes	Yes
Hum. cap. controls	Yes	Yes
Male partner controls	Yes	Yes
Avg. dep. var. (y, level)	5.454	5.728
Obs.	2316021	2396584

Details on Part Time back

- The part time value of $l_{PT} = 0.75$ is motivated by
  - Statistics Denmark's definition of part time in work experience statistics
  - Close to typical hours in Denmark
    - A normal full-time week is 37 hours in Denmark
    - part time is typically 30 or 32 hours per week ($81\% - 87\%$ of the full-time hours)
- The value affects the human capital accumulation process and the wage/income process
- Utility function is independent of the exact value
- Results are not overly sensitive to this choice.

Details on the Age of Youngest back

- The age of the youngest child aged 0–6, o_t, evolves as

$$o_{t+1} = \begin{cases} 0 & \text{if } b_{t+1} = 1 \\ o_t + 1 & \text{if } b_{t+1} = 0 \text{ and } o_{t+1} \in \{0, 1, 2, 3, 4, 5\} \\ o_t & \text{if } b_{t+1} = 0 \text{ and } o_t \in \{6+\} \\ NC & \text{if } b_{t+1} = 0 \text{ and } o_t \in \{NC\}. \end{cases} \tag{1}$$

Details on the Fertility Process back

- The number of children evolves as

$$n_{t+1} = n_t + b_{t+1}(e_t) - x_{t+1} \tag{2}$$

where x_{t+1} refers to a child moving out, as is given by

$$x_{t+1} = \begin{cases} 1 & \text{with probability } q_t(n_t, o_t) \\ 0 & \text{with probability } 1 - q_t(n_t, o_t) \end{cases} \tag{3}$$

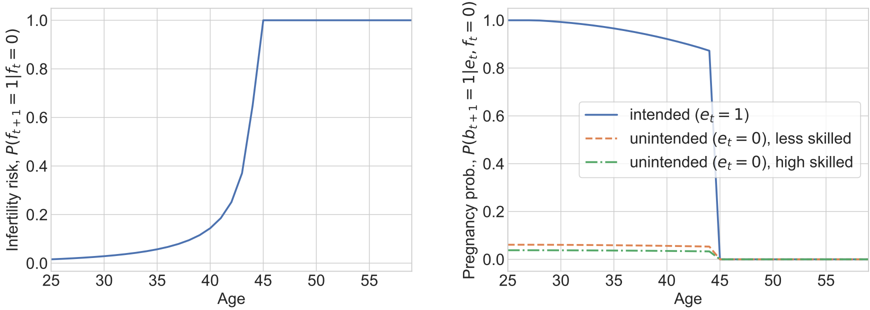
- Children can move out once the fertile period ends at T_f
- x_{t+1} is a realization of a Binomial distribution with

$$q_t(n_t, o_t) = \begin{cases} P_{bin}(1, p_x | n_t - o_t) & \text{if } n_t > 0,\ t > T_f \text{ and } o_t \in \{6+\} \\ 0 & \text{else} \end{cases}$$

where

$$P_{bin}(1, p_x | n) = \frac{n!}{(n-1)!} p_x (1 - p_x)^{n-1}$$

Details on Fertility back



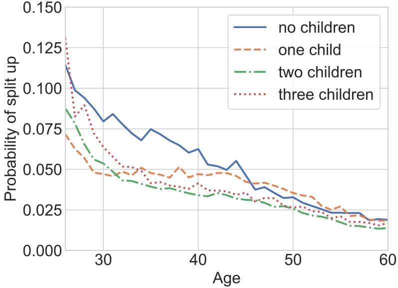
(c) Infertility Probability.

(d) Biological Fecundity, $\overline{\wp}_t$.

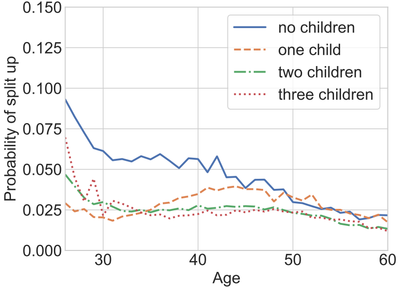
Figure: Biological Fecundity.

Notes: Figure 5 shows in panel (a) the probability of permanent infertility, $P(f_{t+1} = 1|f_t = 0)$, based on Sommer (2016). Panel (b) shows the probability of a pregnancy conditional on being fertile. The likelihood of an intended pregnancy, $P(b_{t+1} = 1|e_t = 1, f_t = 0)$, is calibrated following the biological fecundity in Leridon (2004) together with the infertility probabilities in Sommer (2016). The likelihood of unintended pregnancies,

Details on Divorce back



(a) Less skilled.



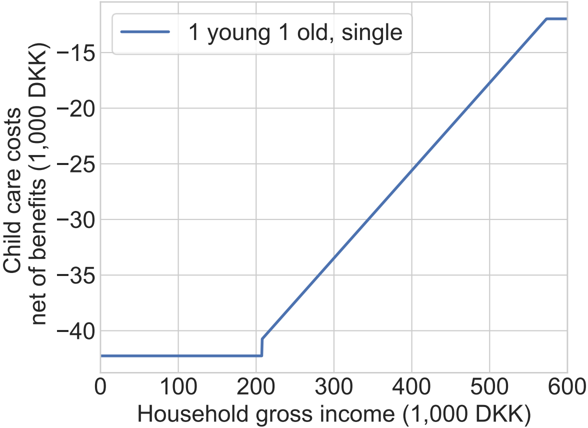
(b) High skilled.

Figure: Dissolution Probabilities.

Notes: Figure 6 shows the likelihood of a couple splitting up in the following period based on the current number of children present in the household, based on Danish register data for less and high skilled couples.

Details on Child Costs and Transfers back

Figure: Costs net of Benefits, $\mathcal{C}(n_t, o_t, Y_t, s_t)$.



Details on Parental Leave back

- Women take all parental leave in year of childbirth
- Income in the year of birth is

$$\tilde{Y}_{w,t}|_{b_t=1} = \begin{cases} \overline{w}_t \cdot \pi + \underline{w}_t \cdot (1-\pi) & \text{if } l_{w,t} = 0 \\ \overline{w}_t \cdot \pi + \tilde{w}_t \cdot (1-\pi) & \text{if } l_{w,t} = l_{PT} \\ \tilde{w}_t & \text{if } l_{w,t} = 1 \end{cases} \tag{4}$$

where $\tilde{w}_t$ is the wage-level associated with full-time work and

$$\overline{w}_t = \overline{\tau}\tilde{w}_t$$

$$\underline{w}_t = \underline{\tau}\tilde{w}_t$$

is the level of parental leave benefits in a period with high replacement rate, $\overline{\tau}$, and low replacement rate $\underline{\tau}$, respectively.

- We set $\pi = 25/52$, $\overline{\tau} = 1$ and estimate $\underline{\tau}$.

Details on Preferences back

- Dis-utility of work is linked to fertility through

$$q_j(\bullet) = \begin{cases} \tilde{\alpha}_{PT,child,j} + \tilde{\alpha}_{PT,more,j}(n_t - 1) + \tilde{\alpha}_{PT,young,j}\mathbf{1}(o_t \leq 3) + \tilde{\alpha}_{PT,work}\mathbf{1}(l_{-j,t} > 0) & \text{if } l_j \\ \tilde{\alpha}_{FT,child,j} + \tilde{\alpha}_{FT,more,j}(n_t - 1) + \tilde{\alpha}_{FT,young,j}\mathbf{1}(o_t o_t \leq 3) + \tilde{\alpha}_{FT,work}\mathbf{1}(l_{-j,t} > 0) & \text{if } l_j \end{cases}$$

where $l_{j,t} = 0$ is the reference alternative.

- The baseline dis-utility of work also varies with age,

$$g_j(\bullet) = \begin{cases} \mu_{PT,j} + \tilde{\mu}_{PT,age,j}(age_{j,t} - 25) & \text{if } l_{j,t} > 0 \\ \mu_{FT,j} + \tilde{\mu}_{FT,age,j}(age_{j,t} - 25) + \tilde{\mu}_{FT,age2,j}(age_{j,t} - 25)^2 & \text{if } l_{j,t} = 1 \end{cases}$$

Parameter Estimates back

Parameter		Less skilled estimate	Less skilled se	High skilled estimate	High skilled se
Utility from children					
ω_1	Value of having at least one child	8.479	(0.016)	-12.675	(0.019)
$\omega_{1,age}$	Value of having at least one child, age.	0.333	(0.001)	1.769	(0.002)
ω_2	Value of having at least two children	14.133	(0.019)	19.102	(0.015)
ω_3	Value of having at least three children	5.663	(0.028)	0.001	(0.025)
η_0	Value of fertility effort when 1st child aged 0	-0.352	(0.007)	-0.450	(0.007)
η_1	Value of fertility effort when 1st child aged 1	-0.023	(0.001)	-0.005	(0.001)
η_2	Value of fertility effort when 1st child aged 2	0.025	(0.001)	0.070	(0.001)
Utility from market work, $g_w(\bullet)$ and $g_m(\bullet)$. Relative to not working.					
$\mu_{PT,w}$	Value of working, women	-0.227	(0.001)	-0.442	(0.001)
$\mu_{PT,age,w}$	Value of working wrt. age, women	-1.826	(0.005)	-1.887	(0.002)
$\mu_{FT,w}$	Additional value of full time work, women	-0.319	(0.000)	-0.395	(0.000)
$\mu_{FT,age^2,w}$	Additional value of full time work wrt. age squared, women	-0.665	(0.002)	-0.865	(0.001)
$\mu_{FT,age,w}$	Additional value of full time work wrt. age, women	-0.040	(0.004)	-1.291	(0.004)
$\mu_{PT,m}$	Value of working, men	-0.392	(0.001)	-0.562	(0.001)
$\mu_{PT,age,m}$	Value of working wrt. age, men	-1.417	(0.004)	-2.057	(0.004)
$\mu_{FT,m}$	Additional value of full time work, men	-0.374	(0.000)	-0.439	(0.000)
$\mu_{FT,age,m}$	Additional value of full time work wrt. age squared, men	-0.706	(0.003)	-0.749	(0.002)
$\mu_{FT,age^2,m}$	Additional value of full time work wrt. age, men	-3.471	(0.016)	-2.471	(0.006)
Utility from market work with children, $q_w(\bullet)$ and $q_m(\bullet)$. Relative to not working.					
$\alpha_{PT,child,w}$	Value of working with children, women	14.712	(0.050)	-0.873	(0.007)
$\alpha_{PT,young,w}$	Value of working with young children, women	3.450	(0.034)	0.150	(0.012)
$\alpha_{PT,more,w}$	Value of working with more children, women	-1.409	(0.010)	9.834	(0.015)
$\alpha_{PT,birth,w}$	Value of working at birth, women	49.787	(0.167)	31.055	(0.080)
$\alpha_{FT,child,w}$	Additional value of full time work with children, women	-0.430	(0.006)	4.345	(0.014)
$\alpha_{FT,young,w}$	Additional value of full time work with young children, women	0.609	(0.016)	0.570	(0.009)
$\alpha_{FT,more,w}$	Additional value of full time work with more children, women	-0.059	(0.009)	0.040	(0.007)
$\alpha_{PT,child,m}$	Value of working with children, men	4.137	(0.024)	3.182	(0.016)
$\alpha_{PT,young,m}$	Value of working with young children, men	0.001	(0.010)	-0.017	(0.015)
$\alpha_{PT,more,m}$	Value of working with more children, men	0.073	(0.011)	0.035	(0.012)
$\alpha_{PT,birth,m}$	Value of working at birth, men	0.000	(0.015)	-0.030	(0.034)
$\alpha_{FT,child,m}$	Additional value of full time work with children, men	4.101	(0.027)	5.629	(0.015)
$\alpha_{FT,young,m}$	Additional value of full time work with young children, men	-1.745	(0.014)	0.254	(0.007)
$\alpha_{FT,more,m}$	Additional value of full time work with more children, men	3.602	(0.020)	0.277	(0.009)
$\alpha_{FT,birth,m}$	Additional value of full time work at birth, men	-1.221	(0.030)	-0.334	(0.013)
$\alpha_{PT,work}$	Value of working with children. Partner working.	0.420	(0.008)	-0.425	(0.006)
$\alpha_{FT,work}$	Value of working full time with children. Partner working.	0.726	(0.008)	-0.045	(0.006)
Wage and human capital process					
$\gamma_{0,w}$	Wage: constant, women	0.563	(0.000)	0.783	(0.001)
$\gamma_{1,w}$	Wage: human capital, women	0.091	(0.000)	0.095	(0.000)
$\gamma_{0,m}$	Wage: constant, men	0.649	(0.001)	0.721	(0.001)
$\gamma_{1,m}$	Wage: human capital, men	0.099	(0.000)	0.110	(0.000)
σ_w	Human capital: shock variance (std), women	0.141	(0.000)	0.158	(0.000)
σ_m	Human capital: shock variance (std), men	0.171	(0.000)	0.183	(0.000)
κ_w	Human capital: initial factor, women	0.280	(0.003)	0.631	(0.003)

Informativeness of Estimation Moments back

- Based on M_4 in Honoré, Jørgensen and de Paula (2020)
- The percentage change in the asymptotic variance of elements of $\hat{\boldsymbol{\theta}}$ from removing groups of moments in $g(\boldsymbol{\theta})$

$$I_k = \text{diag}(\tilde{\Sigma}_k - \Sigma)/\text{diag}(\Sigma) \cdot 100 \tag{5}$$

where

$$\tilde{\Sigma}_k = (G'\tilde{W}_k G)^{-1} G'\tilde{W}_k S \tilde{W}_k G (G'\tilde{W}_k G)^{-1}$$
$$\tilde{W}_k = W \odot (\iota_k \iota_k')$$

and $\odot$ is element-wise multiplication and ι_k is a $J \times 1$ vector with ones in all elements except the kth *group* of moments being zeros.

1. Share working and the share working full time conditional on working, split by age and gender.
2. Average labor income when working, split by age and gender.
3. Share with at least 1, 2 or 3 children, split by age.
4. Distribution of years between first and second childbirths.
5. Share working and share working full time after first and second childbirth, split by gender.
6. Average wealth split by age.

Sensitivity: Change in the Marginal Dis-Utility of Work

back

- Based on the approximation (Jørgensen, 2023)

$$\frac{\partial \hat{\boldsymbol{\theta}}}{\partial \boldsymbol{\phi}'} \approx -(G'WG)^{-1}G'D$$

in which
$G = \frac{\partial g(\hat{\boldsymbol{\theta}}|\boldsymbol{\phi})}{\partial \hat{\boldsymbol{\theta}}'}$
$D = \frac{\partial g(\hat{\boldsymbol{\theta}}|\boldsymbol{\phi})}{\partial \boldsymbol{\phi}'}$

- We calculate

$$\begin{aligned}\frac{d\Delta_j(l,n)}{d\boldsymbol{\phi}'} &= \frac{\partial \Delta_j(l,n)}{\partial \boldsymbol{\theta}'}\frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{\phi}'} \\ &\approx -\frac{\partial \Delta_j(l,n)}{\partial \boldsymbol{\theta}'}(G'WG)^{-1}G'D\end{aligned}$$

and report elasticities

Simulation Details back

- Simulate 500,000 synthetic households from age 25 through 60
- Initialize all households as couples with zero net wealth and the empirical joint distribution of number of children, age of youngest and human capital.
- The effect at age t of a wage increase is

$$\Delta y_t = y_t - \tilde{y}_t$$

where $y_t = n_t^{-1} \sum_i y_{i,t}$ is the average simulated optimal outcome under the baseline estimated model and $\tilde{y}_t^{(s_1:s_2)} = n_t^{-1} \sum_i \tilde{y}_{i,t}^{(s_1:s_2)}$ is the average simulated optimal outcome under the counterfactual setting in which wages are scaled by μ percent in periods s_1 through s_2.

- Formally, wages in the alternative model are given as

$$\tilde{w}_{i,t}^{(s_1:s_2)} = \begin{cases} (1+\mu)w_{i,t} & \text{if } s_1 \leq t \leq s_2 \\ w_{i,t} & \text{else.} \end{cases}$$

Unless otherwise explicitly stated, we use a five percent increase, $\mu = 0.05$.

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DYNAMICS OF WELFARE REFORM

HAMISH LOW

(joint with Costas Meghir, Luigi Pistaferri, Alessandra Voena

May 9th, 2023

DEPARTMENT OF ECONOMICS

UNIVERSITY OF OXFORD

Welfare Support

- Families face different shocks through their life-times - wages, employment, health
- Low-income families insure shocks through the welfare system
  - Different components (US): TANF, EITC, SNAP, (also SSI/DI)
  - In the U.S., focus on single mothers as the primary beneficiaries
- Focus of reform debates: limiting incentive costs
  - Disincentives to work
  - Incentives to be a single parent

1. Marital status and welfare eligibility are closely tied
2. Broader issue of trading off incentive costs against insurance value

1. **Changes in Welfare**
2. Reduced Form Evidence
3. Modeling Welfare Programs
4. Reform

Welfare Reforms in the US since the 1990s

- 10* increase in Earned Income Tax Credit (EITC)
  - Only applied to working families
- Expansion of Medicaid
  - Increased coverage for non-welfare families near poverty line
  - Further increases with Obamacare
- Cash assistance for parents:
  Aid to Families with Dependent Children (AFDC) replaced by
  Temporary Aid to Needy Families (TANF)
  - Work requirements, time limits and other measures to encourage work
  - Substantial declines in take-up (32% of low educated single mothers in 1996)
- Contrast to UK: increase of tax credits, but simultaneous increase of out-of-work payments

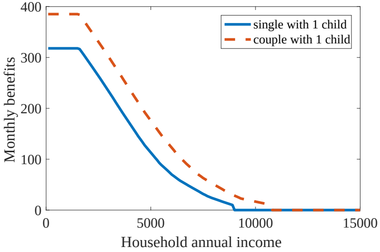
Welfare Reform in US since 1990s

- Multiple reforms at the same time: EITC increase, but also decline in welfare
- Key question: how to think about the dynamic and long run impacts of the welfare system
- More generally: how forward looking are welfare claimants
- Impacts on single parents versus married mothers
  Role of marriage

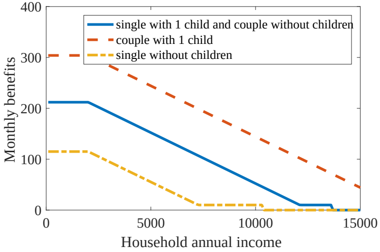
The 1996 Welfare Reform and Time Limits

- PRWORA signed in August 1996
- Shift from welfare entitlement to time limited support
- Childcare support
- Work requirements
- Federal block grants covering benefits for up to 60 months
  - Pre-reform (AFDC): eligible if youngest child under 18
  - Post reform (TANF): federal funding covers 5 years max
- States could impose their own rules: from 21 to 60 months
- State-level variation in timing of adoption (1995-1998)

Welfare generosity and household structure



(a) AFDC



(b) Food stamps

Notes: Average monthly AFDC and food stamps benefits by household annual income

Different Approaches

- Reduced-form evidence
  - Welfare program participation
  - Labour supply
  - Marital status
- Most evaluations study only single mothers and one outcome
- Use life-cycle structural model:
  - Incorporates dynamic incentives to claim and to work
  - Accounts for changes in household formation and dissolution
- Use structural model to simulate welfare reform

1. Changes in Welfare

2. **Reduced Form Evidence**

3. Modeling Welfare Programs

4. Reform

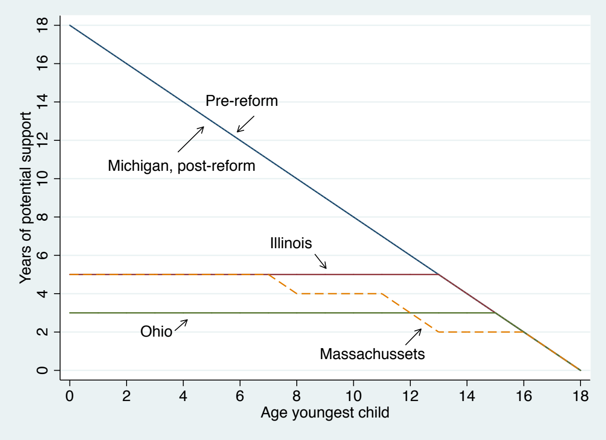
Reduced form evidence on the Effect of Time limits

- On single mothers:
  - Effect on welfare use:
    Grogger (2003)
  - Banking of benefits:
    Grogger and Michalopoulos (2003), Mazzolari and Ragusa (2012)
  - Effect on employment:
    Fang and Keane (2004) Kline and Tartari (2016)
  - Surveys:
    Blank (2002) Grogger and Karoly (2005)
- On married women, marriage and broader impacts:
  Bitler et al. (2004), Moffitt, Phelan, and Winkler (2020) Schoeni and Blank (2000)

Variation for Identification of the Effect of Time Limits

- Identification of the effect of time limits
- Variation in strictness across states
- Variation in the year when implemented
- Importance depends on age of the youngest child: if age over 13, then no impact.

Variation Across States and Age of Child



Sources: Grogger and Michalopoulos (2003), Mazzolari and Ragusa (2012)

Examples

- Youngest child is 10 in year t and the time limit is 5 years:
  $Exposed = 1$
- Youngest child is 13 in year t and the time limit is 5 years:
  $Exposed = 0$
- Youngest child is 13 in year t and the time limit is 2 years:
  $Exposed = 1$
- Youngest child is 17 in year t and the time limit is 2 years:
  $Exposed = 0$

Empirical strategy

Household i, demographic characteristics d, state s, year t:

$$y_{idst} = \alpha Expsd_{dst} Post_{st} + \beta' X_{idst} + f_{st} + f_{ds} + f_s + f_t + f_d + \varepsilon_{idst}$$

- $Exposed = 0$ unaffected households
- $Post = 1$ after the reform
- X controls, f fixed effects
  - Age dummies and Household structure controls
  - EITC and unemployment rate controls
  - Month-by-year fixed effects
  - Year-by-state fixed effects
  - State-by-demographic group fixed effects
- Identification is within state - between demographic groups: unaffected by work requirements

Datasets

- Survey of Income and Program Participation (SIPP)
  - Rolling Panel 1985-2008 (years 1984-2011)
  - Start with 1990 panel (after 1988 FSA)
  - Information on our outcomes of interest
- Current Population Survey
  - March survey
  - Data frame: 1990-2011
- Focus on women who did not complete college

Source: Low, Meghir, Pistaferri, and Voena (2022)

Welfare Utilization and Employment

	AFDC/TANF Utilization					
	Whole sample		Married women		Unmarried women	
	SIPP	CPS	SIPP	CPS	SIPP	CPS
$Exposed_{dst}Post_{st}$	-0.030***	-0.016***	-0.011***	-0.003**	-0.087***	-0.084***
	(0.004)	(0.003)	(0.003)	(0.002)	(0.015)	(0.013)
Mean pre-reform	0.098	0.077	0.035	0.019	0.297	0.304
Obs	254,627	112,128	188,483	88,522	66,144	23,606
R^2	0.12	0.07	0.08	0.03	0.26	0.15
	Employment					
	Whole sample		Married women		Unmarried women	
	SIPP	CPS	SIPP	CPS	SIPP	CPS
$Exposed_{dst}Post_{st}$	0.014	-0.002	-0.001	-0.017	0.050***	0.054**
	(0.012)	(0.011)	(0.014)	(0.011)	(0.014)	(0.026)
Mean pre-reform	0.640	0.647	0.643	0.654	0.631	0.620
Obs	254,627	112,128	188,483	88,522	66,144	23,606
R^2	0.12	0.06	0.11	0.05	0.21	0.13

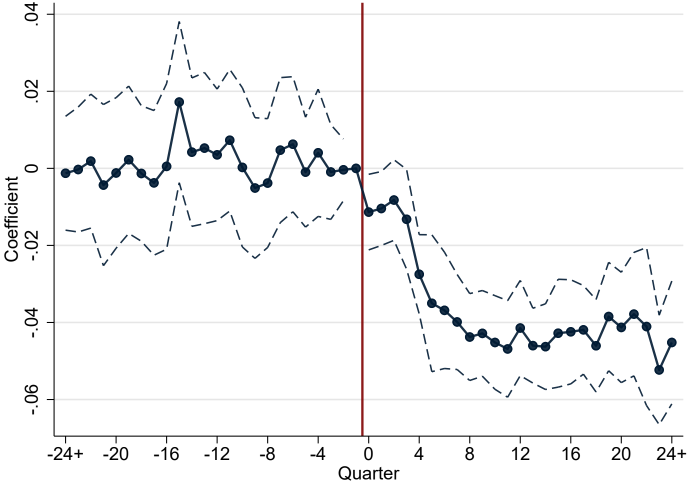
Notes: Standard errors in parentheses clustered at the state level.

Marriage

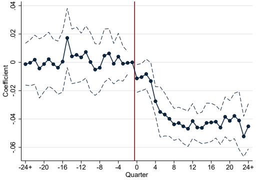
	Divorce/Separated		*Married*		*Newborn*$_{t+1}$ (*Annual*)	
	Whole sample		Whole sample		Whole sample	
	SIPP	CPS	SIPP	CPS	SIPP	CPS
$Exposed_{dst} Post_{st}$	-0.027***	-0.015*	0.004	-0.007	-0.001	-0.002
	(0.007)	(0.008)	(0.007)	(0.010)	(0.005)	(0.004)
Mean pre-reform	0.150	0.126	0.758	0.796	0.059	0.049
Obs	254,627	112,128	254,627	112,128	55,142	47,344
R^2	0.03	0.01	0.05	0.05	0.08	0.04

Notes: Standard errors in parentheses clustered at the state level.

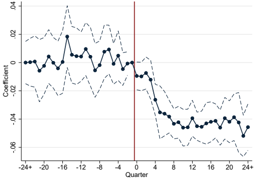
Program Participation



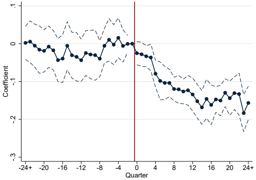
Program Participation and Employment Dynamics



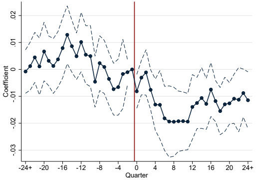
(a) Full sample



(b) Time limits above 24 months



(c) Single



(d) Married

Summary: Reduced Form Evidence

Effect of time limits:

1. Welfare utilization declined
2. Employment increased among single women
3. Decline in divorce
4. No robust effects on marriage (or fertility)

Modeling Welfare Programs

- For single mothers:

  Chan (2013): dynamic discrete choice, large welfare effects

  Ziliak (2015)

  Moffitt, Phelan, and Winkler (2020)

- With marriage:

  Swann (2005)

Banking of Benefits and the role of marriage

- Time limits cause women to bank benefits.
- Banking takes place as precaution: future reduced insurance.
- Family formation and asset accumulation crucial to understand the insurance channel.
- Shutting down marriage makes the prospects for singles worse because of possible source of insurance disappears.
- Shutting down divorce makes married mothers better insured but makes marriage harder.

Life-cycle Model of Marriage and Welfare

Low, Meghir, Pistaferri, and Voena (2022)

- Develop and estimate a dynamic model:
  - Incorporates dynamic incentives
  - Accounts for household formation and dissolution
- Use model to understand how marriage and divorce interact with the social safety net
- Need dynamics to understand the effects of time limits

Related literature

- Collective model and dynamic household decision making
  - Chiappori (1992), Blundell, Chiappori, and Meghir (2005), Mazzocco (2007)
  - Voena (2015), Fernández and Wong (2014)
- Dynamic models of labor supply
  - Keane and Wolpin (2010)
    Low, Meghir, and Pistaferri (2010)
    Blundell et al. (2016)

The model

- Life cycle setup
- Choices
  - Marriage and divorce
  - Participation in AFDC/TANF
  - Female labor supply
  - Consumption and savings
- Resources
  - AFDC/TANF, Food stamps, EITC
  - Stochastic wages
- Markets
  - Imperfect capital markets
  - Marriage market
  - Stochastic fertility

Welfare benefits and time limits

- Before welfare reform

$$b_t(k_t, w_t^W P_t^W, m_t y_t^M, A_t)$$

- After welfare reform

$$b_t(\cdot, TB_t) = \begin{cases} = 0 & if \;\; TB_t > \text{time limit} \\ = b_t(\cdot) & if \;\; TB_t \leq \text{time limit} \end{cases}$$

- Additional exogenous programs: food stamps, EITC

The problem of a single woman

- Chooses consumption, work and welfare participation
- Evaluates marital offers if they arrive (with prob. λ_t)
- May or may not have children

$$V_t^{Ws} = max \ \Big\{ u(c_t^{Ws}, P_t^{Ws}, B_t^{Ws})$$
$$+\beta E_t[\lambda_{t+1}[(1-m_{t+1})V_{t+1}^{Ws} + m_{t+1}V_{t+1}^{Wm}] + (1-\lambda_{t+1})V_{t+1}^{Ws}]\Big\}$$

$s.t.$

$$\frac{A_{t+1}^{Ws}}{1+r} = A_t^{Ws} - \frac{c_t^{Ws}}{e(k_t)} + (w_t^{Ws} - CC^a)P_t^{Ws} + B_t^{Ws}b_t + G_t^{Ws}$$

The problem of a couple

- Chooses consumption, work and welfare participation
- Collective decision with limited commitment
- Anticipates possible future divorce

$$V_t^m = max \ \Big\{\theta_t^W \ u(c_t^{Wm}, P_t^{Wm}, B_t^m) + \theta_t^M \ \ u(c_t^{Mm}, P_t^{Mm}) + L^\tau \ [1.em]$$
$$+\beta E_t \left[(1-d_{t+1})V_{t+1}^m + d_{t+1}\left(\theta_t^W V_{t+1}^{Ws} + \theta_t^M V_{t+1}^{Ms}\right)\right]\Big\}$$

$s.t.$

$$\frac{A_{t+1}}{1+r} = A_t - \frac{F(c_t^{Wm}, c_t^{Mm})}{e(k_t)} + (w_t^{Ws} - CC^a)P_t^{Wm} + y_t^M + B_t^m b_t + G_t$$

Marriage decision

- Singles meet a potential match with probability λ_t
  - Draw from singles' empirical distribution of $\{A_t^j, y_t^j, [k_t, TB_t]\}$
  - Draw match quality L^0
- Marriage decision
  - Get married $(m_t = 1)$ iff

$$\exists \ feasible \ \ \boldsymbol{\theta_t} \ \ s.t. \ \ V_t^{jm}(\theta_t^j) \geq V_t^{js} \quad \text{for} \ \ j = H, W$$

  - $\boldsymbol{\theta_t}$ at the time of marriage equates gains from marriage

Divorce decision

- Uncertainty:
  match quality L^{τ}
  spouses' income
- Re-allocation of resources in the marriage
  - Limited commitment (see Mazzocco 2007, Voena 2015)
- Divorce $(d_t = 1)$ iff

$$\nexists \; feasible \;\; \boldsymbol{\theta_t} \;\; s.t. \;\; V_t^{jM}(\theta_t^j) \geq V_t^{jS} \quad \text{for} \;\; j = H, W$$

Fertility, Match quality, and Earnings

- Fertility process
  - $P(newborn_t | k_t^a) = g(t, m_{t-1})$
- Match quality
  - Initial quality L^0
  - After τ years of marriage: $L^\tau = L^{\tau-1} + \xi^\tau$
- Earnings process

$$y_{it}^M \in \{0, w_{it}^M\}$$
$$ln(w_{it}^j) = a_0^j + a_1^j age_t^j + a_2^j \cdot (age_t^j)^2 + z_{it}^j$$
$$z_{it}^j = z_{i,t-1}^j + \zeta_{it}^j$$
$$j \in \{F, M\}$$

Estimation steps

- Estimation on pre-reform data - Unaffected by other changes such as work requirements
- Parameters
  1. Fix a set of parameters exogenously
  2. Estimate directly from data
     - ★ Wage and employment processes; Fertility process
     - ★ Singles' types distribution for men and women
  3. Remaining parameters by MSM targeting pre-reform moments
     - ★ Cost of working; Probability of meeting a partner
     - ★ Distribution of match quality
     - ★ Cost of welfare participation (stigma)
- Moments
  - Draw 4-year simulated panels as in SIPP data
  - Match pre-reform moments from 1960s birth cohort of SIPP

Pre-set parameters of the model and initial conditions

Parameter	Value/source
Panel A - Parameters fixed from other sources	
Relative risk aversion (γ)	1.5
Discount factor (β)	0.98
Childcare costs (CC^a)	CEX
Economies of scale in marriage (ρ)	1.23

Wage offer parameters

Parameter	Value
Panel A - Men	
Variance of fixed effect (earnings in period 1)	0.18
Variance of earnings shocks	0.027
Life cycle profile of log earnings (a_0^M, a_1^M, a_2^M)	9.76, 0.043, -0.001
Panel B - Women	
Variance of fixed effect (earnings in period 1)	0.15
Variance of earnings shocks	0.038
Life cycle profile of log earnings (a_0^W, a_1^W, a_2^W)	1.96, 0.022, -0.0003

Estimated singles' distributions

- Joint distributions of assets and productivity among singles
- Allow mass on zero assets
- Conditionally, $\{\ln(A_t^M), \ln(y_t^M)\} \sim BVN(\boldsymbol{\mu}_t^M, \boldsymbol{\Sigma}_t^M)$
- Conditionally, $\{\ln(A_t^W), \ln(w_t^W)\} \sim BVN(\boldsymbol{\mu}_{ta}^W, \boldsymbol{\Sigma}_{ta}^W)$
    - Include selection correction on women's wages

Model Parametrization

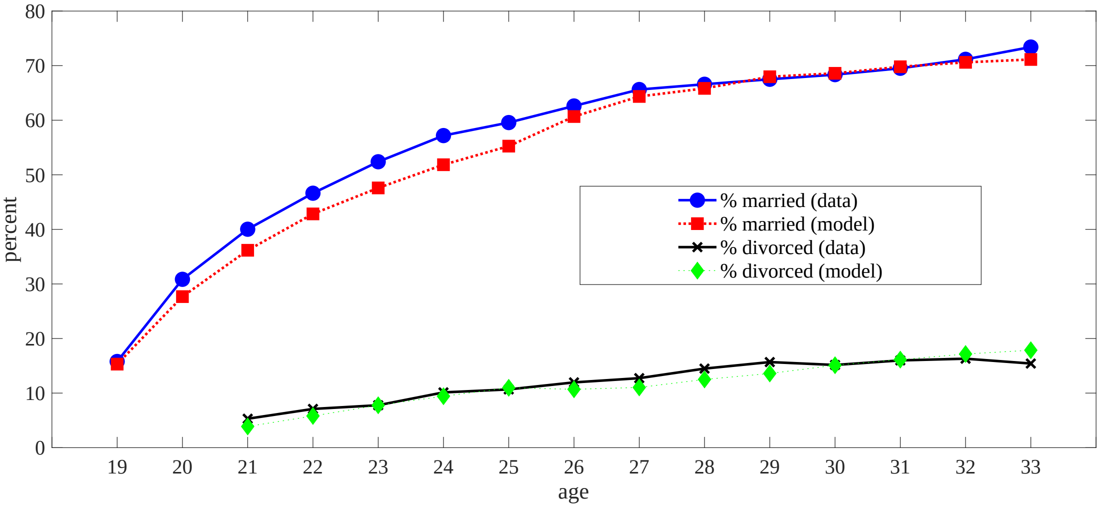
- Preferences: $u(c, P, B) = \frac{\left(c \cdot e^{\psi(M,k^a) \cdot P}\right)^{1-\gamma}}{1-\gamma} - \eta B$

- Meeting probabilities:

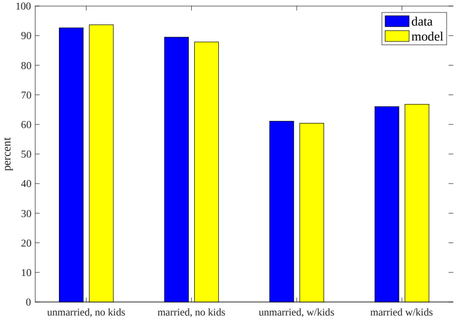
$$\lambda_t = min\{max\{\lambda_0 + \lambda_1 t + \lambda_2 t^2, 0\}, 1\}$$

- Match quality after τ years of marriage: $L^\tau = L^{\tau-1} + \xi^\tau$
  - $\xi_\tau \sim N(0, \sigma_\xi)$
  - $L^0 \sim N(0, \sigma_0)$

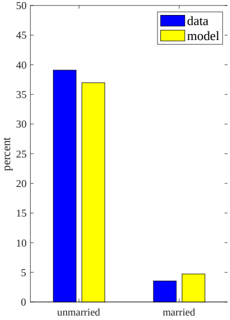
Target moments: Marital status



Target moments: Employment and Welfare Participation



(a) Employment moments



(b) Welfare moments

Parameter estimates

Table: Parameters Estimated by Method of Simulated Moments

Parameter		Estimate	(s.e.)
Cost of work			
Unmarried, no children	$\exp\{\psi^{s0}\}$	0.338	(0.009)
Married, no children	$\exp\{\psi^{m0}\}$	0.584	(0.006)
Unmarried, with child	$\exp\{\psi^{s1}\}$	0.433	(0.014)
Married, with child	$\exp\{\psi^{m1}\}$	0.476	(0.007)
Cost of being on AFDC	η	0.003	(0.0002)
Match quality			
Variance at marriage	σ_0^2	0.097	(0.002)
Variance of innovations	σ_ξ^2	0.031	(0.002)

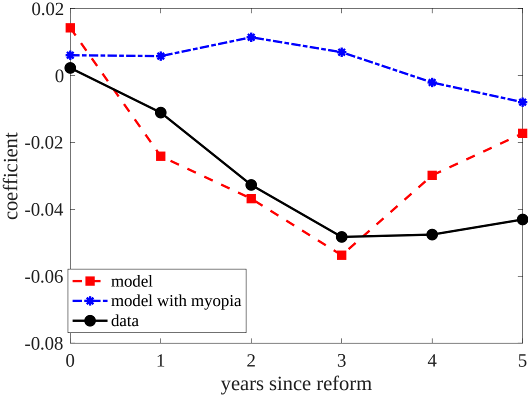
1. Changes in Welfare
2. Reduced Form Evidence
3. Modeling Welfare Programs
4. **Reform**

Simulate the Welfare Reform

Simulate the introduction of time limits, holding other features fixed

- Two issues:
  1. How forward looking is behaviour
  2. The importance of marriage and divorce in understanding behaviour
- Simulate the transition following welfare reform
  - Forward looking behaviour: dynamics of welfare banking
  - Validation: compare to the difference-in-differences estimates
- Long-term steady-state: heterogeneity of effect across the productivity distribution

Dynamic response of welfare utilization to time limits for mothers

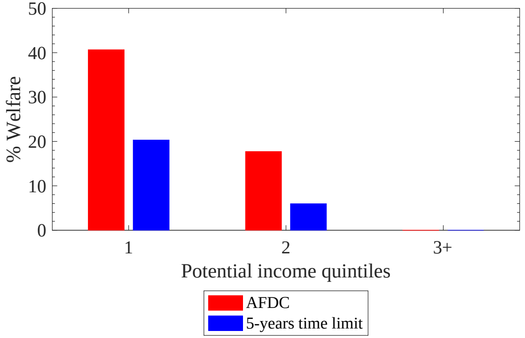


Notes: By *Model with myopia* we mean individuals who behave as if the introduction of time limits had not occurred (until they actually run out of benefits), but are forward looking in terms of other behavior.

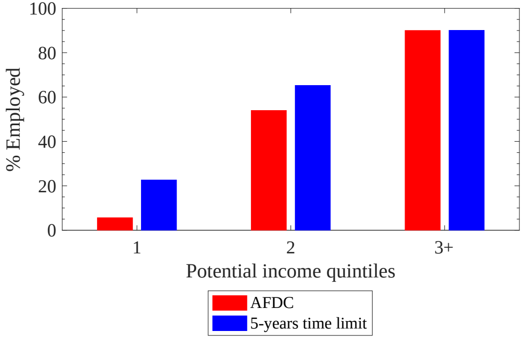
Difference-in-Differences Estimates: Simulated and CPS Data

	(1) Benefits Unmarr.	(2) Benefits Married	(3) Employed Unmarr.	(4) Employed Married	(5) Divorced
Coeff. - SIPP data	-0.093 [-0.131,-0.055]	-0.011 [-0.017,-0.004]	0.066 [0.039,0.093]	0.004 [-0.023,0.030]	-0.021 [-0.041,-0.001]
Baseline:					
Effect of Time limits	-0.078	-0.010	0.072	-0.018	-0.011
Alternatives:					
Time limits in no marriage model	-0.168	-	0.158	-	-
Time limits in no divorce model	-0.153	0.002	0.143	-0.011	-
Time limits in no assets model	-0.117	-0.006	0.117	-0.012	-0.012

Long-term effects of time limits



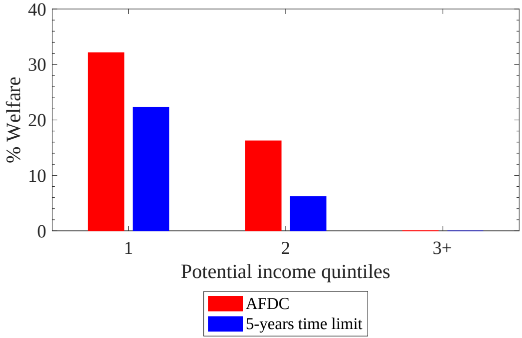
(a) Welfare use



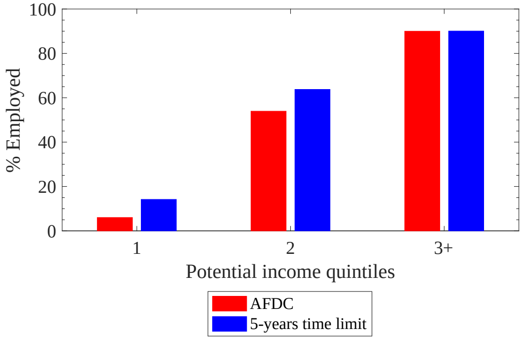
(b) Employment

Notes: Percentage of mothers on welfare (a) and working (b) by policy regime, by age-specific quintiles.

Anticipated Welfare Use and Employment

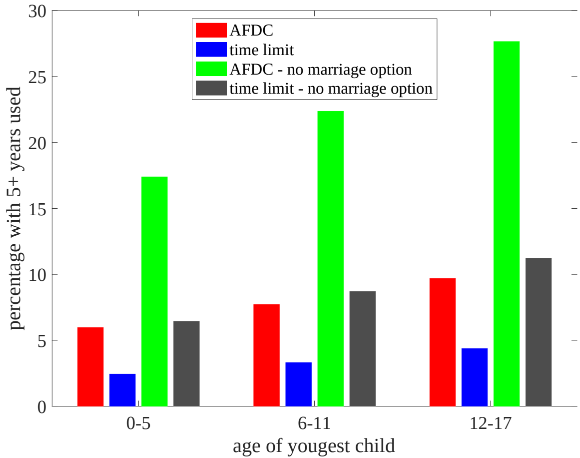


(c) Welfare use - not run out of benefits



(d) Employment - not run out of benefits

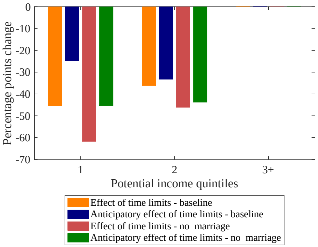
Long-term distribution of lifetime welfare utilization



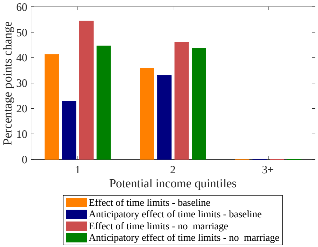
Long term effects of time limits on mothers

Role of the marriage option

Figure: Welfare Use and Employment of Unmarried Mothers with and without Marriage Transitions



(a) Effect of time limit on welfare - decomposed

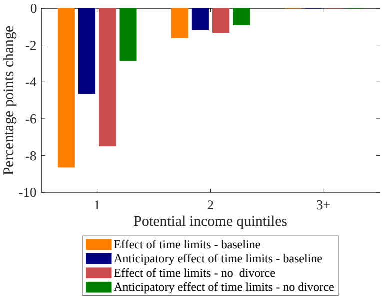


(b) Effect of time limit on employment - decomposed

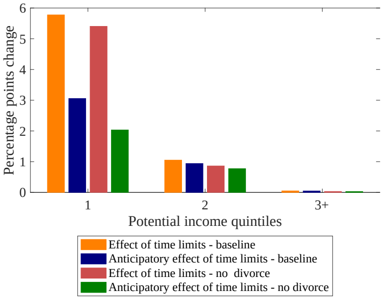
- No marriage: less insurance, more cuts in welfare use in anticipation of limits

Long term effects of time limits on mothers

Role of the divorce option



(a) Welfare use



(b) Employment

- No divorce: more insurance, less cuts in welfare use in anticipation of limits

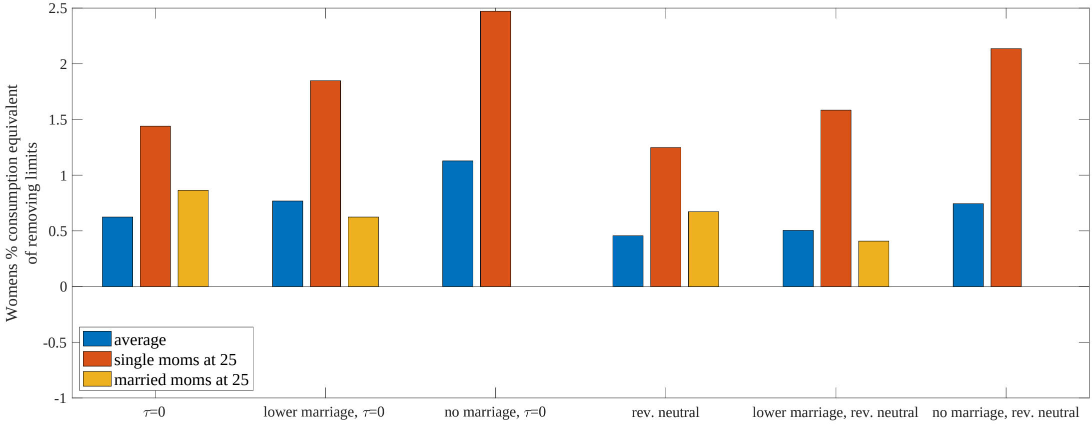
Computing the welfare effects of the reform

- Consider revenue neutral policies
  - Compute government saving from time limits
  - Return the government saving to households
    - **A.** As a negative payroll tax to women
    - **B.** As a negative payroll tax to women and men
- Compute the consumption equivalent
  - % of lifetime consumption that makes agents indifferent between having time limits or not

$$E_0\left[U\left(s,\tau\right)\right]|_{\pi} = \frac{1}{N}\sum_{i=1}^{N}\sum_{t=0}^{T-R}\beta^t\left(\frac{\left(\left(1-\pi^s\right)c_{i,t}^s\cdot e^{\psi\left(m_{i,t},k_{i,t}^a\right)\cdot P_{i,t}^s}\right)^{1-\gamma}}{1-\gamma}-\eta B_{i,t}^s+L_{i,t}m_{i,t}^s\right)$$

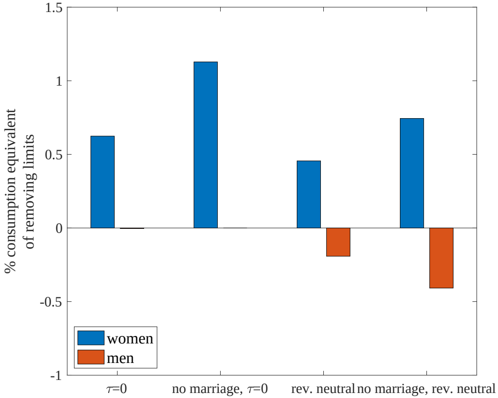
Consumption Equivalents

Figure: Lifetime Utility Costs Of Time Limits on Women



Consumption Equivalents - Women v Men

Figure: Lifetime Utility Costs Of Time Limits on Women and Men



Conclusions

- Reduced form evidence of time limits:
  - fall in welfare use
  - rise in employment
  - fall in divorce
- Joint modelling of decision over working, claiming benefits, marriage and divorce
  1. Forward looking behaviour: stop claiming in anticipation of time limit
  2. Stay married because outside option is worse
  3. Marriage and option of marriage provides insurance.
     Time limits larger reduction in claiming if focus on always single
  4. Time limits affect the lowest quintile of productivity the most
- Satisfies the "objectives" of the policy, but key groups worse off

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Children and Time Allocation

Thomas H. Jørgensen

2025

Plan for today

- Blundell, Pistaferri and Saporta-Eksten (2018): "Children, Time Allocation and Consumption Insurance"
  - Unitary model - Combines several US data sources.

Plan for today

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  - Unitary model - Combines several US data sources.

- **Reading guide:**
  1. What are the main *research questions*?

  2. What is the *(empirical) motivation*?

  3. What are the central *mechanisms in the model*?

  4. What is the *simplest model* in which we could capture these?

Plan for today

- Blundell, Pistaferri and Saporta-Eksten (2018): "Children, Time Allocation and Consumption Insurance"
  - Unitary model - Combines several US data sources.

- **Reading guide:**
  1. What are the main *research questions*?
     - How do couples allocate time and consumption when having children?
     - How do children affect couples' abilities to smooth consumption?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

Empirical Motivation: Siminski and Yetsenga (2022)

- Australian time-use data on panel of couples!

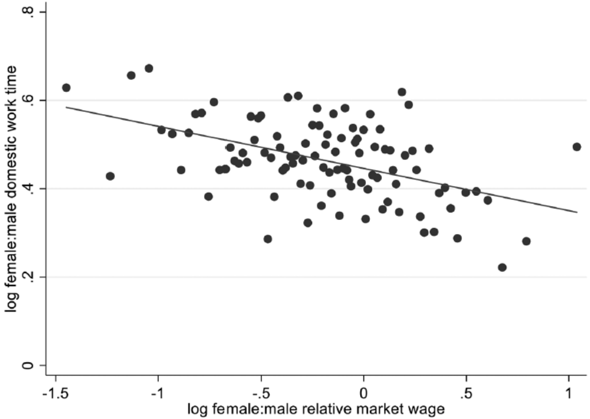

FIG. 1.—Relative domestic work time by relative wage. Each point represents 1 percentile of the female-to-male relative wage distribution among heterosexual couples. A color version of this figure is available online.

Outline

1. Model and Mechanisms

2. Estimation
   - Data
   - First Step: MRS
   - Second Step: SMM

3. Simulations

Model Overview

- **Write** out the *recursive formulation* of the model
  States
  Choices
  (transitions)

Model Overview

- **Choices:**
  $H_{j,t}$: work hours, $j \in \{1, 2\}$ (2=woman)
  $L_{j,t}$: leisure hours, $j \in \{1, 2\}$
  $T_{j,t}$: Parenting hours, $j \in \{1, 2\}$ (child care)
  C_t: Household consumption
- **States:**
  A_t: wealth
  $F_{j,t}$: permanent income shock, $j \in \{1, 2\}$
  $u_{j,t}$: transitory income shock, $j \in \{1, 2\}$
  ε: vector of 5 unobserved time-fixed taste-shifters.
  (only allow for ε_{L_2},wife's leisure, using two-point, fnt 27)
  z_t: child (50/50 prob. at age 28, young for 10years)

State Transitions

- **Budget**

$$A_{t+1} = (1+r)[A_t + \mathcal{T}(z_t, H_{1,t}W_{1,t} + H_{2,t}W_{2,t}) - C_t]$$

where joint taxation gives

$$\mathcal{T}(z_t, H_{1,t}W_{1,t} + H_{2,t}W_{2,t}) = \chi_t(b(z_t) + H_{1,t}W_{1,t} + H_{2,t}W_{2,t})^{1-\mu_t}$$

with $b(z_t)$ being a consumption floor.

- **Hours worked** are

$$H_{j,t} = \overline{L} - L_{j,t} - T_{j,t}$$

- **Wages** are

$$\log W_{j,t} = x'_{j,t}\beta^j_W + F_{j,t} + u_{j,t}$$
$$F_{j,t} = F_{j,t-1} + v_{j,t}$$

Preferences

- **Utility** is

$$\exp(\tilde{\phi}_C(z_t,\varepsilon_t))\frac{[C_t-\gamma(z_t)\mathbf{1}(H_{2,t}>0)]^{1-1/\eta}}{1-1/\eta}$$
$$-\frac{1}{1-\rho_L}\left[\exp(\tilde{\phi}_{L_1}(z_t,\varepsilon_t))L_{1,t}^{1-1/\varphi_{L_1}}+\exp(\tilde{\phi}_{L_2}(z_t,\varepsilon_t))L_{2,t}^{1-1/\varphi_{L_2}}\right]^{1-\rho_L}$$
$$-\frac{1}{1-\rho_T}\left[\exp(\tilde{\phi}_{T_1}(z_t,\varepsilon_t))T_{1,t}^{1-1/\varphi_{T_1}}+\exp(\tilde{\phi}_{T_2}(z_t,\varepsilon_t))T_{2,t}^{1-1/\varphi_{T_2}}\right]^{1-\rho_T}$$

where, for $x\in\{C,L_1,L_2,T_1,T_2\}$,

$$\tilde{\phi}_x(z_t,\varepsilon_t)=\phi_x^{nk}+\phi_x^k z_t+\varepsilon_{x,t}$$

are taste-shifters.
(only $var(\varepsilon_{L_{2},t})>0$ so irrelevant in all other)

Preferences

- **Utility** is

$$\exp(\tilde{\phi}_C(z_t, \varepsilon_t)) \frac{[C_t - \gamma(z_t)\mathbf{1}(H_{2,t} > 0)]^{1-1/\eta}}{1-1/\eta}$$
$$- \frac{1}{1-\rho_L}\left[\exp(\tilde{\phi}_{L_1}(z_t, \varepsilon_t)) L_{1,t}^{1-1/\varphi_{L_1}} + \exp(\tilde{\phi}_{L_2}(z_t, \varepsilon_t)) L_{2,t}^{1-1/\varphi_{L_2}}\right]^{1-\rho_L}$$
$$- \frac{1}{1-\rho_T}\left[\exp(\tilde{\phi}_{T_1}(z_t, \varepsilon_t)) T_{1,t}^{1-1/\varphi_{T_1}} + \exp(\tilde{\phi}_{T_2}(z_t, \varepsilon_t)) T_{2,t}^{1-1/\varphi_{T_2}}\right]^{1-\rho_T}$$

where
$\eta > 0$ is the consumption Frisch elasticity ($1/\eta$ is the CRRA)
$\gamma(z_t)$ is cost of work (for women)
$\varphi_x \in (0, 1)$ is the curvature wrt x. (Governs how sensitive x is to e.g. wage changes.)
Note the negative sign and that $1 - 1/\varphi_x < 0$

Preferences

- **Utility** is

$$
\begin{aligned}
&\exp(\tilde{\phi}_C(z_t,\varepsilon_t))\frac{[C_t-\gamma(z_t)\mathbf{1}(H_{2,t}>0)]^{1-1/\eta}}{1-1/\eta} \\
&-\frac{1}{1-\rho_L}\left[\exp(\tilde{\phi}_{L_1}(z_t,\varepsilon_t))L_{1,t}^{1-1/\varphi_{L_1}}+\exp(\tilde{\phi}_{L_2}(z_t,\varepsilon_t))L_{2,t}^{1-1/\varphi_{L_2}}\right]^{1-\rho_L} \\
&-\frac{1}{1-\rho_T}\left[\exp(\tilde{\phi}_{T_1}(z_t,\varepsilon_t))T_{1,t}^{1-1/\varphi_{T_1}}+\exp(\tilde{\phi}_{T_2}(z_t,\varepsilon_t))T_{2,t}^{1-1/\varphi_{T_2}}\right]^{1-\rho_T}
\end{aligned}
$$

  where they claim that
  $\rho_x < 1$ is the complementarity ($\rho_x > 0$) / substitutability ($\rho_x < 0$) between men and women

- This is not true, I think. See note.
  If $\varphi_{x_1} = \varphi_{x_2} = \varphi_x$, then that parameter is the *elasticity of substitution*.
  $1-\rho_x$ controls the *returns to scale.*

Preferences

- **Utility** is

$$\exp(\tilde{\phi}_C(z_t,\varepsilon_t))\frac{[C_t-\gamma(z_t)\mathbf{1}(H_{2,t}>0)]^{1-1/\eta}}{1-1/\eta}$$
$$-\frac{1}{1-\rho_L}\left[\exp(\tilde{\phi}_{L_1}(z_t,\varepsilon_t))L_{1,t}^{1-1/\varphi_{L_1}}+\exp(\tilde{\phi}_{L_2}(z_t,\varepsilon_t))L_{2,t}^{1-1/\varphi_{L_2}}\right]^{1-\rho_L}$$
$$-\frac{1}{1-\rho_T}\left[\exp(\tilde{\phi}_{T_1}(z_t,\varepsilon_t))\,T_{1,t}^{1-1/\varphi_{T_1}}+\exp(\tilde{\phi}_{T_2}(z_t,\varepsilon_t))\,T_{2,t}^{1-1/\varphi_{T_2}}\right]^{1-\rho_T}$$

where
interpreting the last part as "home production of children"
$\rightarrow$ relative weight on j is their absolute advantage in child production
$\rightarrow$ if $\tilde{\phi}_{T_2}(z_t,\varepsilon_t)>\tilde{\phi}_{T_1}(z_t,\varepsilon_t)$ mothers have an absolute advantage

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- **Zero-step calibration** of some parameters (table 2)

Estimation and Identification Strategy

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- **First-step estimation** of some parameters using MRS conditions using log-linear approximations, without solving the model

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- **First-step estimation** of some parameters using MRS conditions using log-linear approximations, without solving the model
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- **First-step estimation** of some parameters using MRS conditions using log-linear approximations, without solving the model
- **Second-step SMD/SMM estimation** of some parameters using solution/simulation from model.

- I will spend some time on the first-step estimation to give some detail. It takes up a big part of the paper

Estimation and Identification Strategy

- **Zero-step calibration** of some parameters (table 2)
- **First-step estimation** of some parameters using MRS conditions using log-linear approximations, without solving the model
- **Second-step SMD/SMM estimation** of some parameters using solution/simulation from model.

- I will spend some time on the first-step estimation to give some detail. It takes up a big part of the paper
- Illustrates the amount of hoops one could be willing to jump to reduce the parameter space in the SMD...

Data Sources

- **Panel Study of Income Dynamics (PSID)**
  labor income, and hours worked, $H_{j,t}$, $\rightarrow w_{j,t}$
  Non-durable consumption, c_t, and assets, A_t.

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  Time used for leisure, $L_{j,t}$, and child care, $T_{j,t}$
  Only for one respondent (not both partners)
  $\rightarrow$ Use responses of women and *impute* values for their partners:

$$X_{1,t} = f(cohort_1, educ_1), \; X \in \{L, T\}$$

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- **Consumer Expenditure Survey (CEX)**
  Non-durable consumption, c_t.
  (better quality than PSID)

MRS: 1

- **MRS between wife's and husband's leisure** can be found to give *[note: nothing about ρ_L!]*

$$L_2 = \left[\frac{(1-1/\varphi_{L_1})\exp(\tilde{\phi}_{L_1})}{(1-1/\varphi_{L_2})\exp(\tilde{\phi}_{L_2})}\right]^{-\varphi_{L_2}} \left[\frac{W_2}{W_1}\right]^{-\varphi_{L_2}} L_1^{\varphi_{L_2}/\varphi_{L_1}}.$$

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- Taking logs gives ($x = \log(X)$)

$$\begin{aligned} l_2 = &-\varphi_{L_2}[\log(1-1/\varphi_{L_1}) + \tilde{\phi}_{L_1} - \log(1-1/\varphi_{L_1}) - \tilde{\phi}_{L_2}] \\ &-\varphi_{L_2}(w_2 - w_1) + \frac{\varphi_{L_2}}{\varphi_{L_1}} l_1 \end{aligned}$$

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- We can write this as

$$l_2 = K_0 + \varphi_{L_2}(w_1 - w_2) + \frac{\varphi_{L_2}}{\varphi_{L_1}} l_1 + \varphi_{L_2}(\varepsilon_1 - \varepsilon_2)$$

where $\varepsilon_1 - \varepsilon_2$ is unobserved (random) and the constant is

$$K_0 = -\varphi_{L_2}[\log(1-1/\varphi_{L_1}) - \log(1-1/\varphi_{L_1}) + \phi_{L_1}^{nk} + \phi_{L_1}^{k} z - \phi_{L_2}^{nk} - \phi_{L_2}^{k} z]$$

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- **MRS between wife's and husband's leisure** (e.q. 7, $x = \log(X)$)

$$\mathbb{E}[l_{2,t} - K_0 - \varphi_{L_2}(w_{1,t} - w_{2,t}) - \frac{\varphi_{L_2}}{\varphi_{L_1}} l_{1,t} | I_t] = 0$$

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  can give *three* moments to identify K_0, φ_{L_2} and φ_{L_1}.
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MRS: 2

- **MRS between wife's leisure and consumption** (e.q. 8)

$$\mathbb{E}[l_{2,t} - K_1 + \varphi_{L_2} w_{2,t} - \mu \varphi_{L_2} y - \frac{\varphi_{L_2}}{\eta} c_t - \frac{\varphi_{L_2}}{\varphi_{L_1}} \rho_L (1 - \varphi_{L_1}) l_{1,t}$$
$$+ \varphi_{L_2} \rho_L \frac{\varphi_{L_2}(1 - \varphi_{L_2})}{\varphi_{L_1}(1 - \varphi_{L_1})} \frac{W_{2,t} L_{2,t}}{W_{1,t} L_{1,t}} | I_t] = 0$$

where μ is "known" tax parameter and y is household income. Can likewise give *three* moments to identify K_1, η and ρ_L.

MRS: 2

- **MRS between wife's leisure and consumption** (e.q. 8)

$$\mathbb{E}[l_{2,t} - K_1 + \varphi_{L_2} w_{2,t} - \mu \varphi_{L_2} y - \frac{\varphi_{L_2}}{\eta} c_t - \frac{\varphi_{L_2}}{\varphi_{L_1}} \rho_L (1 - \varphi_{L_1}) l_{1,t}$$
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- **MRS between wife's and husband's parental time** (e.q. 9)

$$\mathbb{E}[t_{2,t} - K_2 - \varphi_{T_2}(w_{1,t} - w_{2,t}) - \frac{\varphi_{T_2}}{\varphi_{T_1}} t_{1,t} | I_t] = 0$$

- **MRS between wife's parental time and consumption** (e.q. 10)

$$\mathbb{E}[t_{2,t} - K_3 + \varphi_{T_2} w_{2,t} - \mu\varphi_{T_2} y - \frac{\varphi_{T_2}}{\eta} c_t - \frac{\varphi_{T_2}}{\varphi_{T_1}} \rho_T (1 - \varphi_{T_1}) t_{1,t}$$

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- Similarly for the time-use of men (as discussed above)

Parameter Estimates

TABLE 3
PARAMETER ESTIMATES

	A. MRS Estimates		
	Leisure and Consumption (1)		Parental Time (2)
φ_{L_1}	.211	φ_{T_1}	.115
	(.037)		(.081)
φ_{L_2}	.162	φ_{T_2}	.503
	(.025)		(.201)
ρ_L	.535	ρ_T	−.197
	(.099)		(.123)
η	.903		
	(.049)		
Observations	11,195		2,901

	B. Preference Shifters	
	With Children	Without Children
ϕ_{L_1}	−8.925	−7.680
	(1.108)	(1.013)
ϕ_{L_2}	−9.397	−8.816
	(1.036)	(1.024)
ϕ_{T_1}	−23.993	N/A
	(10.245)	
ϕ_{T_2}	−3.957	N/A
	(1.201)	
$\sigma^2_{\varepsilon_{L_2}}$	1.476	.700
	(.174)	(.087)
γ	(see table 2)	4,794
		(438)
ϕ_C	.132	Normalized to 0
	(.024)	

Parameter Estimates

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leisure does not respond alot to wage-changes

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Child-care time of mothers reponds a bit to wages

B. PREFERENCE SHIFTERS

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decreasing returns to scale
(them: lesure complements)

	B. Preference Shifters	
	With Children	Without Children
ϕ_{L_1}	−8.925 (1.108)	−7.680 (1.013)
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increasing returns to scale
(them: substitutes)

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children decrease the value of leisure

Parameter Estimates

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women have a large abs. advantage in child-care

Parameter Estimates

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random pref. shocks have more variance when children are present

Parameter Estimates

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ϕ_C	$\boxed{.132}$	Normalized to 0
	$\boxed{(.024)}$	

marg. util. of cons. is 13% higher when children present

Outline

1. Model and Mechanisms
2. Estimation
   - Data
   - First Step: MRS
   - Second Step: SMM
3. Simulations

Simulations

- Simulate **transitory and permanent** wage changes.
  Men and women separately
- **Transitory:** Approximate Frisch (since little income effect)
- **Permanent:** Approximate Marshall

Consumption and Labor Supply Responses

- Age 30 response from 10% increase in wage in two models
  With child from age 28 + Without child from age 28 (elasticities)

TABLE 5
CONSUMPTION AND LABOR SUPPLY RESPONSES TO TRANSITORY AND PERMANENT SHOCKS

	TOTAL RESPONSE						EXTENSIVE VS. INTENSIVE MARGIN			
	C		H_1		H_2		E_2		H_2|Employed	
	With Kids (1)	Without Kids (2)	With Kids (3)	Without Kids (4)	With Kids (5)	Without Kids (6)	With Kids (7)	Without Kids (8)	With Kids (9)	Without Kids (10)
Transitory:										
Husband	.119	.123	.180	.222	−.076	.001	−.051	.005	−.041	.006
Wife	.130	.135	.000	−.006	.703	.394	.574	.280	.329	.167
Permanent:										
Husband	.393	.410	.105	.116	−.296	−.140	−.193	−.065	−.170	−.088
Wife	.353	.375	−.070	−.106	.531	.304	.491	.266	.208	.086

NOTE.—Model-simulated responses for transitory and permanent shocks.

1. Consumption response consistent with buffer-stock theory: transitory shocks have little effect
2. Women have larger responses than men
3. Children increases response for women
4. Extensive margin important (for women)

Leisure and Parental Time Responses

TABLE 6
LEISURE AND PARENTAL TIME RESPONSES TO TRANSITORY AND PERMANENT SHOCKS

	L_1		L_2		T_1	T_2
	With Kids (1)	Without Kids (2)	With Kids (3)	Without Kids (4)	With Kids (5)	With Kids (6)
Transitory:						
Husband	−.230	−.231	−.003	−.001	−.095	.131
Wife	−.007	.006	−.217	−.309	.033	−.538
Permanent:						
Husband	−.131	−.120	.078	.110	−.067	.261
Wife	.085	.110	−.151	−.238	.058	−.443

NOTE.—Model-simulated responses for transitory and permanent shocks.

1. Leisure elasticities similar between men/women w/w.o. kids and compliments (same-sign cross trans ela)
2. Permanent → reduction in both own leisure and child care time and opposite sign cross elasticity → specialization.
3. Women have large responses on child-care time from own and male wages.

Consumption Insurance

TABLE 7
INSURANCE EFFECTS

Consumption	−3.9%	
After-tax and transfers household earnings	−5.0%	
Before-tax (after-transfers) household earnings	−5.6%	
	Husband	Wife
Earner's average share of before-tax earnings	.66	.34
Earner's before-tax and transfers earnings response:	−10.7%	+2.0%
Hours	−1.0%	+3.0%
Leisure	+1.3%	−.8%
Parental time	+.7%	−2.6%

NOTE.—Insurance decomposition calculations based on model-simulated responses to a 10 percent permanent decline in the husband's wage.

1. Some consumption insurance (3.9% drop from 10% drop in wages)
2. Substitution effect dominates (-1% in hours worked)
3. Sizable cross-effect (+3% in work hours of women)
4. Leisure margin most active for men, parent time most for women.

Counterfactual Simulations

Two counterfactuals with same budget effects:

1. unconditional child-subsidy, $b(z) \uparrow$
2. employment subsidy, $\gamma(z) \downarrow$

TABLE 10
POLICY EXPERIMENTS

	C (1)	H_1 (2)	H_2 (3)	E_2 (4)	L_1 (5)	L_2 (6)	T_1 (7)	T_2 (8)
	A. Experiment 1: Unconditional Subsidy for Families with Young Children							
Total	.6%	−.4%	−.7%	−.4%	.4%	.3%		
Before young children	.9%	−.4%	−.5%	−.2%	.4%	.4%		
With young children	1.3%	−.6%	−1.8%	−1.0%	.8%	.7%	.2%	1.0%
After young children	.1%	−.1%	−.1%	−.1%	.1%	.1%		
Consumption equivalent utility value	.95%							
	B. Experiment 2: Employment Subsidy for Wives with Young Children							
Total	.1%	−.2%	1.9%	4.6%	.2%	−.5%		
Before young children	.9%	−.4%	−.5%	−.1%	.4%	.4%		
With young children	−.3%	−.3%	6.5%	13.1%	.3%	−1.7%	.3%	−5.6%
After young children	.1%	−.1%	−.1%	~0%	.1%	.1%		
Consumption equivalent utility value	.17%							

Toy Model I

Unitary model, couples jointly **choose**

- labor supply of both, $l_{j,t} \in [0, 24]$, $j \in \{f, m\}$,
- home production of both, $h_{j,t} \in [0, 24]$, $j \in \{f, m\}$, subject to time-constraint $T_{j,t} = l_{j,t} + h_{j,t} \leq 24$

State variables

- human capital of both, $K_{j,t} \geq 0$
- child present, $n_t \in \{0, 1\}$

Toy Model II

- Couples care about market goods, C_t, and home produced goods, H_t, through the composite good

$$Q_t = C_t^{\omega} H_t^{1-\omega}$$

where $\omega = \omega_0 + \omega_n n_t \in (0, 1)$ is the relative weight on market goods.

- Home production is a (CES) function,

$$H_t = \left(\alpha h_{f,t}^{\frac{\sigma-1}{\sigma}} + (1-\alpha) h_{m,t}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

where $\sigma \in (0, 1)$ is the elasticity of substitution and $\frac{\alpha}{1-\alpha} \in (0, 1)$ is the productivity in home production of females relative to males (her absolute advantage in home production, AAH).

Toy Model II

- Household preferences over work and consumption are

$$U(T_{f,t}, T_{m,t}, Q_t) = \frac{Q_t^{1-\rho}}{1-\rho} - \nu \left(\frac{T_{f,t}^{1+\frac{1}{\epsilon_f}}}{1+\frac{1}{\epsilon_f}} + \frac{T_{m,t}^{1+\frac{1}{\epsilon_m}}}{1+\frac{1}{\epsilon_m}} \right) \tag{1}$$

  in which the last term is the dis-utility from total work hours ($\nu > 0$) and $\epsilon_j > 0$ controls the curvature of the dis-utility. $\rho > 1$ is the constant relative risk aversion coefficient.

- There are no bequest motive and thus

$$V_{T+1}(n_{t+1}, K_{f,T+1}, K_{m,T+1}) = 0. \tag{2}$$

Toy Model III

- The recursive formulation of the model is for $t \leq T$:

$$
\begin{aligned}
V_t(n_t, K_{f,t}, K_{m,t}) &= \max_{l_{f,t}, h_{f,t}, l_{m,t}, h_{m,t}} U(T_{f,t}, T_{m,t}, Q_t) + \beta \mathbb{E}_t[V_{t+1}(n_{t+1}, K_{f,t+1}, K_{m,t+1})] && (3)\\
&\text{s.t.} && (4)\\
C_t &= w_{f,t} l_{f,t} + w_{m,t} l_{m,t} + X_t && (5)\\
H_t &= \left(\alpha h_{f,t}^{\frac{\sigma-1}{\sigma}} + (1-\alpha) h_{m,t}^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}} && (6)\\
Q_t &= C_t^{\omega} H_t^{1-\omega} && (7)\\
\log w_{j,t} &= \gamma_{j,0} + \gamma_{j,1} K_{j,t},\ j \in \{f, m\} && (8)\\
K_{j,t+1} &= (1-\delta) K_{j,t} + l_{j,t}/24,\ j \in \{f, m\} && (9)\\
n_{t+1} &= \begin{cases} n_t + 1 & \text{with prob. } p \cdot (1 - n_t) \\ n_t & \text{with prob. } 1 - p \cdot (1 - n_t) \end{cases} && (10)\\
T_{j,t} &= l_{j,t} + h_{j,t},\ j \in \{f, m\} && (11)\\
l_{j,t}, h_{j,t} &\geq 0,\ j \in \{f, m\} && (12)\\
T_{j,t} &\leq 24,\ j \in \{f, m\} && (13)
\end{aligned}
$$

Exam: Upload

You should hand in a single zip-file with all assignments and the exam.
The zip-file should be named after your KU username (e.g. abs123) and have the following folder and file structure:

Assignment_1\
Assignment_1.pdf - with text and all results
files for reproducing the results
Assignment_2\
Assignment_2.pdf - with text and all results
files for reproducing the results
Assignment_3\
Assignment_3.pdf - with text and all results
files for reproducing the results
Exam\
Exam.pdf - with text and all results
files for reproducing the results

- **Individual exam!**

Exam: Tips

- **Try to answer all questions**
  48 hours, but thought of as 2×9 work days
  Make sure that your computer+Python works!
  Similar flavor as assignments

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- **Thanks for now - Good luck!**

References I

Blundell, R., L. Pistaferri and I. Saporta-Eksten (2018): "Children, Time Allocation, and Consumption Insurance," *Journal of Political Economy*, 126(S1), S73–S115.

Siminski, P. and R. Yetsenga (2022): "Specialization, ComparativeAdvantage, and the Sexual Divisionof Labor," *Journal of Labor Economics*, 40(4), 851–887.

CES and the Elasticity of Substitution.

Thomas Jørgensen

May 14, 2024

Motivated by the functional forms in Blundell, Pistaferri and Saporta-Eksten (2018), let child production be governed by the CES function

$$Q = -\frac{1}{1-\rho}\left[\alpha_1 T_1^{1-1/\phi_1} + \alpha_2 T_2^{1-1/\phi_2}\right]^{1-\rho}$$

where $\rho < 1$, $\phi_1 \in (0,1)$ (and potentially $\alpha_2 = 1-\alpha_1$). The authors state that if $\rho < 0$ the two inputs are substitutes and if $\rho > 0$ the two inputs are complements. We will look into this below through a simple static model.

The budget constraint is

$$C \leq (1-T_1)W_1 + (1-T_2)W_2$$

where C is consumption and hours worked is normalized to one. The budget constraint can be written as

$$Z \leq T_1 W_1 + T_2 W_2$$

where $Z = W_1 + W_2 - C$ is potential income, which we will condition on and thus ignore consumption maximization here.

The Lagrangian is (in which λ is the shadow price on potential income)

$$\mathcal{L} = -\frac{1}{1-\rho}\left[\alpha_1 T_1^{1-1/\phi_1} + \alpha_2 T_1^{1-1/\phi_2}\right]^{1-\rho} + \lambda[Z - T_1 W_1 - T_2 W_2]$$

with FOCS for $j \in \{1,2\}$

$$\frac{\partial \mathcal{L}}{\partial T_j} = -\left[\alpha_1 T_1^{1-1/\phi_1} + \alpha_2 T_1^{1-1/\phi_2}\right]^{-\rho}(1-1/\phi_j)\alpha_j T_j^{-1/\phi_j} - \lambda W_j = 0.$$

Combining the FOCS we get

$$\frac{W_1}{W_2} = \frac{\left[\alpha_1 T_1^{1-1/\phi_1} + \alpha_2 T_1^{1-1/\phi_2}\right]^{-\rho}(1-1/\phi_1)\alpha_1 T_1^{-1/\phi_1}}{\left[\alpha_1 T_1^{1-1/\phi_1} + \alpha_2 T_1^{1-1/\phi_2}\right]^{-\rho}(1-1/\phi_2)\alpha_2 T_2^{-1/\phi_2}}$$

which gives

$$T_1 = \left[\frac{(1-1/\phi_2)\alpha_2}{(1-1/\phi_1)\alpha_1}\right]^{-\phi_1} \left[\frac{W_1}{W_2}\right]^{-\phi_1} T_2^{\phi_1/\phi_2}. \tag{1}$$

Already here we see that the relative inputs do not depend on ρ. Let the **elasticity of substitution** be the percentage change in the relative inputs from a percentage change in the (inverse) relative prices,

$$\varepsilon = \frac{\partial T_1/T_2}{\partial W_2/W_1}\frac{W_2/W_1}{T_1/T_2}$$

where eq. (1) gives us

$$\begin{aligned}\frac{\partial T_1/T_2}{\partial W_2/W_1} &= \frac{\partial}{\partial W_2/W_1}\left[\frac{(1-1/\phi_2)\alpha_2}{(1-1/\phi_1)\alpha_1}\right]^{-\phi_1} \left[\frac{W_2}{W_1}\right]^{\phi_1} T_2^{\phi_1/\phi_2-1}\\ &= \phi_1\left[\frac{(1-1/\phi_2)\alpha_2}{(1-1/\phi_1)\alpha_1}\right]^{-\phi_1} \left[\frac{W_2}{W_1}\right]^{\phi_1-1} T_2^{\phi_1/\phi_2-1}\end{aligned}$$

and we get

$$\varepsilon = \phi_1\left[\frac{(1-1/\phi_2)\alpha_2}{(1-1/\phi_1)\alpha_1}\right]^{-\phi_1} \left[\frac{W_2}{W_1}\right]^{\phi_1} T_2^{\phi_1/\phi_2-1}\frac{T_2}{T_1}$$

where inserting

$$\frac{T_2}{T_1} = \left[\frac{(1-1/\phi_1)\alpha_1}{(1-1/\phi_2)\alpha_2}\right]^{-\phi_2} \left[\frac{W_2}{W_1}\right]^{-\phi_2} T_1^{\phi_2/\phi_1-1}$$

finally gives

$$\varepsilon = \phi_1\left[\frac{(1-1/\phi_1)\alpha_1}{(1-1/\phi_2)\alpha_2}\frac{W_2}{W_1}\right]^{\phi_1-\phi_2} T_2^{\phi_1/\phi_2-1}T_1^{\phi_2/\phi_1-1} \tag{2}$$

which, if $\phi_1 = \phi_2 = \phi$, simplifies to

$$\varepsilon = \phi.$$

This is not ρ... It could be interesting to insert the estimated parameters from Blundell, Pistaferri and Saporta-Eksten (2018) in e.q. (2) for the parental time and the leisure time to see if it aligns with the story that leisure time is complements (they estimate $\rho_L > 0$) and parental time is substitutes ($\rho_T < 0$). I think it will because this is also what the simulations suggest.

References

Blundell, R., L. Pistaferri and I. Saporta-Eksten (2018): "Children, Time Allocation, and Consumption Insurance," *Journal of Political Economy*, 126(S1), S73–S115.